

3

Multiple Integrals and its Applications

MULTIPLE INTEGRALS

3.1. Introduction

Integration is an important concept in mathematics together with its inverse, differentiation.

It is one of the two main operations in calculus. Given a function f of a real variable x and an

interval $[a, b]$ of the real line, the **definite integral** $\int_a^b f(x) dx$ is defined informally to be the signed area of the region in the xy -plane bounded by the graph of f , the x -axis, and the vertical lines $x = a$ and $x = b$, such that area above the x -axis adds to the total, and that below the x -axis subtracts from the total.

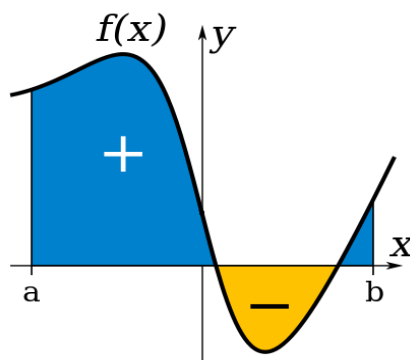


Fig. 3.1.

Definite integral of a function can be represented as the signed area of the region bounded by its graph.

Consider a function F whose derivative is the given function f . In this case, it is called an indefinite integral and is written as $F = \int f(x)dx$.

In calculus of a single variable the definite integral $\int_a^b f(x)dx$ for $f(x) \geq 0$ is the area under the curve $f(x)$ from $x = a$ to $x = b$. For general $f(x)$, the definite integral is equal to the area above the x -axis minus the area below the x -axis.

The definite integral can be extended to functions of more than one variable. Consider a function of two variables $z = f(x,y)$. The definite integral is denoted by $\iint_R f(x,y)dA$ where R

is the region of integration in the xy -plane. For positive $f(x,y)$, the definite integral is equal to the volume under the surface $z = f(x,y)$ and above xy -plane for x and y in the region R . This is shown in the figure below.

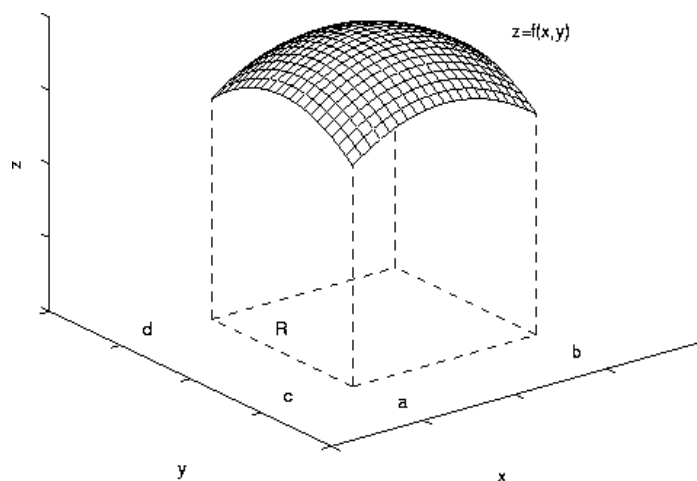


Fig. 3.2.

For general $f(x,y)$, the definite integral is equal to the volume above the xy -plane minus the volume below the xy -plane.

3.2. Engineering Applications of Integration

Double integrals arise in a number of areas of science and engineering, including computations of

- Area of a 2D region
- Volume
- Mass of 2D plates
- Force on a 2D plate
- Average of a function
- Center of Mass and Moment of Inertia
- Surface Area

3.3. Double Integral

Suppose that $f(x,y)$ is continuous on a rectangular region in the xy plane as shown below. The double integral $\iint_R f(x,y)dA$ represents the volume under the surface.

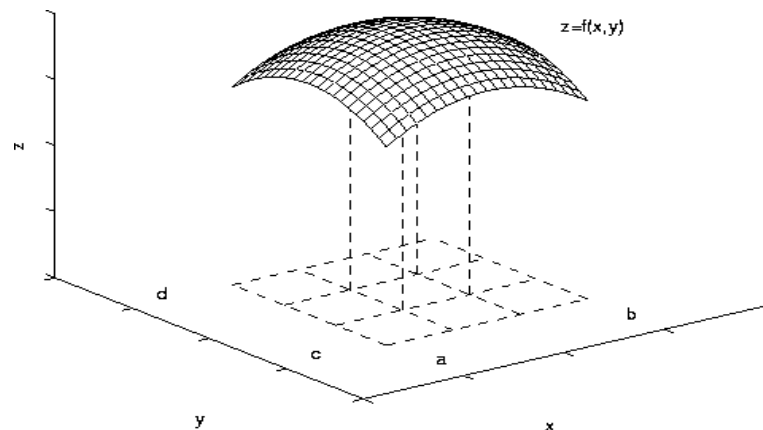


Fig. 3.3.

Definite integrals: The definite integral $\int_a^b f(x) dx$ represents the function $f(x)$ is integrated with respect to x between the limits a and b .

Double integrals: The double integral of a function $f(x, y)$ over a region D in R^2 is denoted by $\iint_D f(x, y) dx dy$

Note1

The order in which the integration are performed in the double integral is $\int_{y_1}^{y_2} \int_{x_1}^{x_2} f(x, y) dx dy$

3.3.1. Double integrals in polar co-ordinates

The double integral in polar co-ordinates of a function $f(r, \theta)$ is $\int_{\theta_1}^{\theta_2} \int_{r_1}^{r_2} f(r, \theta) dr d\theta$. We

first integrate $f(r, \theta)$ with respect to r , keeping θ is constant, between the limits r_1 and r_2 then the resulting expression is to be integrated with respect to θ between the limits θ_1 and θ_2 , r_1 and r_2 will be functions of θ and θ_1, θ_2 will be constants. The triple integral in

polar co-ordinates of a function $f(r, \theta, \phi)$ is $\int_{\phi_1}^{\phi_2} \int_{\theta_1}^{\theta_2} \int_{r_1}^{r_2} f(r, \theta, \phi) dr d\theta d\phi$. We first integrate

$f(r, \theta, \phi)$ with respect to r , between the limits r_1 and r_2 , keeping θ and ϕ is constant. Then the resulting expression is to be integrated with respect to θ between the limits θ_1 and θ_2 , keeping ϕ is constant. The result thus obtained is finally integrated with respect to ϕ between the limits ϕ_1 and ϕ_2 . In the integral ϕ_1 and ϕ_2 are constants, θ_1 and θ_2 are either constants or functions of ϕ and r_1 and r_2 are either constants or functions of θ and ϕ .

3.3.2. Geometrical Meaning of Double Integrals:

Consider $I_1 = \int_a^b \int_{f_1(y)}^{f_2(y)} f(x, y) dy dx$

Let APB and DQC be the two curves whose equations are $y = f_1(x)$ and $y = f_2(x)$ respectively. PQ is a vertical strip with thickness dx .

The first integration $\int_{f_1(y)}^{f_2(y)} f(x, y) dy$ is along one edge of the strip PQ from P to Q, x

remaining constant, while the second integration, after multiplying by the breadth dx , corresponds to the sliding of the edge from the AD to the position BC. The whole region of integration is the area ABCD. (Refer Fig. 3.4)

Fig 3.4.

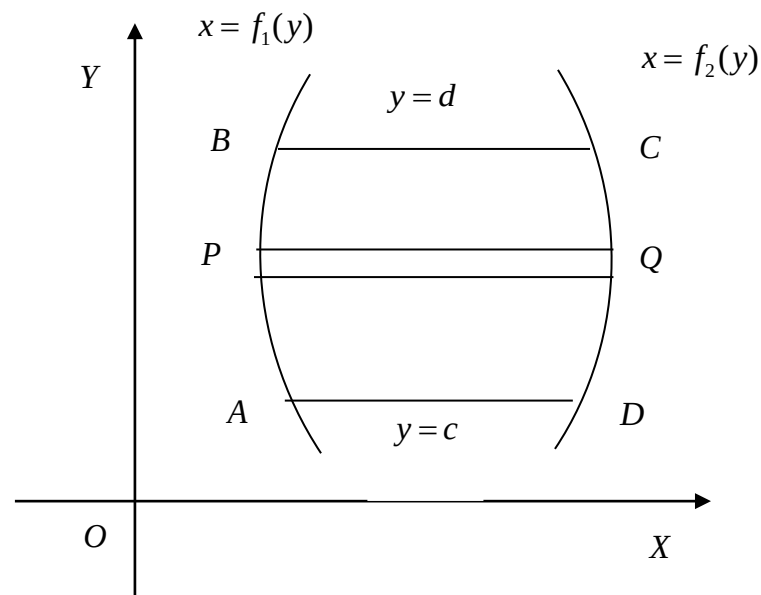
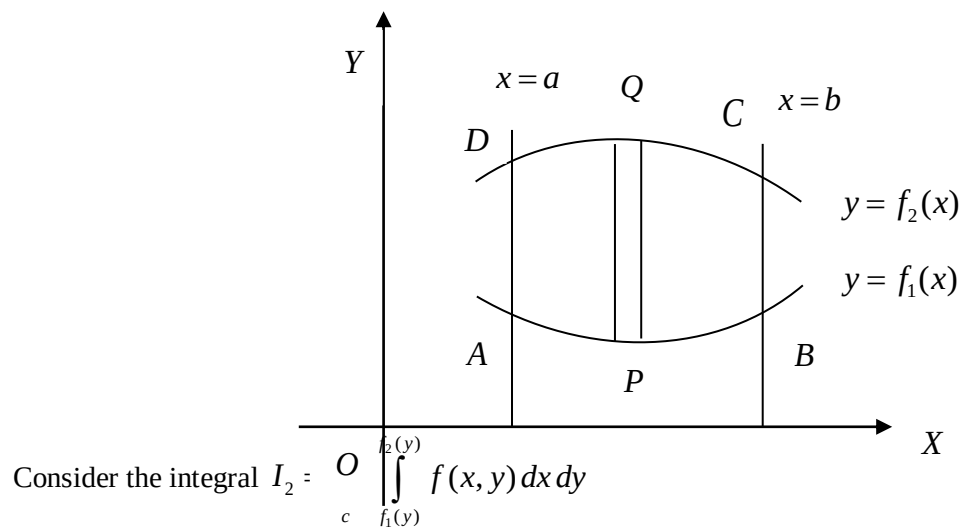


Fig. 3.5.

Here APB and DQC are two curves whose equations are $x = f_1(y)$ and $x = f_2(y)$ respectively. PQ is a horizontal strip, width dy . The first integration is with respect to x keeping y constant and so it is along one edge of the strip PQ and the second integration after multiplying by dy corresponds to the sliding of this edge from the position AD to the position BC. The whole region of integration is the area ABCD. Suppose both pairs of limits are constants. The two integrals now become,

$$I_1 = \int_a^b \int_c^d f(x, y) dy dx \quad \dots(1)$$

$$\text{and } I_2 = \int_c^d \int_a^b f(x, y) dx dy \quad \dots(2)$$

The region of integration is the rectangle ABCD Shown in Fig. 3.6.

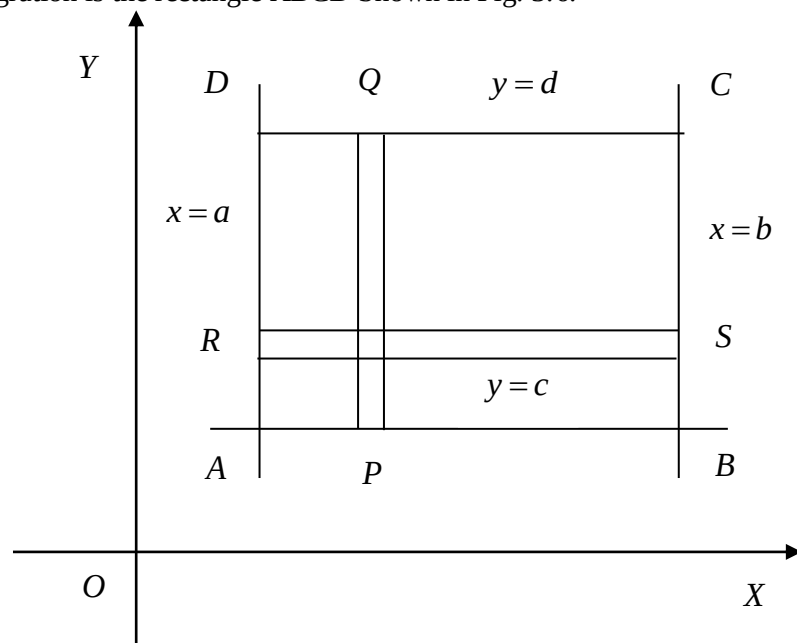


Fig. 3.6

To find (1) , first we integrate along an edge of the vertical strip PQ and then slide this edge from AD to BC. To find (2), first we integrate along an edge of the vertical strip RS and then slide this edge from AB to DC. Clearly $I_1 = I_2$

$$(i.e) \int_a^b \int_c^d f(x, y) dx dy = \int_c^d \int_a^b f(x, y) dx dy$$

In other words, for constant limits, the order of integration is immaterial.

Example 1: Evaluate $\int_0^a \int_0^b xy(x-y)dydx$ **(Jan -2010)**

Solution:

$$\begin{aligned} \int_0^a \int_0^b xy(x-y)dydx &= \int_0^a \int_0^b (x^2y - xy^2) dydx \\ &= \int_0^a \left[\frac{x^2y^2}{2} - \frac{xy^3}{3} \right]_0^b dx = \int_0^a \left(\frac{x^2b^2}{2} - \frac{xb^3}{3} \right) dx \\ &= \left[\frac{x^3b^2}{6} - \frac{x^2b^3}{6} \right]_0^a = \frac{a^3b^2}{6} - \frac{a^2b^3}{6} \\ &= \frac{a^2b^2(a-b)}{6} \end{aligned}$$

Example 2: Evaluate $\int_0^1 \int_1^2 (x^2 + y^2) dx dy$

Solution:

$$\begin{aligned}
 \int_0^1 \int_1^2 (x^2 + y^2) dx dy &= \int_0^1 \left[\frac{x^3}{3} + y^2 x \right]_1^2 dy \\
 &= \int_0^1 \left(\frac{8}{3} + 2y^2 - \frac{1}{3} - y^2 \right) dy = \left[\frac{8}{3}y + \frac{2y^3}{3} - \frac{1}{3}y - \frac{y^3}{3} \right]_0^1 \\
 &= \frac{8}{3} + \frac{2}{3} - \frac{1}{3} - \frac{1}{3} = \frac{10}{3} - \frac{2}{3} = \frac{8}{3}
 \end{aligned}$$

Example 3: Evaluate $\int_0^1 \int_x^1 x^2 y dx dy$

Solution:

$$\begin{aligned}
 \int_0^1 \int_x^1 x^2 y dx dy &= \int_0^1 \left(\frac{x^2 y^2}{2} \right)_x^1 dx \\
 &= \int_0^1 \left(\frac{x^2}{2} - \frac{x^4}{2} \right) dx = \left[\frac{x^3}{6} - \frac{x^5}{10} \right]_0^1 \\
 &= \frac{1}{6} - \frac{1}{10} = \frac{4}{60} = \frac{1}{15}
 \end{aligned}$$

Example 4: Evaluate $\iint xy(x+y) dx dy$ **over the region R bounded by** $y = x^2$ **and** $y = x$

(May-2011)

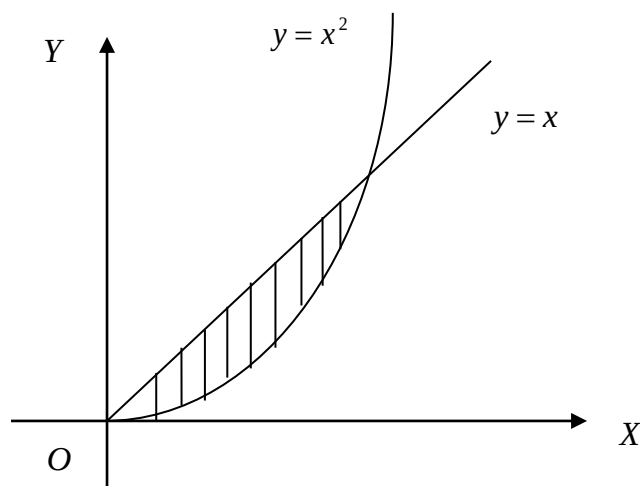
Solution:

The point of intersection at the curves are given by (0,0) and (1,1). The required region D is as shown in Fig. 3.7.

In this region x varies from 0 to 1

y varies from x^2 to x .

$$\iint xy(x+y) dx dy = \int \int (x^2 y + xy^2) dx dy$$



$$\begin{aligned}
&= \int_0^1 \int_{x^2}^x (x^2 y + xy^2) dy dx \\
&= \int_0^1 \left[\frac{x^2 y^2}{2} + \frac{xy^3}{3} \right]_{x^2}^x dx \\
&= \int_0^1 \left[\frac{x^4}{2} + \frac{x^4}{3} - \frac{x^6}{2} - \frac{x^7}{3} \right] dx \\
&= \int_0^1 \frac{3x^4 + 2x^4 - 3x^6 - 2x^7}{6} dx
\end{aligned}$$

Fig.3.7

$$\begin{aligned}
&= \frac{1}{6} \left[\frac{5x^5}{5} - \frac{3x^7}{7} - \frac{2x^8}{8} \right]_0^1 = \frac{1}{6} \left[1 - \frac{3}{7} - \frac{2}{8} \right] \\
&= \frac{1}{6} \left[\frac{56 - 24 - 14}{56} \right] \\
&= \frac{1}{6} \times \frac{18}{56} = \frac{3}{56}
\end{aligned}$$

Example 5: Evaluate $\int_1^b \int_1^a \frac{dx dy}{xy}$

(May-2011)

Solution:

$$\begin{aligned}
\int_1^b \int_1^a \frac{dx dy}{xy} &= \int_1^b \frac{dy}{y} \int_1^a \frac{dx}{x} \\
&= \int_1^b \frac{dy}{y} [\log x]_1^a \\
&= (\log a - \log 1) \int_1^b \frac{dy}{y}
\end{aligned}$$

$$\begin{aligned}
 &= \log a [\log y]_1^b \\
 &= \log a \cdot \log b = \log a + b
 \end{aligned}$$

Example 6: Evaluate $\int_0^a \int_0^{\sqrt{a^2-x^2}} dy dx$

Solution:

$$\begin{aligned}
 \int_0^a \int_0^{\sqrt{a^2-x^2}} dy dx &= \int_0^a [y]_0^{\sqrt{a^2-x^2}} dx \\
 &= \int_0^a \sqrt{a^2-x^2} dx \\
 &= \left[\frac{a^2}{2} \sin^{-1}\left(\frac{x}{a}\right) + \frac{x}{2} \sqrt{a^2-x^2} \right]_0^a \\
 &= \frac{a^2}{2} \sin^{-1}\left(\frac{a}{a}\right) + \frac{a}{2} \sqrt{a^2-a^2} \\
 &= \frac{a^2}{2} \frac{\pi}{2} = \frac{a^2 \pi}{4}
 \end{aligned}$$

Example 7: Evaluate $\int_0^1 \int_0^{\sqrt{1+x^2}} \frac{dy dx}{1+x^2+y^2}$

Solution:

$$\begin{aligned}
 \int_0^1 \int_0^{\sqrt{1+x^2}} \frac{dy dx}{1+x^2+y^2} &= \int_0^1 \left(\int_0^{\sqrt{1+x^2}} \frac{dy}{(\sqrt{1+x^2})^2 + y^2} \right) dx \\
 &\quad \left(\because \int_0^1 \frac{dy}{a^2 + y^2} = \frac{1}{a} \tan^{-1}\left(\frac{y}{a}\right) \right)
 \end{aligned}$$

$$\begin{aligned}
&= \int_0^1 \left[\frac{1}{\sqrt{1+x^2}} \tan^{-1} \left(\frac{y}{\sqrt{1+x^2}} \right) \right]_0^{\sqrt{1+x^2}} dx \\
&= \int_0^1 \left[\frac{1}{\sqrt{1+x^2}} \tan^{-1} 1 - \tan^{-1} 0 \right] dx \\
&= \int_0^1 \frac{1}{\sqrt{1+x^2}} \frac{\pi}{4} dx \\
&= \frac{\pi}{4} \left[\log \left(x + \sqrt{1+x^2} \right) \right]_0^1 \\
&= \frac{\pi}{4} \left[\log(1 + \sqrt{2}) - \log 1 \right] \\
&= \frac{\pi}{4} \log(1 + \sqrt{2})
\end{aligned}$$

Example 8: Evaluate $\int_0^a \int_0^{\sqrt{a^2-x^2}} \frac{dydx}{\sqrt{a^2-x^2-y^2}}$

Solution:

$$\begin{aligned}
\int_0^a \int_0^{\sqrt{a^2-x^2}} \frac{dydx}{\sqrt{a^2-x^2-y^2}} &= \int_0^a \left[\int_0^{\sqrt{a^2-x^2}} \frac{dy}{\sqrt{a^2-x^2-y^2}} \right] dx \\
&= \int_0^a \left[\int_0^{\sqrt{a^2-x^2}} \frac{dy}{\sqrt{\left(\sqrt{a^2-x^2}\right)^2 - y^2}} \right] dx \\
&= \int_0^a \left[\sin^{-1} \frac{y}{\sqrt{a^2-x^2}} \right]_0^{\sqrt{a^2-x^2}} dx \quad \left(\because \int \frac{dy}{\sqrt{a^2-y^2}} = \sin^{-1} \frac{y}{a} \right)
\end{aligned}$$

$$\begin{aligned}
&= \int_0^a \left[\sin^{-1} \frac{\sqrt{a^2 - x^2}}{\sqrt{a^2 - x^2}} - \sin^{-1} 0 \right] dx \\
&= \int_0^a (\sin^{-1} 1 - 0) dx \\
&= \int_0^a \frac{\pi}{2} dx = \frac{\pi}{2} (x)_0^a \\
&= \frac{\pi}{2} a
\end{aligned}$$

Example 9: Evaluate $\int_0^a \int_0^{\sqrt{a^2 - y^2}} \sqrt{a^2 - x^2 - y^2} dy dx$

Solution:

$$\begin{aligned}
\int_0^a \int_0^{\sqrt{a^2 - y^2}} \sqrt{a^2 - x^2 - y^2} dy dx &= \int_0^a \left[\int_0^{\sqrt{a^2 - y^2}} \sqrt{\left(\sqrt{a^2 - y^2}\right)^2 - x^2} dx \right] dy \\
&= \int_0^a \left[\frac{\left(\sqrt{a^2 - y^2}\right)^2}{2} \sin^{-1} \frac{x}{\sqrt{a^2 - y^2}} + \frac{x}{2} \sqrt{\left(\sqrt{a^2 - y^2}\right)^2 - x^2} \right]_0^{\sqrt{a^2 - y^2}} dy \\
&= \int_0^a \left[\frac{\left(\sqrt{a^2 - y^2}\right)^2}{2} \sin^{-1}(1) + \frac{\sqrt{a^2 - y^2}}{2} \sqrt{\left(\sqrt{a^2 - y^2}\right)^2 - a^2 - y^2} \right] dy \\
&= \int_0^a \left(\frac{a^2 - y^2}{2} \frac{\pi}{2} + 0 \right) dy = \frac{\pi}{4} \left[a^2 y - \frac{y^3}{3} \right]_0^a
\end{aligned}$$

$$= \frac{\pi}{4} \left[a^3 - \frac{a^3}{3} \right]$$

$$= \frac{\pi}{4} \left(\frac{2a^3}{3} \right) = \frac{\pi a^3}{6}$$

Example 10: Evaluate $\iint_R xy dx dy$ over the positive quadrant of the circle

$$x^2 + y^2 = a^2$$

Solution:

$$x^2 + y^2 = a^2$$

x varies from 0 to $\sqrt{a^2 - y^2}$

y varies from 0 to a

The region of the integration is shown in Fig. 3.8

$$\begin{aligned} \iint_R xy dx dy &= \int_0^a \int_0^{\sqrt{a^2 - y^2}} xy dy \\ &= \int_0^a y \left(\frac{x^2}{2} \right)_0^{\sqrt{a^2 - y^2}} dy \\ &= \int_0^a y \left[\frac{a^2 - y^2}{2} \right] dy \\ &= \frac{1}{2} \int_0^a (a^2 y - y^3) dy \\ &= \frac{1}{2} \left[\frac{a^2 y^2}{2} - \frac{y^4}{4} \right]_0^a \end{aligned}$$

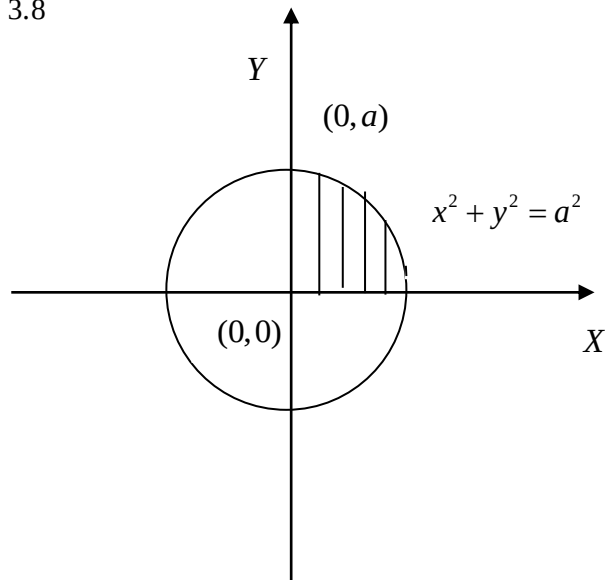


Fig. 3.8

$$= \frac{1}{2} \left[\frac{a^4}{2} - \frac{a^4}{4} \right] = \frac{1}{2} \left[\frac{a^4}{4} \right] = \frac{a^4}{8}$$

Example 11: Evaluate $\int_0^\pi \int_0^{a(1+\cos\theta)} r dr d\theta$

Solution:

$$\int_0^\pi \int_0^{a(1+\cos\theta)} r dr d\theta = \int_0^\pi \left[\frac{r^2}{2} \right]_0^{a(1+\cos\theta)} d\theta$$

$$= \int_0^\pi \frac{a^2 (1+\cos\theta)^2}{2} d\theta$$

$$= \int_0^\pi (1 + 2\cos\theta + \cos^2\theta) d\theta$$

$$\left(\because \cos^2\theta = \frac{1+\cos 2\theta}{2} \right)$$

$$= \frac{a^2}{2} \left[\theta + 2\sin\theta + \frac{\theta}{2} + \frac{\sin 2\theta}{4} \right]_0^\pi$$

$$= \frac{a^2}{2} \left[\pi + 2\sin\pi + \frac{\pi}{2} + \frac{\sin 2\pi}{4} \right] = \frac{a^2}{2} \left[\pi + \frac{\pi}{2} \right]$$

$$= \frac{a^2}{2} \left(\frac{3\pi}{2} \right)$$

$$\int_0^\pi \int_0^{a(1+\cos\theta)} r dr d\theta = \frac{3\pi a^2}{4}$$

Example 12: Evaluate $\int_{-\pi/2}^{\pi/2} \int_0^{2\cos\theta} r^2 dr d\theta$

Solution :

$$\begin{aligned}
 \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \int_0^{2\cos\theta} r^2 dr d\theta &= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \left(\frac{r^3}{3} \right)_0^{2\cos\theta} d\theta \\
 &= \frac{1}{3} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} (2\cos\theta)^3 d\theta = \frac{1}{3} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} 8\cos^3\theta d\theta \\
 &= \frac{8}{3} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos^3\theta d\theta = \frac{16}{3} \int_0^{\frac{\pi}{2}} \cos^3\theta d\theta \\
 &= \frac{32}{9}
 \end{aligned}$$

Example 13: Evaluate $\int_0^{\frac{\pi}{4}} \int_0^{\sqrt{\cos 2\theta}} \frac{r}{(1+r^2)^2} dr d\theta$

Solution:

$$\int_0^{\sqrt{\cos 2\theta}} \frac{r}{(1+r^2)^2} dr$$

Put $(1+r^2) = t$ $r = 0, t = 1$

$2rdr = dt$ $r = \sqrt{\cos 2\theta}, t = 1 + \cos 2\theta = 2\cos^2\theta$

$$\begin{aligned}
 \int_0^{\sqrt{\cos 2\theta}} \frac{r}{(1+r^2)^2} dr &= \int_1^{2\cos^2\theta} \frac{1}{t^2} \frac{dt}{2} \\
 &= \frac{1}{2} \left[-\frac{1}{t} \right]_1^{2\cos^2\theta} \\
 &= \frac{1}{2} \left[-\frac{1}{2\cos^2\theta} + \frac{1}{1} \right] = \frac{1}{2} \left[1 - \frac{1}{2\cos^2\theta} \right]
 \end{aligned}$$

$$\begin{aligned}
\therefore \int_0^{\frac{\pi}{4}} \int_0^{\sqrt{\cos 2\theta}} \frac{r}{(1+r^2)^2} dr d\theta &= \int_0^{\frac{\pi}{4}} \frac{1}{2} \left[1 - \frac{1}{2 \cos^2 \theta} \right] d\theta \\
&= \frac{1}{2} \left[\int_0^{\frac{\pi}{4}} d\theta - \frac{1}{2} \int_0^{\frac{\pi}{4}} \sec^2 \theta d\theta \right] \\
&= \frac{1}{2} \left[(\theta)_0^{\frac{\pi}{4}} - \frac{1}{2} (\tan \theta)_0^{\frac{\pi}{4}} \right] \\
&= \frac{1}{2} \left[\frac{\pi}{4} - \frac{1}{2} \right] = \frac{1}{2} \left[\frac{\pi - 2}{4} \right] \\
&= \frac{\pi - 2}{2}
\end{aligned}$$

Example 14: Evaluate $I = \int_0^a \int_0^{a \cos \theta} r \sin \theta dr d\theta$

Solution:

$$\begin{aligned}
I &= \int_0^{\pi} \sin \theta \int_0^{a \cos \theta} r dr d\theta \\
&= \int_0^{\pi} \sin \theta \left(\frac{r^2}{2} \right)_0^{a \cos \theta} d\theta = \int_0^{\pi} \sin \theta \frac{a^2 \cos^2 \theta}{2} d\theta \\
&= \frac{a^2}{2} \int_0^{\pi} \sin \theta \cos^2 \theta d\theta = \frac{-a^2}{2} \int_0^{\pi} \cos^2 \theta d(\cos \theta) \\
&= \frac{-a^2}{2} \left[\frac{\cos^3 \theta}{3} \right]_0^{\pi} \\
&= \frac{-a^2}{2} \left[\frac{\cos^3 \pi}{3} - \frac{\cos 0}{3} \right] = \frac{-a^2}{2} \left(\frac{-2}{3} \right)
\end{aligned}$$

$$= \frac{a^2}{3}$$

Example 15: Evaluate $I = \int_0^{\frac{\pi}{2}} \int_0^{\infty} \frac{r}{(r^2 + a^2)^2} dr d\theta$

Solution:

$$I = \int_0^{\frac{\pi}{2}} \int_0^{\infty} \frac{r}{(r^2 + a^2)^2} dr d\theta$$

Put $(r^2 + a^2) = t$ when $r = 0$, $t = a^2$

$2r dr = dt$ $r = \infty$, $t = \infty$

$$r dr = \frac{dt}{2}$$

$$\begin{aligned} I &= \int_0^{\frac{\pi}{2}} \int_{a^2}^{\infty} \frac{\left(\frac{dt}{2}\right)}{t^2} d\theta = \frac{1}{2} \int_0^{\frac{\pi}{2}} \int_{a^2}^{\infty} \frac{dt}{t^2} d\theta \\ &= \frac{1}{2} \int_0^{\frac{\pi}{2}} \left(\frac{t^{-1}}{-1} \right)_{a^2}^{\infty} d\theta = \frac{1}{2} \int_0^{\frac{\pi}{2}} \left(\frac{-1}{t^1} \right)_{a^2}^{\infty} d\theta \\ &= \frac{1}{2} \int_0^{\frac{\pi}{2}} \left(0 + \frac{1}{a^2} \right) d\theta = \frac{1}{2a^2} \int_0^{\frac{\pi}{2}} d\theta \\ &= \frac{1}{2a^2} [\theta]_0^{\frac{\pi}{2}} = \frac{1}{2a^2} \left(\frac{\pi}{2} \right) \\ &= \frac{\pi}{4a^2} \end{aligned}$$

Example 16: Evaluate $I = \int_0^{1/2} \int_0^1 \frac{x}{\sqrt{1-x^2y^2}} dy dx$

Solution:

$$\text{Given that } I = \int_0^{\frac{1}{2}} x \left[\frac{\sin^{-1}(xy)}{x} \right]_0^1 dx = \int_0^{\frac{1}{2}} \sin^{-1} x dx$$

$$\text{Let } u = \sin^{-1} x \quad dv = dx$$

$$du = \frac{1}{\sqrt{1-x^2}} \quad v = x$$

$$= \left[x \sin^{-1} x + \sqrt{1-x^2} \right]_0^{\frac{1}{2}} = \frac{\pi}{4} + \sqrt{\frac{3}{4}} - 1$$

$$= \frac{\pi}{4} + \frac{\sqrt{3}}{2} - 1$$

Example 17: Evaluate $\iint_R xy dx dy$ where R is the region taken over the positive quadrant

of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

Solution:

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

$$\frac{x^2}{a^2} = 1 - \frac{y^2}{b^2}$$

$$\frac{x^2}{a^2} = \frac{b^2 - y^2}{b^2}$$

The region of integration is shown in Fig. 3.9.

$$x^2 = a^2 \left(\frac{b^2 - y^2}{b^2} \right)$$

$$x = \pm \frac{a}{b} \sqrt{b^2 - y^2}$$

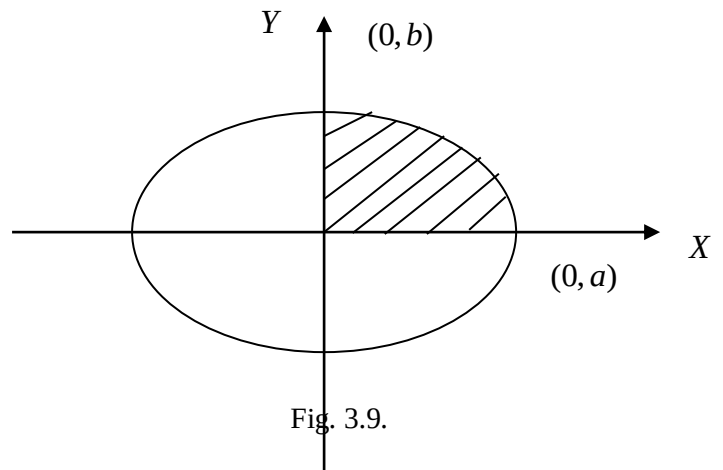


Fig. 3.9.

In positive quadrant, x varies from 0 to $\frac{a}{b}\sqrt{b^2 - y^2}$

y varies from 0 to b .

$$\begin{aligned}\iint_R xy dx dy &= \int_0^b \int_0^{\frac{a}{b}\sqrt{b^2 - y^2}} xy dx dy \\ &= \int_0^b y \left(\frac{x^2}{2} \right)_{x=0}^{\frac{a}{b}\sqrt{b^2 - y^2}} dy\end{aligned}$$

$$\begin{aligned}&= \frac{a^2}{2b^2} \left[\frac{b^2 y^2}{2} - \frac{y^4}{4} \right]_0^b = \frac{a^2}{2b^2} \left[\frac{b^4}{2} - \frac{b^4}{4} \right] \\ &= \frac{a^2}{2b^2} \left[\frac{2b^4 - b^4}{4} \right] = \frac{a^2}{2b^2} \left(\frac{b^4}{4} \right) \\ &= \frac{a^2 b^2}{8}\end{aligned}$$

Exercise

Evaluate the following double integrals

1.

$$\text{a. } \int_1^2 \int_0^x \frac{dx dy}{x^2 + y^2} \quad \text{b. } \int_4^2 \int_1^2 \frac{dy dy}{(x+y)^2} \quad \text{c. } \int_0^1 \int_0^x e^x dy dx \quad \text{d. } \int_0^a \int_0^{\sqrt{a^2 - x^2}} \frac{dx dy}{\sqrt{a^2 - x^2}}$$

$$\text{e. } \int_0^a \int_0^b (x^2 + y^2) dx dy \quad \text{f. } \int_0^a dx \int_0^{b\sqrt{1 - \left(\frac{x^2}{a^2}\right)}} x^3 y dy \quad \text{g. } \int_0^{\frac{1}{2}} \int_0^1 \frac{x}{\sqrt{1 - x^2 - y^2}} dy dx$$

$$\text{h. } \int_0^{\frac{\pi}{2}} \int_0^{\frac{\pi}{2}} (a \cos 2\theta + b \sin 2\theta) d\theta d\phi \quad \text{i. } \int_0^{\frac{\pi}{2}} \int_0^{\frac{\pi}{2}} \sin(x+y) dx dy \quad \text{j. } \int_{\frac{2}{2}}^a \int_{\frac{2}{2}}^b \frac{dx dy}{xy}$$

2. Find the value of $\iint x^2 y^2 dx dy$ over the circular area $x^2 + y^2 \leq 1$
3. Evaluate $\iint x^2 dx dy$ over the area of the circle $x^2 + y^2 = 1$ in the positive quadrant.
4. Evaluate $\iint_R xy dx dy$ taken over the region bounded by the co – ordinate axis and the line $x + 2y = 2$.
5. Evaluate $\iint (x^2 + y^2) dx dy$ over the area bounded by the curve $y = a^4, x + y = 3, y = 0, y = 2$
6. Evaluate a. $\int_0^{2a} r dr d\theta$ b. $\int_0^{\frac{\pi}{2}} \int_0^{a \cos \theta} r \sqrt{a^2 - r^2} dr d\theta$ c. $\int_0^{\frac{\pi}{2}} \int_0^{4 \sin \theta} r dr d\theta$.

d. $\int_0^{\frac{\pi}{2}} \int_0^{a \cos \theta} r^4 dr d\theta$ e. $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \int_0^{a \cos \theta} r^4 dr d\theta$

Answers

- 1.
- a. $\frac{\pi}{4} \log 2$ b. $\log \frac{25}{24}$ c. 0.859 d. a e. $\frac{1}{3} ab(a^2 + b^2)$
- f. $\frac{a^4 b^2}{24}$ g. $\frac{\pi}{12} + \frac{\sqrt{3}}{2} - 1$ h. $\frac{b\pi}{2}$ i. 3 j. $\log \frac{a}{2} \log \frac{b}{2}$
2. $\frac{\pi}{96}$ 3. $\frac{\pi}{16}$ 4. $\frac{1}{6}$ 5. $\frac{463}{48}$

6. a) πa^2 b) $\frac{a^3}{3} \left(\frac{\pi}{2} - \frac{2}{3} \right)$ c) 2π d) $\frac{8a^5}{75}$ e) $\frac{32}{9}$

3.4. Triple integral

The Triple integral of a function $f(x, y, z)$ over a region D in R^3 is denoted by

$$\iiint_D f(x, y, z) dx dy dz$$

Note2

The order in which the integration are performed is the Triple integral is

$$\int_{z_1}^{z_2} \int_{y_1}^{y_2} \int_{x_1}^{x_2} f(x, y, z) dx dy dz$$

3.4.1. Triple integrals in polar co-ordinates

The triple integral in polar co-ordinates of a function $f(r, \theta, \phi)$ is

$$\int_{\phi_1}^{\phi_2} \int_{\theta_1}^{\theta_2} \int_{r_1}^{r_2} f(r, \theta, \phi) dr d\theta d\phi .$$

We first integrate $f(r, \theta, \phi)$ with respect to r , between the

limits r_1 and r_2 , keeping θ and ϕ is constant . Then the resulting expression is to be integrated with respect to θ between the limits θ_1 and θ_2 , keeping ϕ is constant. The result thus obtained is finally integrated with respect to ϕ between the limits ϕ_1 and ϕ_2 . In the integral ϕ_1 and ϕ_2 are constants, θ_1 and θ_2 are either constants or functions of ϕ alone and r_1 and r_2 are either constants or functions of θ and ϕ .

Example 1: Evaluate $\int_0^1 \int_0^2 \int_1^2 xyz^2 dz dy dx$

(Jan-2010)

Solution:

$$\begin{aligned} \int_0^1 \int_0^2 \int_1^2 xyz^2 dz dy dx &= \int_0^1 \int_0^2 \left(\frac{xyz^3}{3} \right)_1^2 dy dx \\ &= \int_0^1 \int_0^2 \left(\frac{8xy}{3} - \frac{xy}{3} \right) dy dx \\ &= \int_0^1 \left(\frac{8xy^2}{6} - \frac{xy^2}{6} \right)_0^2 dx \\ &= \int_0^1 \left(\frac{32x}{6} - \frac{4x}{6} \right) dx = \int_0^1 \frac{28}{6} x dx \\ &= \frac{14}{3} \left[\frac{x^2}{2} \right]_0^1 = \frac{14}{3} \times \frac{1}{2} \\ &= \frac{7}{3} \end{aligned}$$

Example 2: Evaluate $\int_0^1 \int_0^2 \int_0^3 xyz dz dy dx$

(May-2011)

Solution:

$$\begin{aligned} \int_0^1 \int_0^2 \int_0^3 xyz dz dy dx &= \int_0^1 \int_0^2 \left(\frac{xyz^2}{2} \right)_0^3 dy dx \\ &= \int_0^1 \int_0^2 \frac{9xy}{2} dy dx = \frac{9}{2} \int_0^1 \left(\frac{xy^2}{2} \right)_0^2 dx \\ &= \frac{9}{2} \int_0^1 \frac{4}{2} x dx = 9 \left(\frac{x^2}{2} \right)_0^1 = \frac{9}{2} \end{aligned}$$

Example 3: Evaluate $\int_0^a \int_0^b \int_0^c (x + y + z) dx dy dz$

(Nov-2011)

Solution:

$$\begin{aligned}
 \int_0^a \int_0^b \int_0^c (x + y + z) dx dy dz &= \int_0^a \int_0^b \left[\left(\frac{x^2}{2} + xy + zx \right) \right]_0^c dy dz \\
 &= \int_0^a \int_0^b \left(\frac{c^2}{2} + cy + zc \right) dy dz \\
 &= \int_0^a \int_0^b \left(\frac{c^2 y}{2} + \frac{cy^2}{2} + zyc \right)_0^b dz \\
 &= \int_0^a \left(\frac{c^2 b}{2} + \frac{cb^2}{2} + zbc \right) dz \\
 &= \left(\frac{c^2 b}{2} z + \frac{cb^2}{2} z + \frac{z^2 bc}{2} \right)_0^a \\
 &= \frac{abc^2}{2} + \frac{ab^2c}{2} + \frac{a^2bc}{2} \\
 &= \frac{abc}{2} (a + b + c)
 \end{aligned}$$

Example 4: Evaluate $\int_0^1 \int_0^{1-x} \int_0^{1-x-y} \frac{dz dy dx}{(x + y + z + 1)^3}$

Solution:

$$\int_0^1 \int_0^{1-x} \int_0^{1-x-y} (x + y + z + 1)^{-3} dz dy dx = \int_0^1 \int_0^{1-x} \left[\frac{(x + y + z + 1)^{-2}}{-2} \right]_0^{1-x-y} dy dx$$

$$\begin{aligned}
&= \int_0^1 \int_0^{1-x} \left[\frac{(x+y+1-x-y+1)^{-2}}{-2} + \frac{(x+y+1)^{-2}}{2} \right] dy dx \\
&= \int_0^1 \int_0^{1-x} \left(\frac{2^{-2}}{-2} + \frac{(1+x+y)^{-2}}{2} \right) dy dx \\
&= \int_0^1 \left(-\frac{1}{8}y + \frac{(1+x+y)^{-1}}{-2} \right) \Big|_0^{1-x} dx \\
&= \int_0^1 \left(-\frac{(1-x)}{8} + \frac{(1+x+1-x)^{-1}}{-2} \right) - \left[0 + \frac{(1+x)^{-1}}{+2} \right] dx \\
&= - \int_0^1 \left(\frac{1-x}{8} + \frac{1}{4} + \frac{(1+x)^{-1}}{2} \right) dx \\
&= - \left[\frac{\left(x - \frac{x^2}{2} \right)}{8} + \frac{1}{4}x + \frac{\log(1+x)}{2} \right] \Big|_0^1 \\
&= - \left[\left(\frac{1}{8} - \frac{1}{16} + \frac{1}{4} + \frac{\log 2}{2} \right) - \left(0 + \frac{\log 1}{2} \right) \right] \\
&= - \left[\frac{2-1+4-8\log 2}{16} \right] = \left[\frac{5-8\log 2}{16} \right] = \frac{1}{16} [8\log 2 - 5]
\end{aligned}$$

Example 5: Evaluate $\int_0^1 \int_0^{\sqrt{1-x^2}} \int_0^{\sqrt{1-x^2-y^2}} \frac{dx dy dz}{\sqrt{1-x^2-y^2-z^2}} = \frac{\pi^2}{8}$

(May-2002)

Solution:

$$\begin{aligned}
& \int_0^1 \int_0^{\sqrt{1-x^2}} \int_0^{\sqrt{1-x^2-y^2}} \frac{dx dy dz}{\sqrt{1-x^2-y^2-z^2}} = \int_0^1 \int_0^{\sqrt{1-x^2}} \left(\int_0^{\sqrt{1-x^2-y^2}} \frac{dz}{\sqrt{1-x^2-y^2-z^2}} \right) dy dx \\
&= \int_0^1 \int_0^{\sqrt{1-x^2}} \left[\int_0^{\sqrt{1-x^2-y^2}} \frac{dz}{\sqrt{\left(\left(\sqrt{1-x^2-y^2}\right)^2 - z^2\right)}} \right] dy dx \\
&= \int_0^1 \int_0^{\sqrt{1-x^2}} \left[\sin^{-1} \left[\frac{z}{\sqrt{1-x^2-y^2}} \right] \right]_0^{\sqrt{1-x^2-y^2}} dy dx \\
&= \int_0^1 \int_0^{\sqrt{1-x^2}} \left(\sin^{-1} \left(\frac{\sqrt{1-x^2-y^2}}{\sqrt{1-x^2-y^2}} \right) - 0 \right) dy dx \\
&= \int_0^1 \int_0^{\sqrt{1-x^2}} \frac{\pi}{2} dy dx \quad \left(\because \sin^{-1} x = \frac{\pi}{2} \right) \\
&= \int_0^1 \left(\frac{\pi}{2} y \right)_0^{\sqrt{1-x^2}} dx = \int_0^1 \frac{\pi}{2} \sqrt{1-x^2} dx \\
&= \frac{\pi}{2} \left[\frac{x}{2} \sqrt{1-x^2} + \frac{1}{2} \sin^{-1} x \right]_0^1 \quad \left(\because \left[\frac{x}{2} \sqrt{a^2-x^2} + \frac{a^2}{2} \sin^{-1} \left(\frac{x}{a} \right) \right] \right) \\
&= \frac{\pi}{2} \left[0 + \frac{1}{2} \sin^{-1}(1) \right] = \frac{1}{2} \left(\frac{1}{2} \frac{\pi}{2} \right) = \frac{\pi^2}{8}
\end{aligned}$$

Example 6: Evaluate $\iiint xyz dx dy dz$ through positive spherical octant for which

$$x^2 + y^2 + z^2 \leq a^2$$

(or) Evaluate $\int_0^a \int_0^{\sqrt{a^2-x^2}} \int_0^{\sqrt{a^2-x^2-y^2}} xyz dx dy dz$

(Nov-2008, Jan-2010)

Solution:

$$\begin{aligned} \int_0^a \int_0^{\sqrt{a^2-x^2}} \int_0^{\sqrt{a^2-x^2-y^2}} xyz dx dy dz &= \int_0^a \int_0^{\sqrt{a^2-x^2}} \left(xy \frac{z^2}{2} \right)_0^{\sqrt{a^2-x^2-y^2}} dy dx \\ &= \int_0^a \int_0^{\sqrt{a^2-x^2}} \frac{xy(a^2-x^2-y^2)}{2} dy dx \\ &= \frac{1}{2} \int_0^a \int_0^{\sqrt{a^2-x^2}} (a^2 xy - x^3 y - xy^3) dy dx \\ &= \frac{1}{2} \int_0^a \left[\frac{a^2 xy^2}{2} - \frac{x^3 y^2}{2} - \frac{xy^4}{4} \right]_0^{\sqrt{a^2-x^2}} dx \\ &= \frac{1}{2} \int_0^a \left[\frac{a^2 x(a^2-x^2)}{2} - \frac{x^3(a^2-x^2)}{2} - \frac{x(a^2-x^2)^2}{4} \right] dx \\ &= \frac{1}{2} \int_0^a \left[\frac{a^4 x - a^2 x^3 - a^2 x^3 + x^5}{5} - \frac{x[a^4 + x^4 - 2a^2 x^2]}{4} \right] dx \\ &= \frac{1}{2} \int_0^a \left[\frac{2a^4 x - 2a^2 x^3 - 2a^2 x^3 + 2x^5 - a^4 x - x^5 + 2a^2 x^3}{4} \right] dx \\ &= \frac{1}{8} \int_0^a (x^5 + a^4 x - 2a^2 x^3) dx \end{aligned}$$

$$\begin{aligned}
&= \frac{1}{8} \left[\frac{x^6}{6} + \frac{a^4 x^2}{2} - \frac{2a^2 x^4}{4} \right]_0^a \\
&= \frac{1}{8} \left[\frac{a^6}{6} + \frac{a^6}{2} - \frac{2a^6}{4} \right] \\
&= \frac{1}{8} \left[\frac{2a^6 + 6a^6 - 6a^6}{12} \right] = \frac{1}{8} \left[\frac{2a^6}{12} \right] = \frac{a^6}{48}
\end{aligned}$$

Example 7: Evaluate $\int_0^1 \int_0^{1-z} \int_0^{1-y-z} xyz dx dy dz$

Solution:

$$\begin{aligned}
\int_0^1 \int_0^{1-z} \int_0^{1-y-z} xyz dx dy dz &= \int_0^1 \int_0^{1-z} \left(\frac{x^2 yz}{2} \right)_0^{1-y-z} dy dz \\
&= \frac{1}{2} \int_0^1 \int_0^{1-z} (1-y-z)^2 yz dy dz \\
&= \frac{1}{2} \int_0^1 \int_0^{1-z} (1+y^2+z^2-2y-2z+2yz) dy dz \\
&= \frac{1}{2} \int_0^1 \left(y + \frac{y^3}{3} + z^2 y - \frac{2y^2}{2} - 2yz + \frac{2y^2 z}{2} \right)_0^{1-z} dz \\
&= \frac{1}{2} \int_0^1 \left[1-z + \frac{(1-z)^3}{3} + z^2(1-z) - (1-z)^2 - 2(1-z)z + (1-z)^2 z \right] dz \\
&= \frac{1}{2} \int_0^1 \left(\frac{(1-z)^4}{-4} \right) dz = \frac{1}{2} \left(\frac{1}{4} \right) = \frac{1}{8}
\end{aligned}$$

Example 8: Evaluate $\int_0^{\log a} \int_0^x \int_0^{x+y} e^{x+y+z} dx dy dz$

Solution:

$$\begin{aligned}
 \int_0^{\log a} \int_0^x \int_0^{x+y} e^{x+y+z} dx dy dz &= \int_0^{\log a} \int_0^x \int_0^{x+y} e^{x+y+z} dz dy dx \\
 &= \int_0^{\log a} \int_0^x \left[e^{x+y+z} \right]_0^{x+y} dy dx \\
 &= \int_0^{\log a} \int_0^x \left[e^{2x+2y} - e^{x+y} \right] dy dx \\
 &= \int_0^{\log a} \left[\frac{e^{2x+2y}}{2} - e^{x+y} \right]_0^x dx \\
 &= \int_0^{\log a} \left[\frac{e^{4x}}{2} - e^{2x} - \frac{e^{2x}}{2} + e^x \right] dx \\
 &= \left[\frac{e^{4x}}{8} - \frac{e^{2x}}{2} - \frac{e^{2x}}{4} + e^x \right]_0^{\log a} \\
 &= \left(\frac{e^{4\log a}}{8} - \frac{e^{2\log a}}{2} - \frac{e^{2\log a}}{4} + e^{\log a} \right) - \left(\frac{1}{8} - \frac{1}{2} - \frac{1}{4} + 1 \right) \\
 &= \left[\left(\frac{a^4}{8} - \frac{a^2}{2} - \frac{a^2}{4} + a \right) - \left(\frac{1-4-2+8}{8} \right) \right] \\
 &= \left(\frac{a^4 - 4a^2 - 2a^2 + 8a}{8} \right) - \left(\frac{3}{8} \right) \\
 &= \frac{a^4 - 6a^2 + 8a - 3}{8}
 \end{aligned}$$

Example 9: Evaluate $\int_0^{2\pi} \int_0^{\frac{\pi}{4}} \int_0^a \pi^2 \sin \theta dr d\theta d\varphi$

Solution:

$$\begin{aligned}
 \int_0^{2\pi} \int_0^{\frac{\pi}{4}} \int_0^a r^2 \sin \theta dr d\theta d\varphi &= \int_0^{2\pi} \int_0^{\frac{\pi}{4}} \left(\frac{r^3}{3} \sin \theta \right)_0^a d\theta d\varphi \\
 &= \int_0^{2\pi} \int_0^{\frac{\pi}{4}} \left(\frac{a^3}{3} \sin \theta \right) d\theta d\varphi \\
 &= \frac{a^3}{3} \int_0^{2\pi} \left[-\cos \theta \right]_0^{\frac{\pi}{4}} d\varphi \\
 &= \frac{a^3}{3} \int_0^{2\pi} \left[-\cos \frac{\pi}{4} + \cos \theta \right] d\varphi \\
 &= \frac{a^3}{3} \int_0^{2\pi} \left(-\frac{1}{\sqrt{2}} + 1 \right) d\varphi \\
 &= \frac{a^3}{3} \left[-\frac{\varphi}{\sqrt{2}} + \varphi \right]_0^{2\pi} = \frac{a^3}{3} \left[-\frac{2\pi}{\sqrt{2}} + 2\pi \right] \\
 &= \frac{a^3}{3} [2\pi - \sqrt{2}\pi] = \frac{\pi a^3}{3} [2 - \sqrt{2}]
 \end{aligned}$$

Example 10: Evaluate $\int_0^{\frac{\pi}{2}} \int_0^{a \cos \theta} \int_0^{\sqrt{a^2 - r^2}} r dz dr d\theta$

Solution:

$$\int_0^{\frac{\pi}{2}} \int_0^{a \cos \theta} \int_0^{\sqrt{a^2 - r^2}} r dz dr d\theta = \int_0^{\frac{\pi}{2}} \int_0^{a \cos \theta} [rz]_0^{\sqrt{a^2 - r^2}} dr d\theta$$

$$= \int_0^{\frac{\pi}{2}} \int_0^{a \cos \theta} r \sqrt{a^2 - r^2} dr d\theta$$

$$\text{Put } a^2 - r^2 = t^2$$

$$\text{when } r = 0, t^2 = a^2, t = a$$

$$-2r dr = 2t dt$$

$$r = a \cos \theta, t^2 = a^2 - a^2 \cos^2 \theta$$

$$r dr = -t dt$$

$$t = a \sin \theta$$

$$\int_0^{\frac{\pi}{2}} \int_0^{a \cos \theta} \int_{\sqrt{a^2 - r^2}}^a r dz dr d\theta = \int_0^{\frac{\pi}{2}} \int_a^{a \sin \theta} (-t^2 dt) d\theta$$

$$= \int_0^{\frac{\pi}{2}} \left[-\frac{t^3}{3} \right]_a^{a \sin \theta} d\theta$$

$$= \int_0^{\frac{\pi}{2}} \left(\frac{-a^3 \sin^3 \theta}{3} + \frac{a^3}{3} \right) d\theta$$

$$= \int_0^{\frac{\pi}{2}} \frac{a^3}{3} d\theta - \frac{a^3}{3} \int_0^{\frac{\pi}{2}} \sin^3 \theta d\theta$$

$$= \left(\frac{a^3}{3} \theta \right)_0^{\frac{\pi}{2}} - \frac{a^3}{3} \left[\frac{3 - 1}{3} \right] = \frac{a^3 \pi}{6} - \frac{2a^3}{9}$$

$$= \frac{3a^3 \pi - 4a^3}{18}$$

$$= \frac{a^3}{18} [3\pi - 4]$$

$$\left[\because \int_0^{\frac{\pi}{2}} \sin \theta d\theta = \frac{n-1}{n} \frac{n-3}{n-2} - \frac{2}{3} 1 \right]$$

Example 11: Evaluate $\int_0^{\frac{\pi}{2}} \int_0^{a \sin \theta} \int_0^{\frac{a^2 - r^2}{2}} r dz dr d\theta$

Solution:

$$\begin{aligned}
 \int_0^{\frac{\pi}{2}} \int_0^{a \sin \theta} \int_0^{\frac{a^2 - r^2}{2}} r dz dr d\theta &= \int_0^{\frac{\pi}{2}} \int_0^{a \sin \theta} \left[rz \right]_0^{\frac{a^2 - r^2}{2}} dr d\theta \\
 &= \int_0^{\frac{\pi}{2}} \int_0^{a \sin \theta} r \left(\frac{a^2 - r^2}{2} \right) dr d\theta \\
 &= \frac{1}{2} \int_0^{\frac{\pi}{2}} \int_0^{a \sin \theta} (a^2 r - r^3) dr d\theta \\
 &= \frac{1}{2} \int_0^{\frac{\pi}{2}} \left[\frac{a^2 r^2}{2} - \frac{r^4}{4} \right]_0^{a \sin \theta} d\theta \\
 &= \frac{1}{2} \int_0^{\frac{\pi}{2}} \left[\frac{a^2 a^2 \sin^2 \theta}{2} - \frac{a^4 \sin^4 \theta}{4} \right] d\theta \\
 &= \frac{1}{2} \int_0^{\frac{\pi}{2}} \left[\frac{a^4 \sin^2 \theta}{2} - \frac{a^4 \sin^4 \theta}{4} \right] d\theta = \frac{a^4}{2} \int_0^{\frac{\pi}{2}} \frac{2 \sin^2 \theta - \sin^4 \theta}{4} d\theta \\
 &= \frac{a^4}{8} \left[\int_0^{\frac{\pi}{2}} 2 \sin^2 \theta d\theta - \int_0^{\frac{\pi}{2}} \sin^4 \theta d\theta \right] \\
 &= \frac{a^4}{8} \left[2 \cdot \frac{2-1}{2} \frac{\pi}{2} - \frac{4-1}{4} \frac{4-3}{4-2} \frac{\pi}{2} \right] \\
 &= \frac{a^4 \pi}{8 \times 2} \left[1 - \frac{3}{8} \right] = \frac{a^4}{2 \times 8} \left[\frac{8-3}{8} \right] = \frac{5a^4}{128}
 \end{aligned}$$

Example 12: Evaluate $\int_0^4 \int_0^{\frac{4-x}{2}} \int_0^z dx dy dz$ **where** $z = \frac{1}{3}(4-x-2y)$

Solution:

$$\begin{aligned}
 \int_0^4 \int_0^{\frac{4-x}{2}} \int_0^z dx dy dz &= \int_0^4 \int_0^{\frac{4-x}{2}} (z)_0^z dy dx \\
 &= \int_0^4 \int_0^{\frac{4-x}{2}} \frac{1}{3} (4-x-2y) dy dx \\
 &= \frac{1}{3} \int_0^4 \int_0^{\frac{4-x}{2}} \left(4y - xy - \frac{2y^2}{2} \right) dx \\
 &= \frac{1}{3} \int_0^4 \left[4 \frac{(4-x)}{2} - x \left(\frac{4-x}{2} \right) - \frac{(4-x)^2}{2} \right] dx \\
 &= \frac{1}{3} \int_0^4 \left(8 - 2x - 2x + \frac{x^2}{2} - \frac{16 + x^2 - 8x}{4} \right) dx \\
 &= \frac{1}{3} \int_0^4 \left(8 - 4x + \frac{x^2}{2} - 4 - \frac{x^2}{4} + \frac{8x}{4} \right) dx = \frac{1}{3} \int_0^4 \left(\frac{x^2}{4} - \frac{8x}{4} + 4 \right) dx \\
 &= \frac{1}{3} \left[\frac{x^3}{12} - \frac{8x^2}{8} + 4x \right]_0^4 \\
 &= \frac{1}{3} \left[\frac{64}{12} - 16 + 16 \right] = \frac{16}{9}
 \end{aligned}$$

Example 13: Evaluate $\int_1^e \int_1^{\log y} \int_1^{e^x} \log z \, dz \, dx \, dy$

Solution:

$$\begin{aligned}
\int_1^e \int_1^{\log y} \int_1^{e^x} \log z \, dz \, dx \, dy &= \int_1^e \int_1^{\log y} (z \log z - z) \Big|_1^{e^x} dx \, dy \quad \left[\because \int \log x \, dx = x \log x - x \right] \\
&= \int_1^e \int_1^{\log y} \left[(e^x \log e^x - e^x) - (\log 1 - 1) \right] dx \, dy \\
&= \int_1^e \int_1^{\log y} [e^x x - e^x + 1] dx \, dy \quad [\because \log 1 = 0] \\
&= \int_1^e \int_1^{\log y} (1 - e^x + x e^x) dx \, dy \\
&= \int_1^e \left[x - e^x + x e^x - e^x \right]_1^{\log y} dy \\
&= \int_1^e (\log y - e^{\log y} + \log y e^{\log y} - e^{\log y}) - (1 - e^1 + e - e) dy \\
&= \int_1^e [\log y - y + \log y y - y] - (1 - e) dy \\
&= \int_1^e (y \log y + \log y - 2y + e - 1) dy \\
&= \int_1^e y \log y \, dy + \int_1^e (\log y - 2y + e - 1) dy \\
&= \int_1^e y \log y \, dy + \left[y \log y - y - \frac{2y^2}{2} + ey - y \right]_1^e \\
&= \int_1^e y \log y \, dy + \{ [e \log e - e - e^2 + e^2 - e] - [-1 - 1 + 1 - 1] \} \\
&= \int_1^e y \log y \, dy + 2 - e \quad \dots (1)
\end{aligned}$$

$$\begin{aligned}
\text{Now } \int_1^e y \log y \, dy &= \int_1^e \log y \, d\left(\frac{y^2}{2}\right) \\
&= \left[\log y \frac{y^2}{2} \right]_1^e - \int_1^e \frac{y^2}{2} \cdot \frac{1}{y} \, dy \\
&= \left[\log y \frac{y^2}{2} - \frac{y^2}{4} \right]_1^e \\
&= \frac{e^2}{2} \log e - \frac{1}{2} \log 1 - \frac{e^2}{4} + \frac{1}{4} \\
&= \frac{e^2}{2} - \frac{e^2}{4} + \frac{1}{4} \\
&= \frac{e^2 + 1}{4} \quad \dots (2)
\end{aligned}$$

Substitute (2) in (1)

$$\begin{aligned}
\int_1^e \int_1^{\log y} \int_1^{e^x} \log z \, dz \, dx \, dy &= \frac{e^2 + 1}{4} + 2 - e \\
&= \frac{e^2 + 1 + 8 - 4e}{4} \\
&= \frac{e^2 - 4e + 9}{4}
\end{aligned}$$

Example 14: Evaluate $\int_0^a \int_0^x \int_0^x e^{x+y+z} \, dz \, dy \, dx$

Solution:

$$\int_0^a \int_0^x \int_0^x e^{x+y+z} \, dz \, dy \, dx = \int_0^a \int_0^x \left(e^{x+y+z} \right)_0^{x+y} \, dy \, dx$$

$$\begin{aligned}
&= \int_0^a \int_0^x (e^{2x+2y} - e^{x+y}) dy dx \\
&= \int_0^a \left[\frac{e^{2x+2y}}{2} - e^{x+y} \right]_0^x dx \\
&= \int_0^a \left[\left(\frac{e^{4x}}{2} - e^{2x} \right) - \left(\frac{e^{2x}}{4} - e^x \right) \right] dx = \left[\frac{e^{4x}}{8} - \frac{e^{2x}}{2} - \frac{e^{2x}}{4} + e^x \right]_0^a \\
&= \left(\frac{e^{4a}}{8} - \frac{e^{2a}}{2} - \frac{e^{2a}}{4} + e^a \right) - \left(\frac{1}{8} - \frac{1}{2} - \frac{1}{4} + 1 \right) \\
&= \frac{e^{4a} - 4e^{2a} - 2e^{2a} + 8e^a}{8} - \frac{1 - 4 - 2 + 8}{8} \\
&= \frac{e^{4a} - 6e^{2a} + 8e^a - 3}{8}
\end{aligned}$$

Exercise

Evaluate the following triple integrals

$$1. \int_0^1 \int_0^{1-x} \int_0^{1-x-y} xyz dx dy dz$$

$$2. \int_0^1 \int_0^{1-x} \int_0^{2-x} xyz dx dy dz$$

$$3. \int_0^1 \int_0^2 \int_1^2 xyz dx dy dz$$

$$4. \int_0^{\log 2} \int_0^2 \int_0^{x+\log y} e^{x+y+z} dz dy dx$$

$$5. \int_0^4 \int_0^x \int_0^{\sqrt{x+y}} z dz dy dx$$

$$6. \int_0^a \int_0^{\sqrt{a^2-y^2}} \int_0^{\sqrt{a^2-x^2-y^2}} dz dx dy$$

$$7. \int_0^1 \int_{y^3}^1 \int_0^{1-x} x dz dx dy$$

$$8. \int_0^2 \int_1^3 \int_1^2 xy^2 z dz dy dx$$

$$9. \int_0^a \int_0^x \int_0^y xyz dz dy dx$$

$$10. \int_0^{\pi/2} \int_0^1 \int_0^1 r^2 \sin \theta dr d\theta d\phi$$

$$11. \int_0^{\pi/2} \int_0^2 \int_0^{\sqrt{16-r^2}} r dz dr d\theta$$

$$12. \int_0^{\pi/2} \int_0^{a \cos \theta} \int_0^{\sqrt{a^2-r^2}} dz dr d\theta$$

$$13. \int_0^1 \int_0^1 \int_{\sqrt{x^2+y^2}}^2 xyz dz dy dx$$

$$14. \int_0^{\pi} \int_0^{\pi/2} \int_0^k r^2 \sin \theta dr d\theta d\phi$$

Answers

$$1. \frac{1}{720} \quad 2. \frac{13}{240} \quad 3. 1 \quad 4. \frac{1}{2}(l^2 - 1) \quad 5. 14 \quad 6. \frac{\pi a^3}{6} \quad 7. \frac{4}{35}$$

$$8. 26 \quad 9. \frac{a^6}{48} \quad 10. \frac{\pi}{3} \quad 11. \frac{32}{9} \quad 12. \frac{a^2}{16}(4 + \pi^2) \quad 13. \frac{3}{8} \quad 14. \frac{\mathbb{K}^3 \pi}{3}$$

3.5. Change of order of Integration

The evaluation of some double integrals may be very difficult. In this case, we may evaluate it easily by changing the order of integration in a given double integral. When we change the order of integration the limits are also changed. The following points are very important when the change of order of integration takes place.

(i.e.) $\int_a^b \int_{f_1(x)}^{f_2(x)} f(x, y) dy dx$, here the inner integration is with respect to y and the other is with

respect to x . When the order of integration is changed, it will take the form $\int_c^d \int_{g_1(y)}^{g_2(y)} f(x, y) dx dy$.

In such cases, it will be convenient to draw a figure indicating the original region of integration.

Points to remember:

Consider the integral

$\int_a^b \int_{f_1(x)}^{f_2(x)} f(x, y) dy dx$, here the inner integration is with respect to y and the

other is with respect to x . When the order of integration is changed, it will

take the form $\int_c^d \int_{g_1(y)}^{g_2(y)} f(x, y) dx dy$, that is the inner integration is with

respect to x and the other is with respect to y .

Example 1: Change the order of integration in $\int_0^a \int_x^a (x^2 + y^2) dy dx$ and hence evaluate it.

(April-2010)

Solution:

In the integral $\int_0^a \int_x^a (x^2 + y^2) dy dx$, the limit of y varies from x to a and x varies from 0 to a .

Our region of integration is the area enclosed between these four lines $y = x, y = a, x = 0, x = a$

The given integral is $\int_0^a \int_x^a (x^2 + y^2) dy dx$.

The region of integration is shown in the below figure.

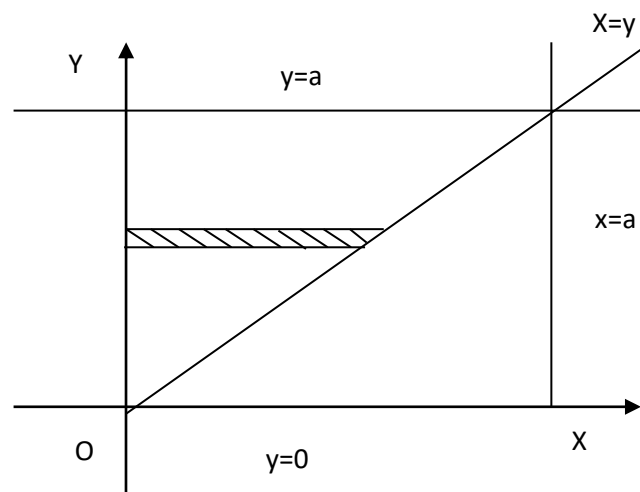


Fig.3.10

Now we have to change the order of integration, (i.e.) our integration should be first with respect to x and then with respect to y . When the first integration is with respect to x , we have to keep y as constant. Whenever the integration is with respect to x we take horizontal strips.

From Fig. 3.10, for the horizontal strips x varies from 0 to y . Now we slide this strips along vertical direction. When we move this strip along vertical direction y varies from 0 to a .

$$\begin{aligned}
 \therefore \int_0^a \int_x^a (x^2 + y^2) dy dx &= \int_0^a \int_0^y (x^2 + y^2) dx dy \\
 &= \int_0^a \left[\frac{x^3}{3} + y^2 x \right]_0^y dy = \int_0^a \left[\frac{y^3}{3} + y^3 \right] dy \\
 &= \left[\frac{y^4}{12} + \frac{y^4}{4} \right]_0^a = \frac{a^4}{12} + \frac{a^4}{4} \\
 &= \frac{4a^4}{12} = \frac{a^4}{3}
 \end{aligned}$$

Example 2: Change the order of integration in $\int_0^a \int_y^a \frac{x \, dx \, dy}{x^2 + y^2}$ and hence evaluate.

(Nov-2006, May- 2012)

Solution:

In the integral $\int_0^a \int_y^a \frac{x \, dx \, dy}{x^2 + y^2}$, x varies from y to a and y varies from 0 to a .

Now we have to change the order of integration. The first integration with respect to y and then with respect to x . Since the first integration with respect to y we have to consider the vertical strip. For this vertical strip the limits for y varies from 0 to x .

Now we move this vertical strip horizontally from $x = 0$ to $x = a$.

$$\begin{aligned} \therefore \int_0^a \int_y^a \frac{x \, dx \, dy}{x^2 + y^2} &= \int_0^a \int_0^x \frac{x \, dy \, dx}{x^2 + y^2} \\ &= \int_0^a \left[\int_0^x \frac{x \, dy}{x^2 + y^2} \right] dx \\ &= \int_0^a \left[x \frac{1}{x} \tan^{-1} \left(\frac{y}{x} \right) \right]_0^x dx \end{aligned}$$

$$\left[\therefore \int_0^a \frac{dx}{x^2 + a^2} = \frac{1}{a} \tan^{-1} \left(\frac{x}{a} \right) \right]$$

$$= \int_0^a \left[\tan^{-1}(1) - \tan^{-1}(0) \right] dx$$

$$= \int_0^a \left[\frac{\pi}{4} - 0 \right] dx = \left[\frac{\pi}{4} x \right]_0^a = \frac{\pi}{4} a$$

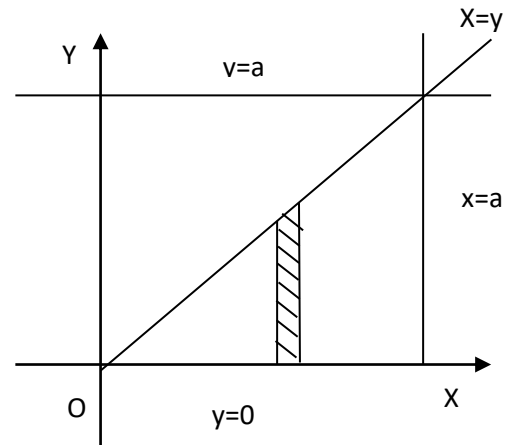


Fig. 3.11

Example 3: Change the order of integration in $\int_0^a \int_0^{2\sqrt{ax}} x^2 dy dx$ and integrate. (Jan-2010)

Solution:

In the integral $\int_0^a \int_0^{2\sqrt{ax}} x^2 dy dx$,

y varies from 0 to $2\sqrt{ax}$

x varies from 0 to a .

We have to change the order of integration.

(i.e.) First integrating with respect to x

and then with respect to y .

The region of integration is shown in Fig. 3.12

We consider the horizontal strip, x varies from $\frac{y^2}{4a}$ to a

Then, we move this strip vertically from $y=0$ to $y=4a$.

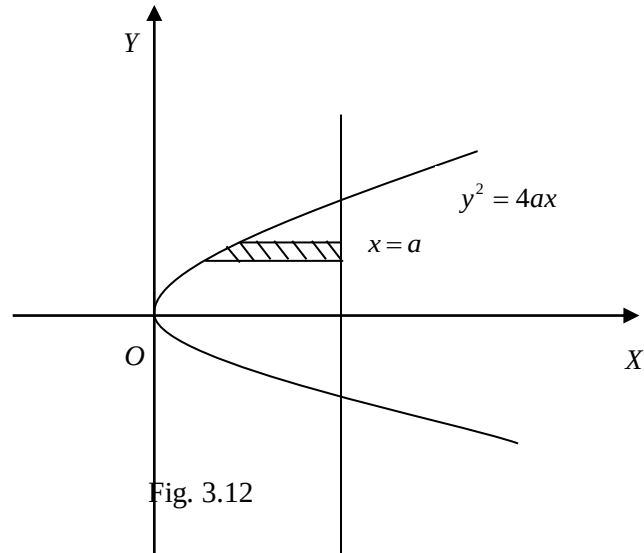


Fig. 3.12

$$\int_0^a \int_0^{2\sqrt{ax}} x^2 dy dx = \int_0^{4a} \int_{\frac{y^2}{4a}}^a x^2 dx dy$$

$$= \int_0^{4a} \left[\frac{x^3}{3} \right]_{\frac{y^2}{4a}}^a dy = \int_0^{4a} \left[\frac{a^3}{3} - \frac{y^6}{192a^3} \right] dy$$

$$= \left[\frac{a^3}{3} y - \frac{y^7}{192 \times 7 \times a^3} \right]_0^{4a}$$

$$= \frac{4a^4}{3} - \frac{4^7 \times a^7}{4^3 \times 3 \times 7 \times a^3} = \frac{4a^4}{3} - \frac{4^4 \times a^4}{21}$$

$$= \frac{28a^4 - 256a^4}{21} = -\frac{228a^4}{21}$$

Example 4: Change the order of integration in $\int_0^{4a} \int_{\frac{x^2}{4a}}^{2\sqrt{ax}} xy \, dy \, dx$ and then evaluate.

(April-2003)

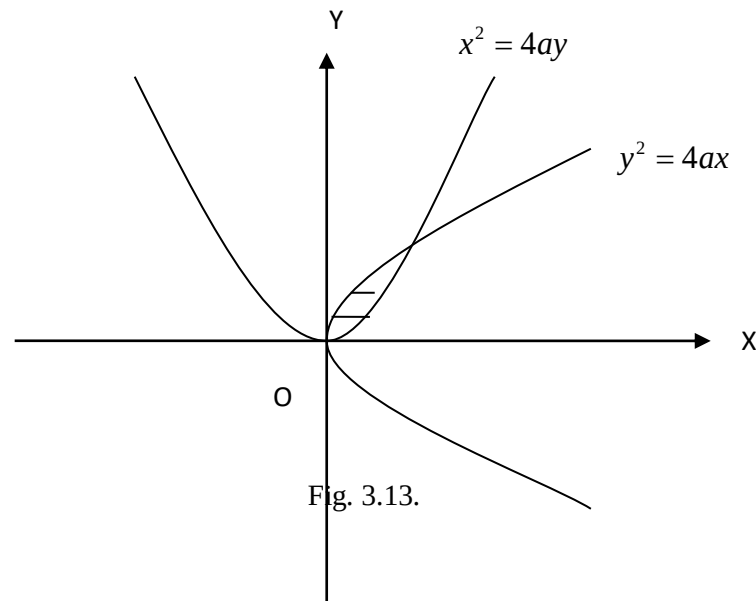
Solution:

For the given integral, the region of integration is $y = \frac{x^2}{4a}$, to $y = 2\sqrt{ax}$ and $x = 0$, to $x = 4a$.

The two curves $x^2 = 4ay$ and $y^2 = 4ax$ meet the point $(4a, 4a)$. Also the line $x = 4a$, passes through the point of intersection of the two parabolas. The region of integration is shown in Fig. 3.13

We have to change the order of integration. The first integration is with respect to x . We have to consider horizontal strip the limits for x varies from $x = \frac{y^2}{4a}$ to $x = \sqrt{4ay}$. Then we move this strip vertically from $y = 0$ to $y = 4a$.

$$\begin{aligned} \int_0^{4a} \int_{\frac{x^2}{4a}}^{2\sqrt{ax}} xy \, dy \, dx &= \int_0^{4a} \int_{\frac{y^2}{4a}}^{2\sqrt{ay}} xy \, dx \, dy \\ &= \int_0^{4a} \left[\frac{x^2}{2} y \right]_{\frac{y^2}{4a}}^{2\sqrt{ay}} dy \\ &= \int_0^{4a} \left[\frac{4ay}{2} y - \frac{y^4}{2 \times 16a^2} y \right] dy \\ &= \int_0^{4a} \left[2ay^2 - \frac{y^5}{32a^2} \right] dy \end{aligned}$$



$$= \left[\frac{2ay^3}{3} - \frac{y^6}{32a^2 \times 6} \right]_0^{4a} = \frac{2a \times 64 \times a^3}{3} - \frac{4096a^6}{192a^2}$$

$$= \frac{128a^4}{3} - \frac{64}{3}a^4 = \frac{64a^4}{3}$$

Example 5: Change the order of integration in $\int_0^1 \int_y^{2-y} xy \, dx \, dy$ and then evaluate.

(May-2002)

Solution:

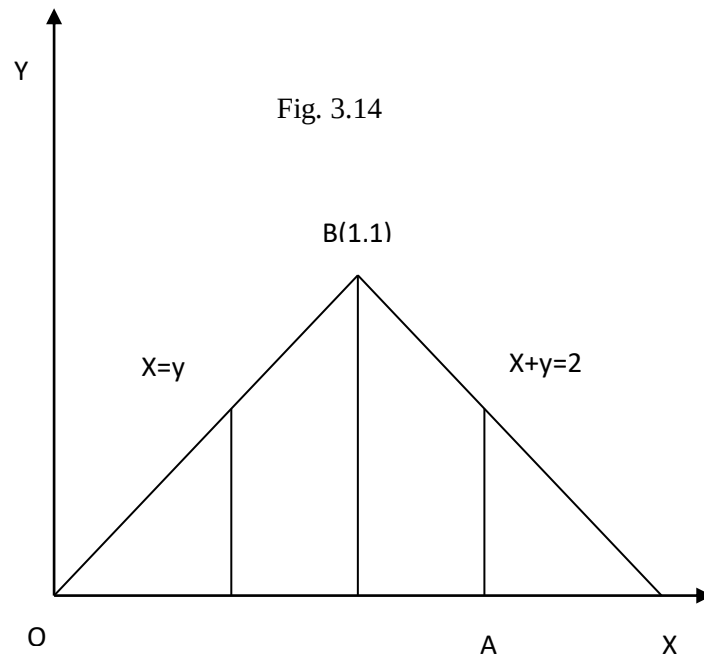
For the given integral, the region of integration is $x=y$, to $x=2-y$ and $y=0$, to $y=1$.

The region of integration R is bounded by the lines $x=y$, $x+y=2$, $y=0$, $y=1$.

The region of integration is shown in the Fig. 3.14

After changing the order of integration, if the line segment is drawn in the region OCB, its upper end lies on the line $y=x$. On the other hand, if it is drawn in the region BCA, its upper end lies on the line $x=y-2$.

$$\begin{aligned} \iint_R xy \, dy \, dx &= \iint_{\Delta OCB} xy \, dy \, dx + \iint_{\Delta BCA} xy \, dy \, dx \\ &= \int_0^1 \int_0^x xy \, dy \, dx + \int_1^2 \int_0^{2-x} xy \, dy \, dx \\ &= \int_0^1 x \left(\frac{y^2}{2} \right)_0^x dx + \int_1^2 x \left(\frac{y^2}{2} \right)_0^{2-x} dx \\ &= \int_0^1 \frac{x^3}{2} dx + \int_1^2 \frac{x}{2} (2-x)^2 dx \\ &= \left(\frac{x^4}{8} \right)_0^1 + \frac{1}{2} \int_1^2 (4x + x^3 - 4x^2) dx \end{aligned}$$



$$= \frac{1}{8} + \frac{1}{2} \left[\frac{4x^2}{2} + \frac{x^4}{4} - \frac{4x^3}{3} \right]_1^2$$

$$= \frac{1}{8} - \frac{1}{6} = \frac{1}{24}$$

Example 6: Change the order of integration and then evaluate $\int_0^1 \int_{x^2}^{2-x} xy \, dy \, dx$

Solution:

In the integral $\int_0^1 \int_{x^2}^{2-x} xy \, dy \, dx$,

The limits of y varies from x^2 to $2-x$ and x varies from 0 to 1.

The region of integration is shown in the Fig. 3.15

Now we have to change the order of integration (i.e.) the integration with respect to x and then with respect to y .

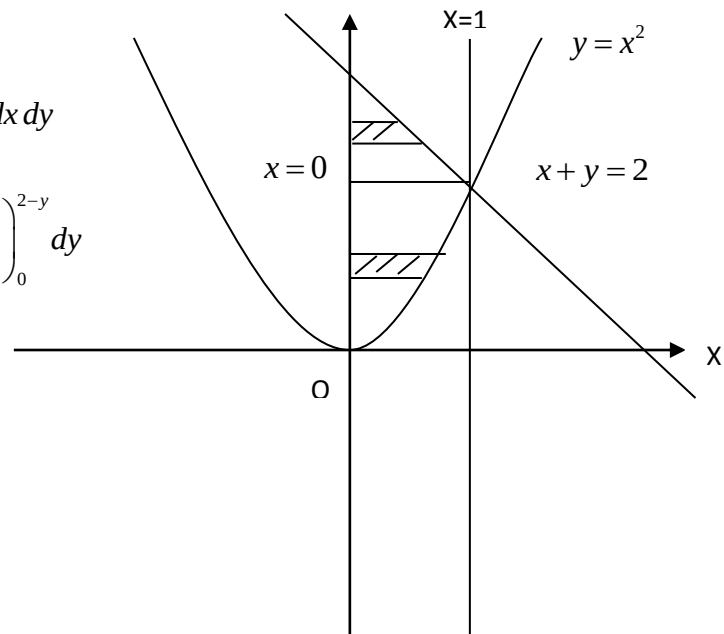
We consider horizontal strips, the region of integration may be divided into two portions A and B .

For the portion A , we have $I_1 = \int_0^1 \int_0^{\sqrt{y}} xy \, dx \, dy$

For the portion B , we have $I_2 = \int_1^2 \int_0^{2-y} xy \, dx \, dy$

$$\begin{aligned} \text{Hence } \int_0^1 \int_{x^2}^{2-x} xy \, dy \, dx &= \int_0^1 \int_0^{\sqrt{y}} xy \, dx \, dy + \int_1^2 \int_0^{2-y} xy \, dx \, dy \\ &= \int_0^1 \left(\frac{x^2}{2} y \right)_0^{\sqrt{y}} dy + \int_1^2 \left(\frac{x^2 y}{2} \right)_0^{2-y} dy \end{aligned}$$

Fig. 3.15.



$$\begin{aligned}
&= \int_0^1 \left(\frac{y^2}{2} \right) dy + \int_1^2 \left(\frac{(2-y)^2}{2} \right) dy \\
&= \int_0^1 \left(\frac{y^2}{2} \right) dy + \int_1^2 \left(\frac{4y + y^3 - 4y^2}{2} \right) dy \\
&= \frac{1}{2} \left(\frac{y^3}{3} \right)_0^1 + \frac{1}{2} \left(\frac{4y^2}{2} + \frac{y^4}{4} - \frac{4y^3}{3} \right)_1^2 \\
&= \frac{1}{2} \cdot \frac{1}{3} + \frac{1}{2} \left(8 + 4 - \frac{32}{3} - 2 - \frac{1}{4} + \frac{4}{3} \right) \\
&= \frac{3}{8}
\end{aligned}$$

Example 7: Change the order of integration and evaluate $\int_0^1 \int_0^x dy dx$.

Solution:

In the given integral $\int_0^1 \int_0^x dy dx$,

y varies from 0 to x

x varies from 0 to 1.

The region of integration is shown in the Fig. 3.16

We change the order of integration by integrating first with respect to x .

Consider the horizontal strip.

y varies from 0 to 1.

x varies from y to 1.

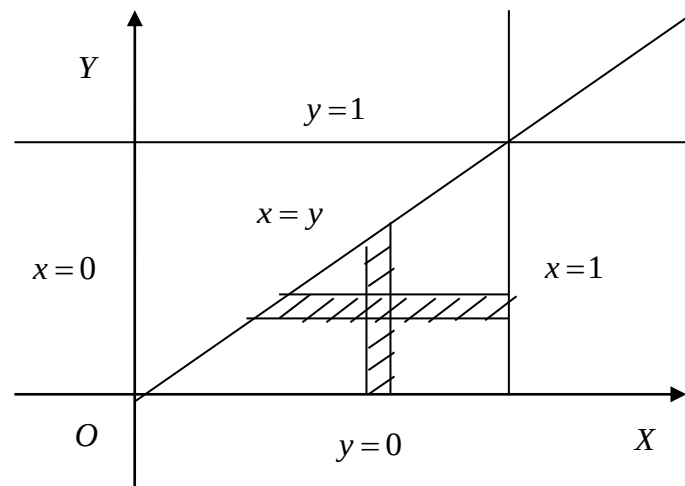


Fig. 3.16.

$$\therefore \int_0^1 \int_y^1 dx dy = \int_0^1 (x)_y^1 dy$$

$$= \int_0^1 (1-y) dy = \left(y - \frac{y^2}{2} \right)_0^1 = 1 - \frac{1}{2} = \frac{1}{2}$$

Example 8: Change the order of integration in $\int_0^a \int_0^{\sqrt{a^2-x^2}} xy \, dy \, dx$ and hence evaluate.

Solution:

In the given integral $\int_0^a \int_0^{\sqrt{a^2-x^2}} xy \, dy \, dx$,

y varies from 0 to $\sqrt{a^2-x^2}$

x varies from 0 to a .

The region of integration is shown in the Fig. 3.17

$$y=0, y=\sqrt{a^2-x^2}$$

$$x=0, x=a$$

We change the order of integration by integrating first with respect to x .

Consider the horizontal strip.

Here x varies from 0 to $\sqrt{a^2-y^2}$ and y varies from 0 to a .

$$\begin{aligned} \therefore \int_0^a \int_0^{\sqrt{a^2-x^2}} xy \, dy \, dx &= \int_0^a \int_0^{\sqrt{a^2-y^2}} xy \, dx \, dy \\ &= \int_0^a \left(\frac{yx^2}{2} \right)_0^{\sqrt{a^2-y^2}} dy \end{aligned}$$

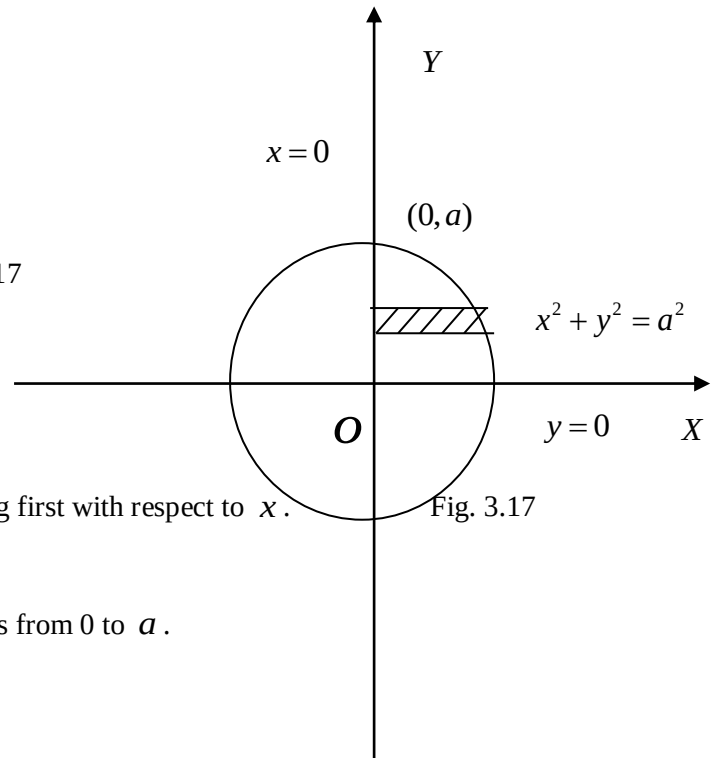


Fig. 3.17

$$\begin{aligned}
&= \int_0^a \frac{y(a^2 - y^2)}{2} dy \\
&= \int_0^a \frac{a^2 y - y^3}{2} dy \\
&= \left[\frac{a^2 y^2}{4} - \frac{y^4}{8} \right]_0^a = \frac{1}{2} \left[\frac{a^4}{2} - \frac{a^4}{4} \right] = \frac{1}{2} \left[\frac{2a^4 - a^4}{4} \right] \\
&= \frac{a^4}{8}
\end{aligned}$$

Example 9: Change the order of integration in $\int_0^1 \int_x^{\sqrt{2-x^2}} \frac{x}{\sqrt{x^2 + y^2}} dy dx$ and hence evaluate.

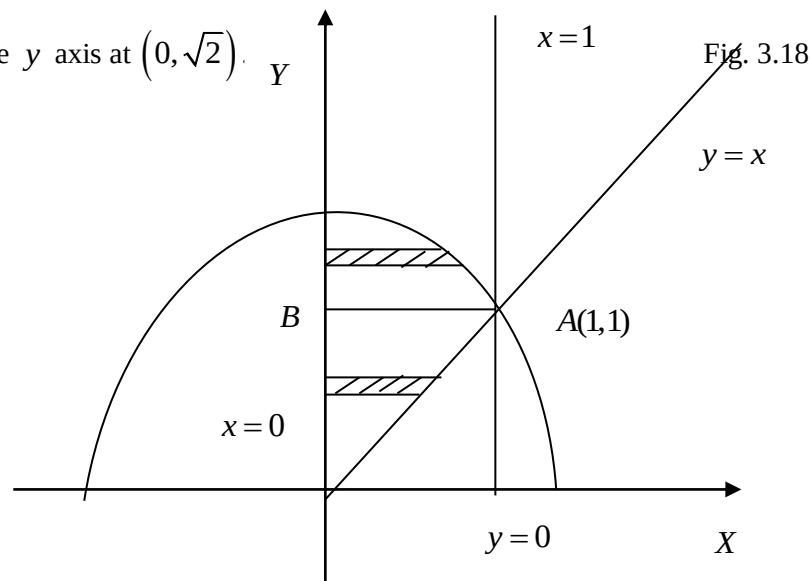
Solution:

In the given integral, y varies from x to $\sqrt{2-x^2}$ and x varies from 0 to 1.

The region of integration is shown in the Fig. 3.18

We change the order of integration by integrating first with respect to x . Consider the horizontal strip.

In Fig. 3.15, the circle $x^2 + y^2 = 2$ and the line $y = x$ intersect at the point $(1,1)$. Also the circle $x^2 + y^2 = 2$ cuts the y axis at $(0, \sqrt{2})$.



The region of integration can be split up into two parts OAB and ABC .

Consider the region OAB , the limits of x varies from 0 to y and y varies from 0 to 1.

In the region ABC , the limits of x varies from 0 to $\sqrt{2-y^2}$ and y varies from 1 to $\sqrt{2}$.

$$\begin{aligned}
 \therefore \int_0^1 \int_0^{\sqrt{2-x^2}} \frac{x}{\sqrt{x^2+y^2}} dy dx &= \iint_{OAB} \frac{x}{\sqrt{x^2+y^2}} dx dy + \iint_{ABC} \frac{x}{\sqrt{x^2+y^2}} dx dy \\
 &= \int_0^1 \int_0^y \frac{x}{\sqrt{x^2+y^2}} dx dy + \int_1^{\sqrt{2}} \int_0^{\sqrt{2-y^2}} \frac{x}{\sqrt{x^2+y^2}} dx dy \\
 &= \int_0^1 \int_0^y \frac{d(x^2+y^2)}{2\sqrt{x^2+y^2}} dy + \int_1^{\sqrt{2}} \int_0^{\sqrt{2-y^2}} \frac{d(x^2+y^2)}{2\sqrt{x^2+y^2}} dx dy \\
 &= \int_0^1 \left[\sqrt{x^2+y^2} \right]_0^y dy + \int_1^{\sqrt{2}} \left[\sqrt{x^2+y^2} \right]_0^{\sqrt{2-y^2}} dy \\
 &= \int_0^1 \left[\sqrt{2y^2} - \sqrt{y^2} \right] dy + \int_1^{\sqrt{2}} \left[\sqrt{2-y^2+y^2} - \sqrt{y^2} \right] dy \\
 &= \int_0^1 \left(\sqrt{2}y - y \right) dy + \int_1^{\sqrt{2}} \left(\sqrt{2} - y \right) dy \\
 &= \left(\sqrt{2} \frac{y^2}{2} - \frac{y^2}{2} \right)_0^1 + \left(\sqrt{2}y - \frac{y^2}{2} \right)_1^{\sqrt{2}} = \left(\frac{1}{\sqrt{2}} - \frac{1}{2} \right) + \left(1 - \sqrt{2} + \frac{1}{2} \right) \\
 &= \frac{1}{\sqrt{2}} + 1 - \sqrt{2} = \frac{1 + \sqrt{2} - 2}{\sqrt{2}} \\
 &= \frac{\sqrt{2} - 1}{\sqrt{2}}
 \end{aligned}$$

Example 10: Change the order of integration in $\int_0^{\infty} \int_x^{\infty} \frac{e^{-y}}{y} dx dy$ and hence evaluate.

(Nov-2008)

Solution:

In the given integral $\int_0^{\infty} \int_x^{\infty} \frac{e^{-y}}{y} dx dy = \int_0^{\infty} \int_x^{\infty} \frac{e^{-y}}{y} dy dx,$

y varies from x to ∞

x varies from 0 to ∞ .

The region of integration is shown in the Fig. 3.19

We change the order of integration by integrating first with respect to x . Consider the horizontal strip. Along the horizontal strip x varies from 0 to y . Along the vertical strip y varies from 0 to ∞

$$\begin{aligned} \therefore \int_0^{\infty} \int_x^{\infty} \frac{e^{-y}}{y} dy dx &= \int_0^{\infty} \int_0^y \frac{e^{-y}}{y} dx dy \\ &= \int_0^{\infty} \left[\frac{e^{-y}}{y} x \right]_0^y dy \\ &= \int_0^{\infty} \left[\frac{e^{-y}}{y} y \right] dy \\ &= \int_0^{\infty} (e^{-y}) dy = \end{aligned}$$

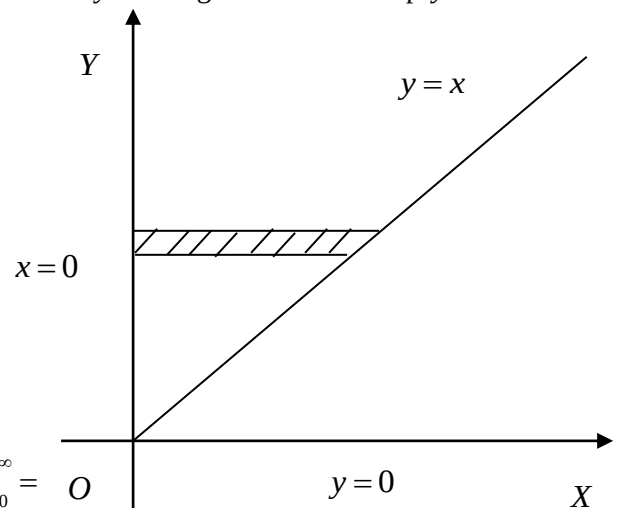


Fig. 3.19

Example 11: Change into polar co-ordinates and hence evaluate $\int_0^{\infty} \int_0^{\infty} e^{-(x^2+y^2)} dy dx$

Solution:

In the given integral, y and x varies from 0 to ∞ . The region of integration is shown in the below figure.

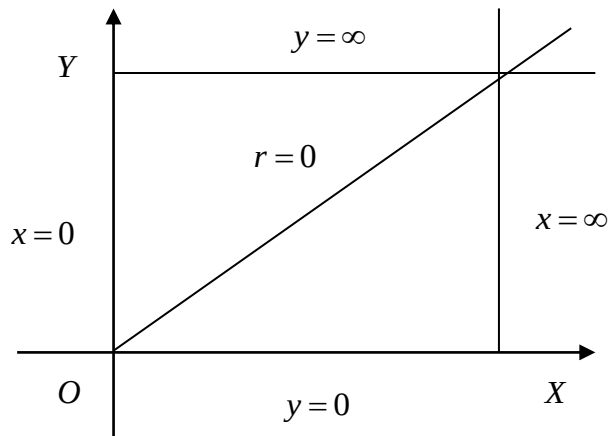


Fig. 3.20

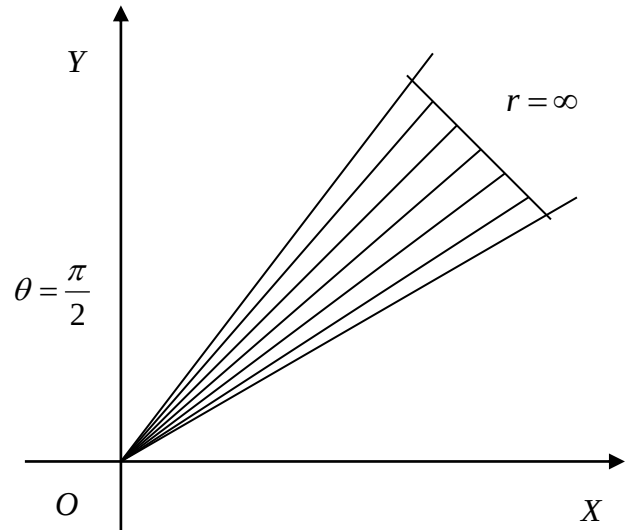


Fig. 3.21

Put $x = r \cos \theta$, $y = r \sin \theta$, $x^2 + y^2 = r^2$

$$dx dy = r dr d\theta$$

Hence $\int_0^\infty \int_0^\infty e^{-(x^2+y^2)} dy dx = \int_0^{\pi/2} \int_0^\infty e^{-r^2} r dr d\theta$

Put $r^2 = t$ When $r = 0$, $\theta = 0$

$2r dr = dt$ When $r = \infty$ $\theta = \infty$

$$\int_0^\infty \int_0^\infty e^{-(x^2+y^2)} dy dx = \int_0^{\pi/2} \left(\int_0^\infty e^{-t} \frac{dt}{2} \right) d\theta$$

$$= \int_0^{\pi/2} \left(\frac{-e^{-t}}{2} \right)_0^\infty d\theta$$

$$= \int_0^{\pi/2} \frac{1}{2} d\theta = \left(\frac{1}{2} \theta \right)_0^{\pi/2} = \frac{\pi}{4}$$

Example 12: Change the order of integration and evaluate $\int_0^{2a} \int_0^{\sqrt{2ax-x^2}} (x^2 + y^2) dy dx$

Solution:

In the given integrals $\int_0^{2a} \int_0^{\sqrt{2ax-x^2}} (x^2 + y^2) dy dx$

y varies from 0 to $\sqrt{2ax-x^2}$

x varies from 0 to $2a$.

The region of integration is shown in the Fig.3.22

$$y = \sqrt{2ax-x^2}, \quad y^2 = 2ax-x^2$$

$x^2 + y^2 - 2ax = 0$ Represents the circle

whose centre is $(a, 0)$ and radius a .

Let $x = r \cos \theta$ and $y = r \sin \theta$

$$x^2 + y^2 = r^2$$

$\therefore x^2 + y^2 - 2ax = 0$ becomes $r^2 - 2ar \cos \theta = 0$

(i.e.) $r = 2a \cos \theta$, r varies from 0 to

$2a \cos \theta$ and θ varies from 0 to $\frac{\pi}{2}$.

$$\int_0^{2a} \int_0^{\sqrt{2ax-x^2}} (x^2 + y^2) dy dx = \int_0^{\frac{\pi}{2}} \int_0^{2a \cos \theta} r^2 r dr d\theta$$

$$= \int_0^{\frac{\pi}{2}} d\theta \int_0^{2a \cos \theta} r^3 dr$$

$$= \int_0^{\frac{\pi}{2}} \left(\frac{r^4}{4} \right)_0^{2a \cos \theta} d\theta$$

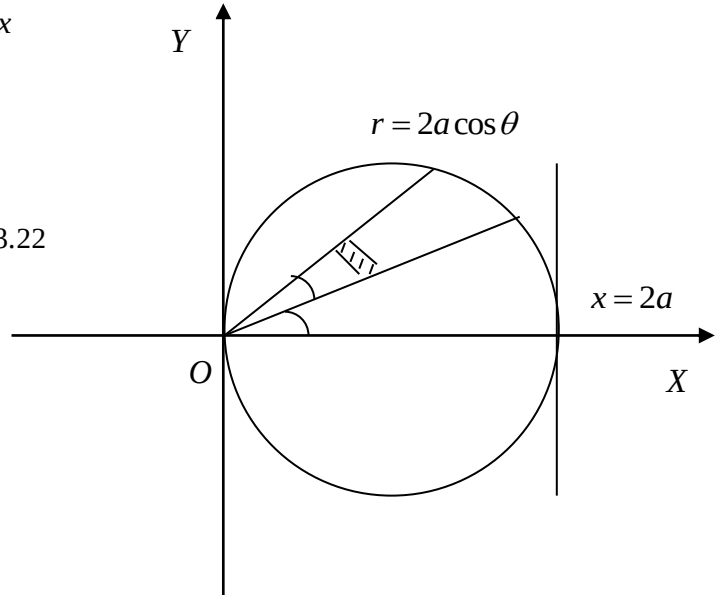


Fig.3.22

$$\begin{aligned}
&= \int_0^{\frac{\pi}{2}} \frac{16a^4 \cos^4 \theta}{4} d\theta \left\{ \because \int_0^{\frac{\pi}{2}} \cos^n \theta d\theta = \frac{n-1}{n} \frac{n-3}{n-2} \frac{n-5}{n-3} \dots \frac{1}{2} \frac{\pi}{2} \right\} \\
&= 4a^4 \int_0^{\frac{\pi}{2}} \cos^4 \theta d\theta = 4a^4 \frac{3}{4} \cdot \frac{1}{2} \cdot \frac{\pi}{2} \\
&= \frac{3\pi a^4}{4}.
\end{aligned}$$

Example 13: Evaluate $\int_1^4 \int_{\sqrt{y}}^2 (x^2 + y^2) dx dy$ by changing the order of integration.

Solution:

In the given integral x varies from $\sqrt{y}$ to 2 and y varies from 1 to 4.

$$x = \sqrt{y}$$

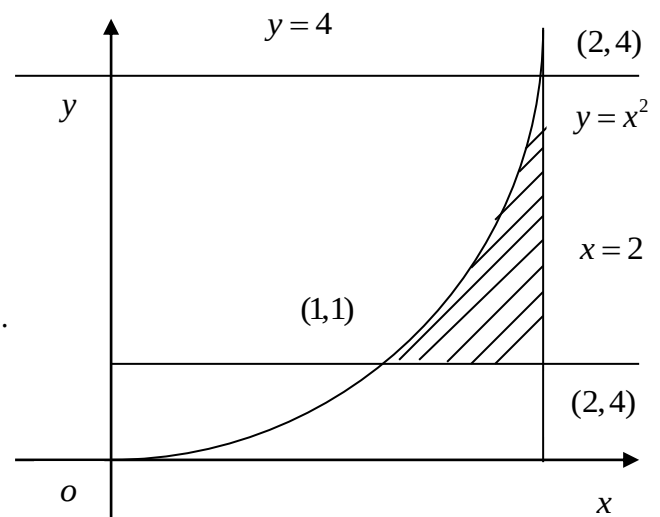
$$y = x^2$$

The region of integration is shown in Fig.3.23

We change the order of integration by integrating first with respect to y . Consider the vertical strip. Along the vertical strip, y varies from 1 to x^2 and x varies from 1 to 2.

$$\begin{aligned}
\therefore \int_1^4 \int_{\sqrt{y}}^2 (x^2 + y^2) dx dy &= \int_1^2 \int_1^{x^2} (x^2 + y^2) dy dx \\
&= \int_1^2 \left(x^2 y + \frac{y^3}{3} \right)_1^{x^2} dx \\
&= \int_1^2 \left(x^4 + \frac{x^6}{3} - x^2 - \frac{1}{3} \right) dx
\end{aligned}$$

Fig. 3.23.



$$\begin{aligned}
&= \left(\frac{x^5}{5} + \frac{x^7}{21} - \frac{x^3}{3} - \frac{1}{3}x \right)_1^2 \\
&= \left(\frac{32}{5} + \frac{128}{21} - \frac{8}{3} - \frac{2}{3} \right) - \left(\frac{1}{5} + \frac{1}{21} - \frac{1}{3} - \frac{1}{3} \right) = \frac{806}{105}
\end{aligned}$$

Exercise

I. Change the order of integration and hence evaluate

$$1. \int_0^a \int_0^x \frac{\cos y}{\sqrt{(a-x)(a-y)}} dx dy \quad 2. \int_0^{4a} \int_{\frac{x^2}{4a}}^{2\sqrt{ax}} dy dx$$

$$3. \int_0^a \int_b^{\sqrt{b^2-y^2}} xy dx dy \quad 4. \int_{-a}^a \int_0^{\sqrt{a^2-y^2}} x dx dy$$

$$5. \int_0^\infty \int_a^\infty \frac{e^{-y}}{y} dy dx \quad 6. \int_0^a \int_{\frac{x^2}{a}}^{2a-x} x^2 y dy dx$$

$$7. \int_0^1 \int_0^{\sqrt{2-x^2}} \frac{x}{\sqrt{x^2+y^2}} dy dx \quad 8. \int_0^a \int_{a-\sqrt{a^2-y^2}}^{a+\sqrt{a^2-y^2}} dx dy$$

$$9. \int_0^3 \int_1^{\sqrt{4-y}} (x+y) dx dy \quad 10. \int_0^4 \int_x^{2\sqrt{x}} (x^2+y^2) dy dx$$

$$11. \int_0^{2a} \int_{\frac{x^2}{4a}}^{3a-x} xy dy dx \quad 12. \int_0^a \int_y^a \frac{x+y}{x^2+y^2} dx dy$$

$$13. \int_0^a \int_y^a \frac{x^2 dx dy}{\sqrt{x^2 + y^2}}$$

14. Evaluate the following integrals by changing into polar co-ordinates.

$$(i) \int_0^1 \int_{x^2}^{2-x} xy dx dy \quad (ii) \int_0^{2a} \int_0^{\sqrt{2ax-x^2}} \frac{x dx dy}{x^2 + y^2} \quad (iii) \int_0^a \int_y^a \frac{x^2 dx dy}{\sqrt{x^2 + y^2}} \quad (iv) \int_{-a}^a \int_0^{\sqrt{a^2-x^2}} (x^2 + y^2) dx dy$$

$$(v) \int_0^a \int_y^a \frac{x dx}{x^2 + y^2} dy$$

Answers

$$\begin{array}{llllll} 1. 2\sin a & 2. 16\frac{a^3}{3} & 3. \frac{a^2 b^2}{8} & 4. \frac{2}{3}a^3 & 5. 1 & 6. \frac{3a^4}{8} \\ 7. \frac{2-\sqrt{2}}{2} & 8. \frac{\pi a^2}{2} & 9. \frac{241}{60} & 10. 21.94 & 11. \frac{8a^4}{3} & \\ 12. \frac{\pi a}{4} + \frac{a}{2} \log 2 & 13. \frac{a^3}{2} \log(1+\sqrt{2}) & 14. (i) \frac{3}{8} & (ii) \frac{\pi a}{2} & (iii) \frac{a^3}{3} \log(1+\sqrt{2}) & \\ (iv) \frac{\pi a^4}{4} & (v) \frac{\pi a}{4}. & & & & \end{array}$$

3.6. Applications of double and triple Integrals

Double and Triple integrals are used in finding the areas under curves, volumes of Solids, centre of gravity of a given mass in a region and moment of inertia of masses about x -axis, y axis and so on.

1. **Area:** The double integral of $f(x, y)$ over a region R is defined as $\iint_R f(x, y) dx dy$.

If $f(x, y)=1$ then $\iint_R dx dy$ represents area of the region R .

2. Area in polar co-ordinate:

The double integral of $f(r, \theta)$ in polar co-ordinates is given by $\iint_R f(r, \theta) r d\theta dr$. If $f(r, \theta) = 1$

then $\iint_R r d\theta dr$ represents the area of the region R .

Point to Remember

Area $\iint_R dx dy$ represents area of the region R

Area in polar co-ordinate $\iint_R r d\theta dr$ represents the area of the region R .

Example 1: Find the area between $y^2 = 4ax$ and $x^2 = 4ay$ by double integration.

(Jan-2010)

Solution:

$$\text{Let } y^2 = 4ax \quad \dots (1)$$

$$x^2 = 4ay \quad \dots (2)$$

$$\text{From (2), } y = \frac{x^2}{4a}$$

$$\text{Substituting } y \text{ in (1), we get } \frac{x^4}{16a^2} = 4ax$$

$$x^4 = 64a^3x$$

$$x^3 = 64a^4$$

$$x = 4a$$

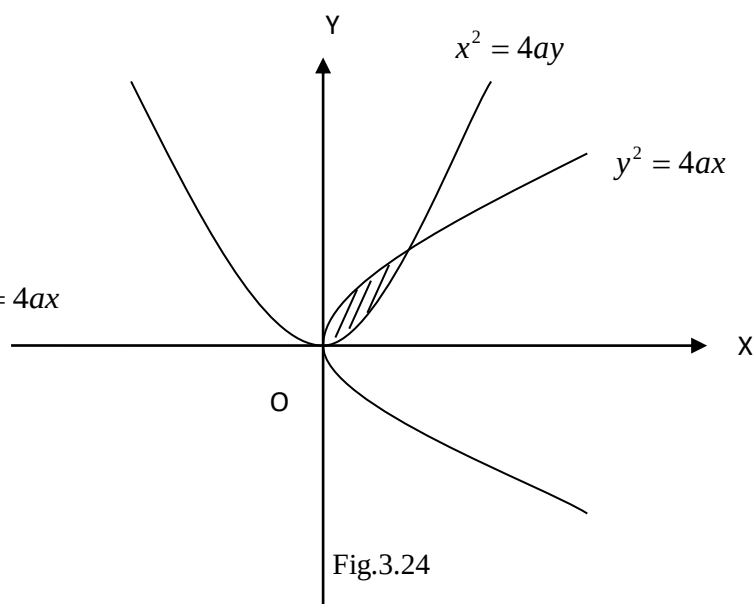


Fig.3.24

Substituting $x = 4a$ in (2), $16a^2 = 4ay \Rightarrow y = 4a$

$\therefore$ The points of intersection are $(4a, 4a)$ and $(0, 0)$.

$$\begin{aligned}
 \text{Required area} &= \int_0^{4a} \int_{\frac{x^2}{4a}}^{\sqrt{4ax}} dy dx \\
 &= \int_0^{4a} \left(y \right)_{\frac{x^2}{4a}}^{\sqrt{4ax}} dx \\
 &= \int_0^{4a} \left(\sqrt{4ax} - \frac{x^2}{4a} \right) dx = \left(\frac{\sqrt{4a} x^{3/2}}{3/2} - \frac{x^3}{12a} \right)_0^{4a} \\
 &= \frac{2\sqrt{4a}(4a)^{3/2}}{3} - \frac{(4a)^3}{12a} = \frac{2 \times 2 \times \sqrt{a} \times 4 \times 2 \times a\sqrt{a}}{3} - \frac{64a^3}{12a} \\
 &= \frac{32a^2}{3} - \frac{32a^2}{6} = \frac{64a^2 - 32a^2}{6} = \frac{32a^2}{6} = \frac{16a^2}{3} \text{ sq. units}
 \end{aligned}$$

Example 2: Find the area enclosed by the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

Solution:

Equation of ellipse is given by $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \quad \dots(1)$

$$\frac{y^2}{b^2} = 1 - \frac{x^2}{a^2} = \frac{a^2 - x^2}{a^2}$$

$$y^2 = \frac{b^2}{a^2} (a^2 - x^2)$$

$$y = \pm \frac{b}{a} \sqrt{a^2 - x^2}$$

In the first quadrant y varies from 0 to $\frac{b}{a} \sqrt{a^2 - x^2}$

Substituting, $y = 0$ in (1),

$$\frac{x^2}{a^2} = 1$$

$$x^2 = a^2, \quad x = \pm a.$$

Area of the ellipse = 4 Area of OAB (see Fig. 3.25)

$$\begin{aligned} &= 4 \int_0^a \int_0^{\frac{b}{a}\sqrt{a^2-x^2}} dy \, dx = 4 \int_0^a (y)_0^{\frac{b}{a}\sqrt{a^2-x^2}} dx \\ &= 4 \int_0^a \frac{b}{a} \sqrt{a^2-x^2} \, dx = \frac{4b}{a} \int_0^a \sqrt{a^2-x^2} \, dx \\ &= \frac{4b}{a} \left[\frac{x}{a} \sqrt{a^2-x^2} + \frac{a^2}{2} \sin^{-1} \left(\frac{x}{a} \right) \right]_0^a \\ &= \frac{4b}{a} \cdot \frac{a^2}{2} \cdot \frac{\pi}{2} \\ &= \frac{\pi ab}{4} \end{aligned}$$

$$\text{Area of the ellipse} = 4 \left(\frac{\pi ab}{4} \right)$$

$$= \pi ab \text{ sq. units}$$

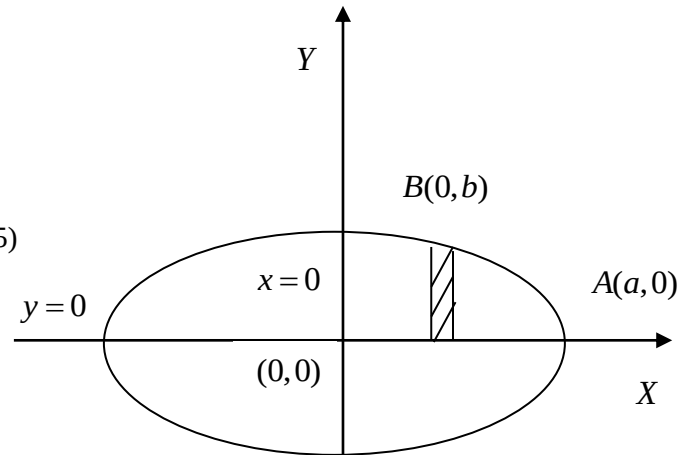


Fig.3.25

Example 3: Find the area between the parabolas $y^2 = x$ and $x^2 = y$ by using double integration. (April 2003)

Solution:

The equations of parabolas are given by

$$y^2 = x \quad \dots(1)$$

$$x^2 = y \quad \dots(2)$$

From (1) and (2), y varies from x^2 to $\sqrt{x}$

x varies from 0 to 1.

$$\begin{aligned} \text{Required Area} &= \int_0^1 \int_{x^2}^{\sqrt{x}} dy \, dx \\ &= \int_0^1 (y)_{x^2}^{\sqrt{x}} dx \\ &= \int_0^1 (\sqrt{x} - x^2) dx \\ &= \left[\frac{x^{\frac{3}{2}}}{\frac{3}{2}} - \frac{x^3}{3} \right]_0^1 \\ &= \frac{2}{3} - \frac{1}{3} = \frac{1}{3} \text{ sq. units.} \end{aligned}$$

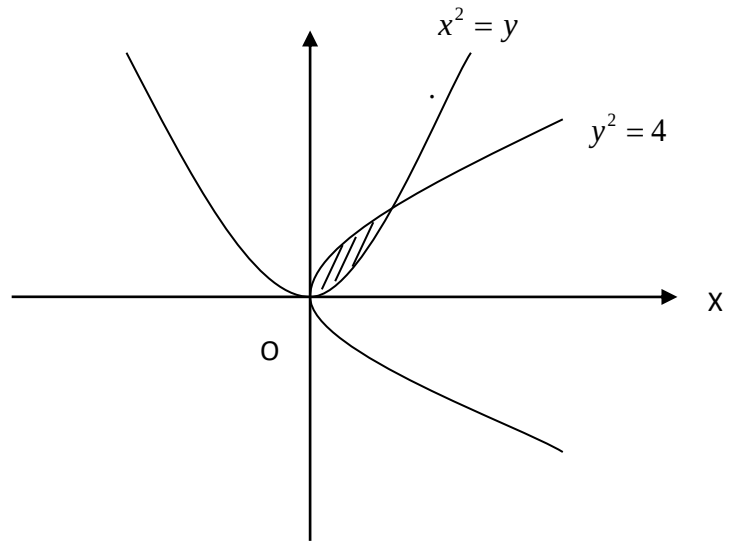


Fig. 3.26

Example 4: Find the area between the parabolas $3y^2 = 25x$ and $5x^2 = 9y$

(May-2009)

Solution:

$$\text{Let } 3y^2 = 25x \quad \dots (1)$$

$$5x^2 = 9y \quad \dots (2)$$

$$\text{From (2) } y = \frac{5x^2}{9}$$

Substituting y in (1), we get

$$3 \left(\frac{5x^2}{9} \right)^2 = 25x$$

$$3 \times \frac{25x^4}{81} = 25x$$

$$x^3 = \frac{25 \times 81}{3 \times 25}$$

$$x^3 = 27 \Rightarrow x = 3$$

$\therefore x$ varies from 0 to 3

y varies from $\frac{5}{9}x^2$ to $\frac{5\sqrt{x}}{\sqrt{3}}$

$$\text{Required Area} = \int_0^3 \int_{\frac{5}{9}x^2}^{\frac{5\sqrt{x}}{\sqrt{3}}} dy \, dx = \int_0^3 (y)_{\frac{5}{9}x^2}^{\frac{5\sqrt{x}}{\sqrt{3}}} dx$$

$$= \int_0^3 \left(\frac{5\sqrt{x}}{\sqrt{3}} - \frac{5}{9}x^2 \right) dx = \left(\frac{5}{\sqrt{3}} \frac{x^{3/2}}{3/2} - \frac{5}{9} \frac{x^3}{3} \right)_0^3$$

$$= \frac{5 \times 2 \times 3\sqrt{3}}{\sqrt{3} \times 3} - \frac{5}{27} 27$$

$$= 10 - 5 = 5 \text{ sq. units.}$$

Example 5: Find the area enclosed between $y = x^2$ and $x + y = 2$

(May- 2007)

Solution:

$$\text{Given that } y = x^2 \quad \dots (1)$$

$$x + y = 2 \quad \dots (2)$$

From (2), $y = 2 - x$.

Substituting y in (1), we get

$$2 - x = x^2$$

$$x^2 + x - 2 = 0 \quad x = 1, -2$$

x varies from -2 to 1

y varies from x^2 to $2-x$

$$\text{Required Area} = \int_{-2}^1 \int_{x^2}^{2-x} dy dx$$

$$= \int_{-2}^1 (y)_{x^2}^{2-x} dx$$

$$= \int_{-2}^1 (2-x-x^2) dx$$

$$= \left(2x - \frac{x^2}{2} - \frac{x^3}{3} \right)_{-2}^1$$

$$= \left(2 - \frac{1}{2} - \frac{1}{3} \right) - \left(-4 - \frac{4}{2} + \frac{8}{3} \right)$$

$$= \left(\frac{12-3-2}{6} \right) - \left(\frac{-24-12+16}{6} \right)$$

$$= \frac{7}{6} + \frac{20}{6} = \frac{27}{6} = \frac{9}{2} \text{ sq. units}$$

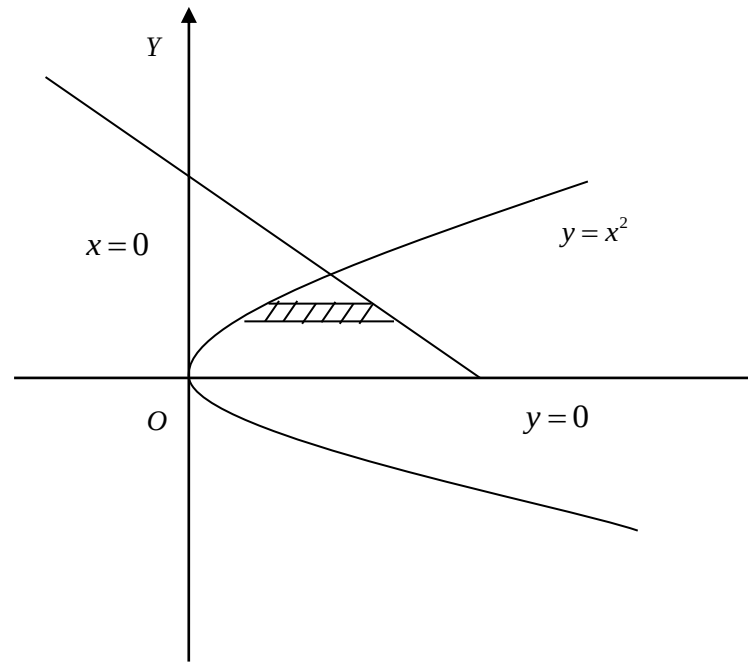


Fig. 3.27

Example 6: Evaluate $\iint_D (x^2 + y^2) dx dy$ where D is the region bounded by $y = x^2$, $x = 2$

and $x = 1$

Solution:

The region of integration is as shown in the figure.

In this region x varies from 1 to 2 and for each fixed x , y varies from 1 to x^2 .

$$\iint_D (x^2 + y^2) dx dy = \int_1^2 \int_1^{x^2} (x^2 + y^2) dy dx$$

$$\begin{aligned}
&= \int_1^2 \left(x^2 y + \frac{y^3}{3} \right)_{x^2} dx \\
&= \int_1^2 \left(x^4 + \frac{x^6}{3} - x^2 - \frac{1}{3} \right) dx \\
&= \left(\frac{x^5}{5} + \frac{x^7}{21} - \frac{x^3}{3} - \frac{x}{3} \right)_1^2 \\
&= \left(\frac{32}{5} + \frac{128}{21} - \frac{8}{3} - \frac{2}{3} \right) - \left(\frac{1}{5} + \frac{1}{21} - \frac{1}{3} - \frac{1}{3} \right) \\
&= \left(\frac{672 + 640 - 280 - 70}{105} \right) - \left(\frac{21 + 5 - 35 - 35}{105} \right) \\
&= \frac{762}{105} + \frac{44}{105} = \frac{806}{105}.
\end{aligned}$$

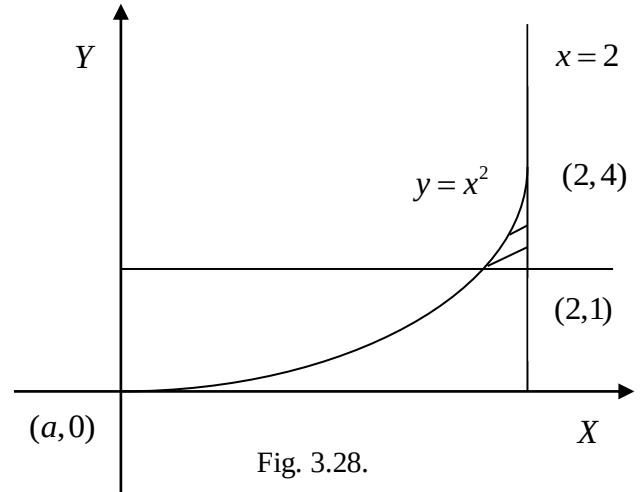


Fig. 3.28.

Example 7: Evaluate $\iint_D (x+y)^2 dx dy$ over the area bounded by the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

Solution:

$$\begin{aligned}
\text{Given that } \frac{x^2}{a^2} + \frac{y^2}{b^2} &= 1 \\
\frac{y^2}{b^2} &= \frac{a^2 - x^2}{a^2} \\
y^2 &= \frac{b^2}{a^2} (a^2 - x^2) \\
y &= \pm \frac{b}{a} \sqrt{a^2 - x^2}
\end{aligned}$$

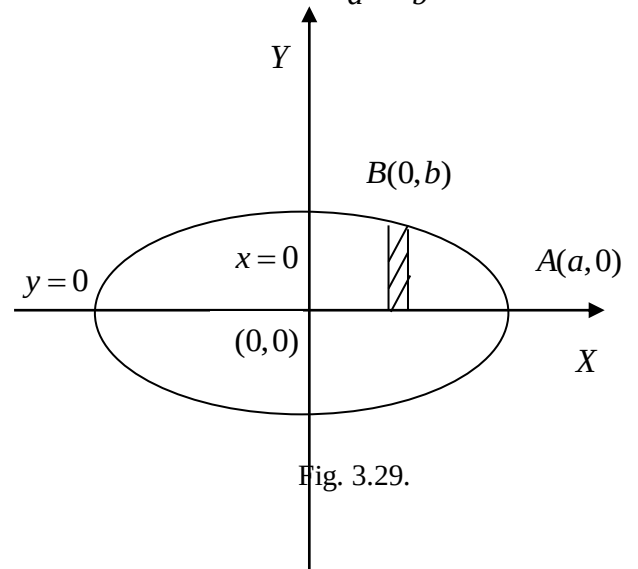


Fig. 3.29.

y varies from $-\frac{b}{a} \sqrt{a^2 - x^2}$ to $\frac{b}{a} \sqrt{a^2 - x^2}$

x varies from $-a$ to a

$$\begin{aligned}
 \iint (x+y)^2 dx dy &= \int_{-a}^a \int_{-b/\sqrt{a^2-x^2}}^{b/\sqrt{a^2-x^2}} (x+y)^2 dy dx \\
 &= \int_{-a}^a \int_{-b/\sqrt{a^2-x^2}}^{b/\sqrt{a^2-x^2}} (x^2 + y^2 + 2xy) dy dx \\
 &= 2 \int_{-a}^a \left(x^2 y + \frac{y^3}{3} \right) \Big|_0^{b/\sqrt{a^2-x^2}} dx \left\{ \because \int_{-b/\sqrt{a^2-x^2}}^{b/\sqrt{a^2-x^2}} 2xy dy = 0, f(y) \text{ is an odd function} \right\} \\
 &= 2 \int_{-a}^a \left(x^2 \frac{b}{a} \sqrt{a^2-x^2} + \frac{b^3}{a^3} \frac{(a^2-x^2)\sqrt{a^2-x^2}}{3} \right) dx \\
 &= \frac{4b}{a} \int_0^a x^2 \sqrt{a^2-x^2} dx + \frac{4b^3}{3a^3} \int_0^a (a^2-x^2)^{3/2} dx
 \end{aligned}$$

Put $x = a \sin \theta$ $x=0, \theta=0$

$$dx = a \cos \theta d\theta \quad x=a, \theta = \frac{\pi}{2}$$

$$\begin{aligned}
 \iint (x+y)^2 dx dy &= \frac{4b}{a} \int_0^{\frac{\pi}{2}} a^2 \sin^2 \theta \sqrt{a^2 - a^2 \sin^2 \theta} a \cos \theta d\theta + \\
 &\quad \frac{4b^3}{3a^3} \int_0^{\frac{\pi}{2}} (a^2 - a^2 \sin^2 \theta)^{3/2} a \cos \theta d\theta \\
 &= \frac{4b}{a} \int_0^{\frac{\pi}{2}} a^4 \sin^2 \theta \cos^2 \theta d\theta + \frac{4b^3}{3a^3} \int_0^{\frac{\pi}{2}} (a^2)^{3/2} (\cos^2 \theta)^{3/2} a \cos \theta d\theta
 \end{aligned}$$

$$\begin{aligned}
&= 4a^3b \int_0^{\frac{\pi}{2}} \sin^2 \theta \cos^2 \theta d\theta + \frac{4b^3}{3a^3} a^4 \int_0^{\frac{\pi}{2}} \cos^4 \theta d\theta \\
&= 4a^3b \int_0^{\frac{\pi}{2}} (\sin^2 \theta - \sin^4 \theta) d\theta + \frac{4b^3}{3a^3} a^4 \int_0^{\frac{\pi}{2}} \cos^4 \theta d\theta \\
&= 4a^3b \left(\frac{1}{2} \cdot \frac{\pi}{2} - \frac{3}{4} \cdot \frac{1}{2} \cdot \frac{\pi}{2} \right) + \frac{4b^3a}{3} \cdot \frac{3}{4} \cdot \frac{1}{2} \cdot \frac{\pi}{2} \\
&= 4a^3b \left(\frac{\pi}{4} - \frac{3\pi}{16} \right) + \frac{4b^3a\pi}{16} \\
&= \frac{a^3b\pi}{4} + \frac{ab^3\pi}{4} \\
&= \frac{ab\pi}{4} (a^2 + b^2)
\end{aligned}$$

Example 8: Find the area of the circle $x^2 + y^2 = r^2$ by using double integral.

Solution:

Area of the circle $A = 4 \iint_R dx dy$ (R is the I quadrant of the circle)

$$\text{Let } x^2 + y^2 = r^2 \quad \dots (1)$$

$$y^2 = r^2 - x^2$$

$$y = \pm \sqrt{r^2 - x^2}$$

$$x = \pm r$$

In first quadrant x varies 0 to r

y varies from 0 to $\sqrt{r^2 - x^2}$

$$\text{Required Area} = 4 \int_0^r \int_0^{\sqrt{r^2 - x^2}} dy dx$$

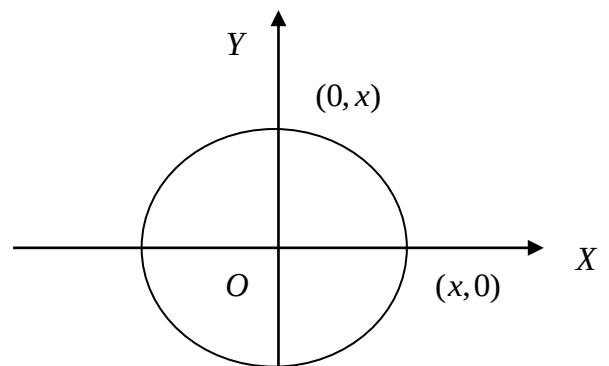


Fig. 3.30

$$\begin{aligned}
&= 4 \int_0^r [y]_0^{\sqrt{r^2-x^2}} dx \\
&= 4 \int_0^r \sqrt{r^2-x^2} dx \\
&= 4 \left[\frac{x}{r} \sqrt{r^2-x^2} + \frac{r^2}{2} \sin^{-1} \left(\frac{x}{r} \right) \right]_0^r \\
&= 4 \left[\frac{r^2}{2} \sin^{-1}(1) \right] = 4 \left[\frac{r^2}{2} \frac{\pi}{2} \right] \\
&= 4 \left[\frac{\pi r^2}{4} \right] = \pi r^2 \text{ sq. units.}
\end{aligned}$$

Example 9: Evaluate $I = \iint_D e^{\frac{y}{x}} dx dy$ where D is the region bounded by the straight lines

$y = x$, $y = 0$ and $x = 1$.

Solution:

The region of integration is shown in the Fig. 3.31.

$$\text{Area} = \int_0^1 \int_0^x e^{\frac{y}{x}} dy dx$$

$$= \int_0^1 \left[\frac{e^{\frac{y}{x}}}{\frac{1}{x}} \right]_0^x dx$$

$$= \int_0^1 (xe^1 - xe^0) dx$$

$$= \int_0^1 [xe - x] dx = \left[\frac{x^2 e}{2} - \frac{x^2}{2} \right]_0^1$$

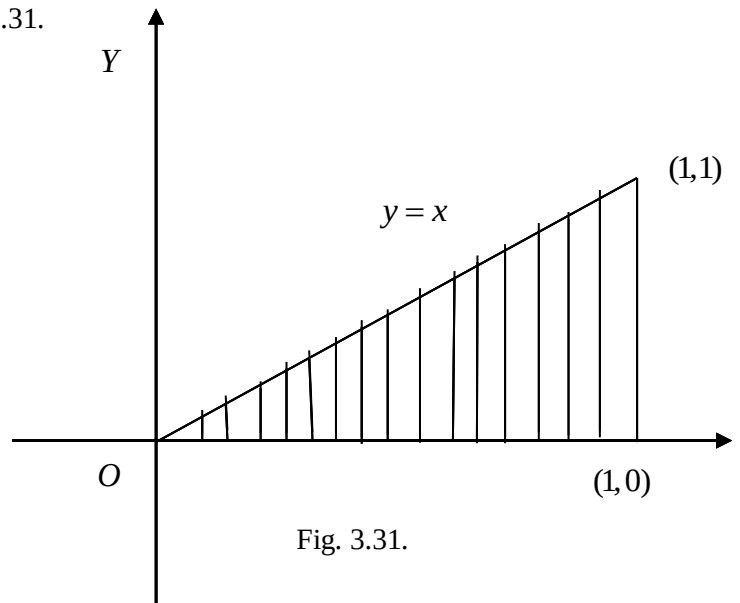


Fig. 3.31.

$$= \frac{e}{2} - \frac{1}{2} = \frac{e-1}{2}$$

Example 10: Find the area situated outside the circle $r = 2a \cos \theta$ and inside the cardioid $r = a(1 + \cos \theta)$.

Solution:

Required Area = $\iint_R r \, dr \, d\theta$, r varies from $2a \cos \theta$ to $a(1 + \cos \theta)$.

θ varies from 0 to π .

$$\begin{aligned} \text{Area} &= 2 \int_0^\pi \int_{2a \cos \theta}^{a(1+\cos \theta)} r \, dr \, d\theta \\ &= 2 \int_0^\pi \left[\frac{r^2}{2} \right]_{2a \cos \theta}^{a(1+\cos \theta)} d\theta \\ &= \frac{2}{2} \int_0^\pi \left[a^2 (1 + \cos \theta)^2 - 4a^2 \cos^2 \theta \right] d\theta \\ &= \int_0^\pi \left[a^2 [1 + \cos^2 \theta + 2 \cos \theta] - 4a^2 \cos^2 \theta \right] d\theta \\ &= \int_0^\pi \left(a^2 + a^2 \cos^2 \theta + 2a^2 \cos \theta - 4a^2 \cos^2 \theta \right) d\theta \\ &= \int_0^\pi \left[a^2 + 2a^2 \cos \theta - 3a^2 \cos^2 \theta \right] d\theta \\ &= \int_0^\pi \left[a^2 + 2a^2 \cos \theta - 3a^2 \left(\frac{1 + \cos 2\theta}{2} \right) \right] d\theta \\ &= \int_0^\pi \left[a^2 + 2a^2 \cos \theta - \frac{3a^2}{2} - \frac{3a^2 \cos 2\theta}{2} \right] d\theta = \left[a^2 \theta + 2a^2 \sin \theta - \frac{3a^2}{2} \theta - \frac{3a^2 \sin 2\theta}{4} \right]_0^\pi \end{aligned}$$

$$= \left[\left(a^2 \pi + 2a^2(0) - \frac{3a^2}{2} \pi - 0 \right) - (0) \right] = \frac{2a^2 \pi - 3a^2 \pi}{2}$$

$$= -\frac{a^2 \pi}{2} = \frac{\pi a^2}{2} \text{ sq. units.}$$

Example 11: Find the area of the cardioid $r = a(1 + \cos \theta)$

(Nov-2006)

Solution:

Given that $r = a(1 + \cos \theta)$

r varies from 0 to $a(1 + \cos \theta)$.

θ varies from $-\pi$ to π .

$$\text{Required Area} = \int_{-\pi}^{\pi} \int_0^{a(1+\cos\theta)} r \, dr \, d\theta$$

$$= \int_{-\pi}^{\pi} \left[\frac{r^2}{2} \right]_0^{a(1+\cos\theta)} d\theta$$

$$= \int_{-\pi}^{\pi} \frac{a^2 (1 + \cos \theta)^2}{2} d\theta$$

$$= \frac{a^2}{2} \int_{-\pi}^{\pi} (1 + \cos^2 \theta + 2 \cos \theta) d\theta$$

$$= \frac{a^2}{2} \int_{-\pi}^{\pi} \left[1 + \frac{1 + \cos 2\theta}{2} + 2 \cos \theta \right] d\theta = \frac{a^2}{2} \left[\theta + \frac{\theta}{2} + \frac{\sin 2\theta}{4} + 2 \sin \theta \right]_{-\pi}^{\pi}$$

$$= \frac{a^2}{2} \left[\left(\pi + \frac{\pi}{2} + 0 + 0 \right) - \left(-\pi - \frac{\pi}{2} + 0 + 0 \right) \right]$$

$$= \frac{a^2}{2} \left[\frac{3\pi}{2} + \frac{3\pi}{2} \right] = \frac{a^2}{2} \left(\frac{6\pi}{2} \right) = \frac{6\pi a^2}{4} \text{ sq. units.}$$

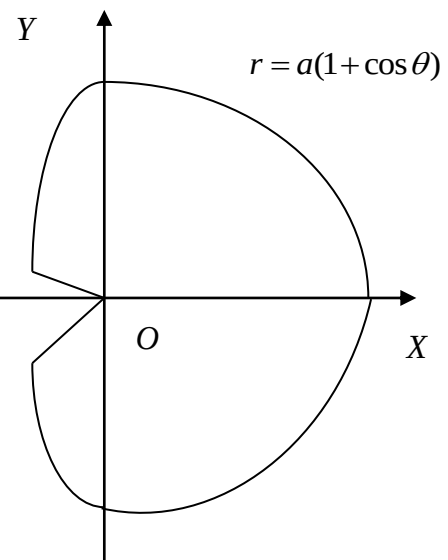


Fig. 3.32.

Exercise

1. Find the area bounded by the circles $x^2 + y^2 = 4$ and $x^2 + y^2 = 4x$.
2. Find the area common to the parabola $y^2 = x$ and the circle $x^2 + y^2 = 4x$.
3. Find the area enclosed by the curve $y^2 = 4ax$ and the lines $x + y = 3a, y = 0$.
4. Find the area bounded by $y = 2 - x$ and $x^2 + y^2 = 4$.
5. Find the area bounded by $y = 2x^2$ and $y^2 = 4x$.
6. Find the area bounded by $x = y - y^2$ and $x + y = 0$.
7. Show that $\iint_D (x + y + a) dx dy$ where D is the region bounded by the circle $x^2 + y^2 = a^2$ is πa^3
8. Evaluate $\iint_D (1 + x + y) dx dy$ where D is the region bounded by the line $y = -x$, $x = \sqrt{y}$, $y = 0$ and $y = 2$.
9. Evaluate $\iint_D (x - y) dx dy$ where D is the region bounded by the line $y = x$ and the parabola $x^2 = y$.
10. Find the area which is inside the circle $r = 3a \cos \theta$ and outside the cardioids $r = a(1 + \cos \theta)$.
11. Evaluate $\iint_R xy(x + y) dx dy$ over the region R bounded by $y = x^2$ and $y = x$

Answers

1. $2\left[\frac{4\pi}{3} - \sqrt{3}\right]$
2. $3\sqrt{3} + \frac{4\pi}{3}$
3. $\frac{10a^2}{3}$
4. $\pi + 2$
5. $\frac{2}{3}$

$$6. \quad \frac{4}{3} \quad 8. \quad \frac{44\sqrt{2}}{15} + \frac{5}{3} \quad 9. \quad \frac{1}{60} \quad 10. \pi a^2$$

3.7. Volume of a solid as a double integral

Let the equation of the surface be given in the form $Z = f(x, y)$.

Let the region R be the projection of the surface in the xy plane. Then the volume of the solid is given

$$\begin{aligned} \text{by volume} &= \iint_R z \, dx \, dy \\ &= \iint_V f(x, y) \, dx \, dy. \end{aligned}$$

In Cartesian co ordinate, the volume of the solid is $\iint 2\pi y \, dy \, dx$.

In Polar co ordinate, the volume of the solid is $\iint 2\pi r^2 \sin \theta \, dr \, d\theta$

Example 1: Find the volume of the tetrahedron bounded by the co-ordinate planes and the planes $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$. (Nov- 2011)

Solution:

$$\text{Let } \frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$$

$$\frac{z}{c} = 1 - \frac{x}{a} - \frac{y}{b}$$

$$z = c \left[1 - \frac{x}{a} - \frac{y}{b} \right]$$

R is the triangular region OAB in the xy -plane.

In $\triangle OAB$, x varies from 0 to a and y varies from 0 to $b \left[1 - \frac{x}{a} \right]$ $\therefore$

$$\text{Volume} = \iint_R z \, dx \, dy$$

$$\begin{aligned}
&= \int_0^a \int_0^{b\left[1-\frac{x}{a}\right]} z \, dy \, dx \\
&= \int_0^a \int_0^{b\left[1-\frac{x}{a}\right]} c\left[1-\frac{x}{a}-\frac{y}{b}\right] \, dy \, dx \\
&= \int_0^a \left[\frac{c\left[1-\frac{x}{a}-\frac{y}{b}\right]^2}{\left(-\frac{2}{b}\right)} \right]_0^{b\left[1-\frac{x}{a}\right]} dx
\end{aligned}$$

$$\begin{aligned}
&= -\int_0^a \frac{bc}{2} \left[\left[1-\frac{x}{a}-\frac{b\left(1-\frac{x}{a}\right)}{b} \right]^2 - \left[1-\frac{x}{a} \right]^2 \right] dx \\
&= -\frac{bc}{2} \int_0^a \left\{ \left[\left(1-\frac{x}{a} \right) - \left(1-\frac{x}{a} \right) \right]^2 - \left[1-\frac{x}{a} \right]^2 \right\} dx \\
&= \frac{bc}{2} \int_0^a \left(1-\frac{x}{a} \right)^2 dx \\
&= \frac{bc}{2} \left[\frac{\left(1-\frac{x}{a} \right)^3}{3\left(-\frac{1}{a}\right)} \right]_0^a = \frac{bc}{2} \left[\frac{\left(1-\frac{a}{a} \right)^3}{\left(-\frac{3}{a}\right)} - \frac{1}{\left(-\frac{3}{a}\right)} \right] \\
&= \frac{bc}{2} \left[\frac{a}{3} \right]
\end{aligned}$$

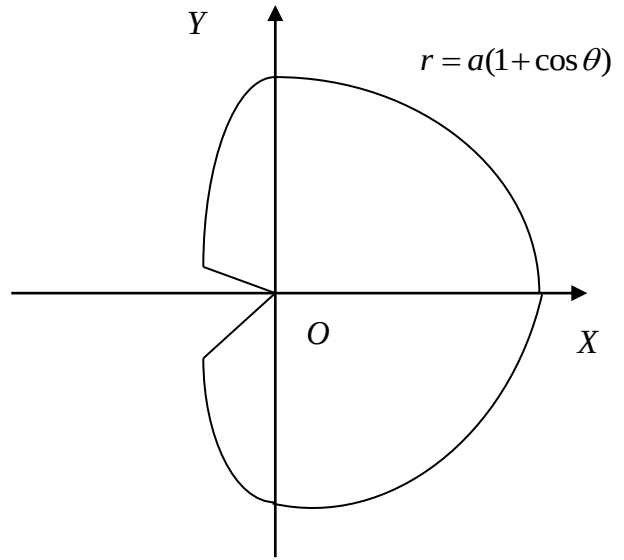


Fig. 3.33.

$$\text{Volume} = \frac{abc}{6} \text{ cubic units.}$$

Example 2: Find the volume of the region bounded by the surface $y^2 = 4ax$ and $x^2 = 4ay$ and the plane $z = 0$ and $z = 3$.

Solution:

$$\text{Let } y^2 = 4ax \quad \dots (1)$$

$$x^2 = 4ay \quad \dots (2)$$

$$\text{From (2), } y = \frac{x^2}{4a}$$

Substituting y in (1) we get,

$$\left(\frac{x^2}{4a} \right)^2 = 4ax$$

$$\frac{x^4}{16a^2} = 4ax$$

$$x^3 = 64a^3$$

$$x^3 = (4a)^3$$

$$x = 4a$$

$$\text{From (1) } y = 2\sqrt{ax}$$

$$\therefore y \text{ varies from } 2\sqrt{ax} \text{ to } \frac{x^2}{4a}$$

$$x \text{ varies from } 0 \text{ to } 4a.$$

$$\text{Volume} = \iint_R z \, dx \, dy$$

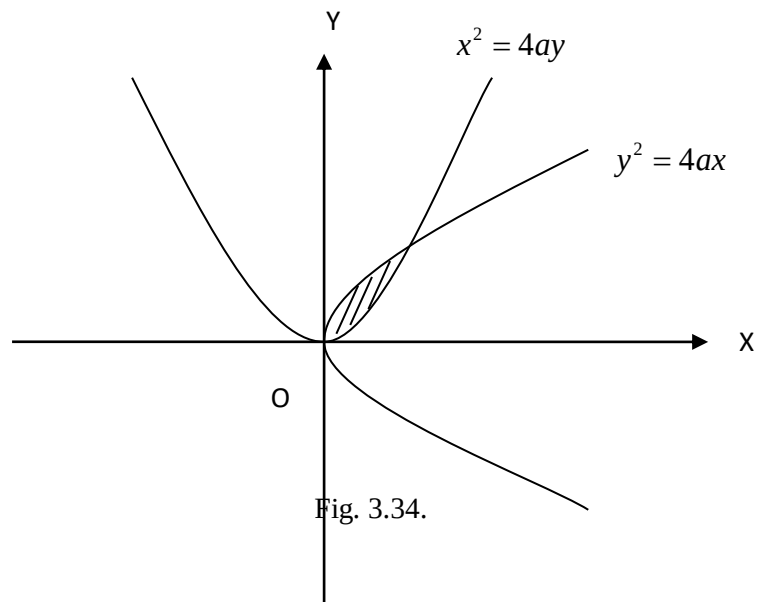


Fig. 3.34.

$$\begin{aligned}
&= \int_0^{4a} \int_{\frac{x^2}{4a}}^{2\sqrt{ax}} z \, dy \, dx \\
&= \int_0^{4a} \int_{\frac{x^2}{4a}}^{2\sqrt{ax}} z \, dy \, dx \\
&= \int_0^{4a} \int_{\frac{x^2}{4a}}^{2\sqrt{ax}} 3 \, dy \, dx \\
&= \int_0^{4a} [3y]_{\frac{x^2}{4a}}^{2\sqrt{ax}} \, dx \\
&= \int_0^{4a} \left[3[2\sqrt{ax}] - 3\left(\frac{x^2}{4a}\right) \right] \, dx \\
&= \int_0^{4a} \left[6\sqrt{ax} - \frac{3x^2}{4a} \right] \, dx \\
&= \left[\frac{6\sqrt{a}x^{3/2}}{3/2} - \frac{3x^3}{12a} \right]_0^{4a} \\
&= \frac{12\sqrt{a}(4a\sqrt{4a})}{3} - \frac{3(64a^3)}{12a} \\
&= 32a^2 - 16a^2 = 16a^2 \text{ cubic units.}
\end{aligned}$$

Example 3: Find the volume of the solid bounded by the surface $x=0, y=0, z=0$ and $x+y+z=1$.

Solution:

$$\begin{aligned}
\text{Let } x+y+z &= 1 & \dots (1) \\
z &= 1-x-y
\end{aligned}$$

In XY plane $z = 0$

Put $z = 0$ in (1) we get $x + y = 1$.

$\therefore$ The region R in the xy plane is a triangle bounded by $x = 0, y = 0$ and the line $x + y + z = 1$.

In the region OAB ,

x varies from 0 to 1.

y varies from 0 to $1 - x$.

$$\begin{aligned}
 \text{Volume} &= \int_0^1 \int_0^{1-x} z \, dy \, dx \\
 &= \int_0^1 \int_0^{1-x} (1 - x - y) \, dy \, dx \\
 &= \int_0^1 \left[y - xy - \frac{y^2}{2} \right]_0^{1-x} dx \\
 &= \int_0^1 \left[1 - x - x(1 - x) - \frac{(1 - x)^2}{2} \right] dx \\
 &= \int_0^1 \left[1 - x - x + x^2 - \frac{1}{2} - \frac{x^2}{2} + \frac{2x}{2} \right] dx \\
 &= \int_0^1 \left(\frac{x^2}{2} - x + \frac{1}{2} \right) dx = \left[\frac{x^3}{6} - \frac{x^2}{2} + \frac{x}{2} \right]_0^1 \\
 &= \frac{1}{6} - \frac{1}{2} + \frac{1}{2} = \frac{1}{6} \text{ cubic units.}
 \end{aligned}$$

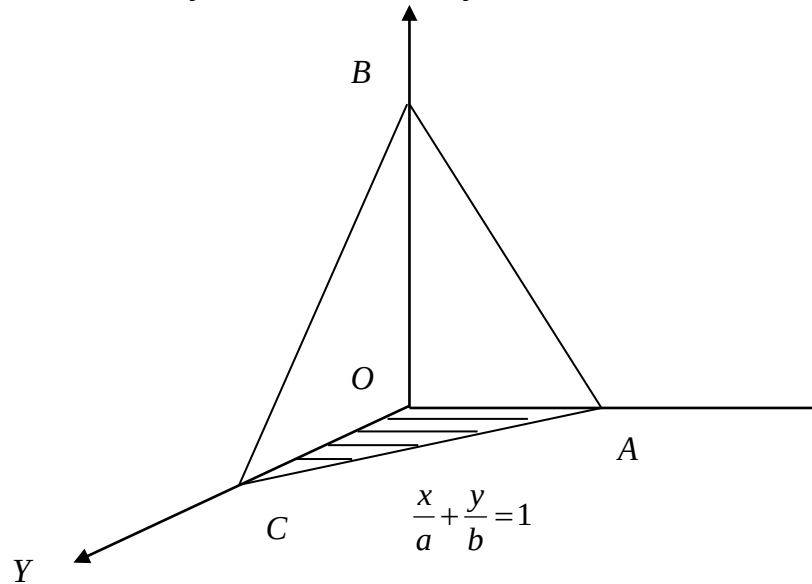


Fig. 3.35.

Example 4: Find the volume bounded by the cylinder $x^2 + y^2 = 4$ and the planes $y + z = 3$ and $z = 0$.

Solution:

$$\text{Let } x^2 + y^2 = 4 \quad \dots (1)$$

$$x^2 = 4 - y^2$$

$$x = \pm \sqrt{4 - y^2}$$

Substituting $x = 0$ in (1),

$$y^2 = 4$$

$$y = \pm 2$$

x varies from $-\sqrt{4 - y^2}$ to $\sqrt{4 - y^2}$.

y varies from -2 to 2 .

$$\text{Volume} = \iint_R z \, dx \, dy$$

$$= \int_{-2}^2 \int_{-\sqrt{4-y^2}}^{\sqrt{4-y^2}} z \, dx \, dy$$

$$= \int_{-2}^2 \int_{-\sqrt{4-y^2}}^{\sqrt{4-y^2}} (3 - y) \, dx \, dy$$

$$= \int_{-2}^2 \left[(3 - y)x \right]_{-\sqrt{4-y^2}}^{\sqrt{4-y^2}} dy$$

$$= \int_{-2}^2 \left[(3 - y) \left(\sqrt{4 - y^2} + \sqrt{4 - y^2} \right) \right] dy$$

$$= 2 \int_{-2}^2 \left((3 - y) \sqrt{4 - y^2} \right) dy$$

$$= 2 \int_{-2}^2 \left(3\sqrt{4 - y^2} - y\sqrt{4 - y^2} \right) dy$$

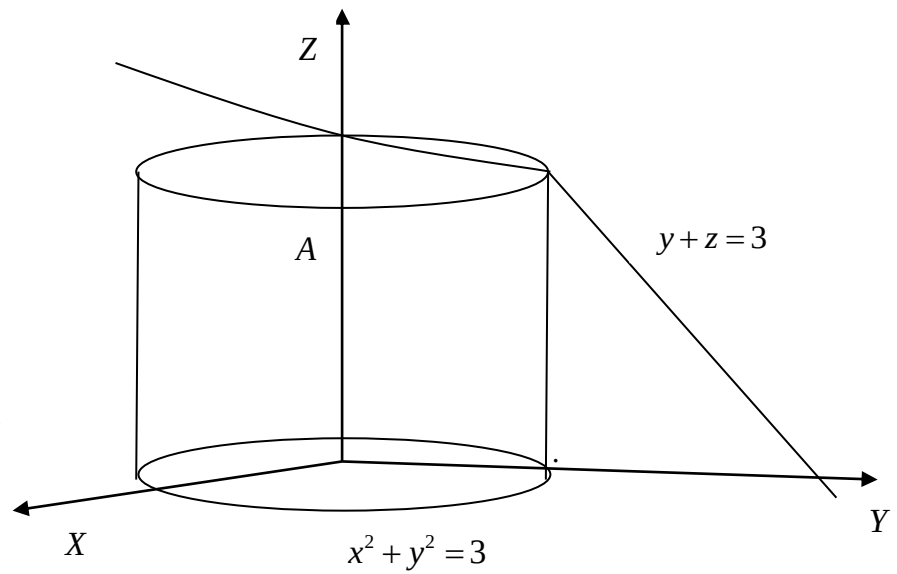


Fig. 3.36

$$\begin{aligned}
&= 2 \int_{-2}^2 3\sqrt{4-y^2} dy = 6 \times 2 \int_0^2 \sqrt{4-y^2} dy \\
&= 12 \left[\frac{y}{2} \sqrt{4-y^2} + \frac{4}{2} \sin^{-1} \left(\frac{y}{2} \right) \right]_0^2 \\
&= 12 \left[\left(0 + \frac{4}{2} \frac{\pi}{2} \right) - 0 \right] = 12 \times \frac{4\pi}{4} \\
&= 12\pi \text{ cubic units.}
\end{aligned}$$

Example 5: Find the volume enclosed by the cylinder $x^2 + y^2 = 2ax$ and $z^2 = 2ax$.

Solution:

Given that $z^2 = 2ax$

$$z = \pm \sqrt{2ax}$$

$$x^2 + y^2 = 2ax$$

$$y^2 = 2ax - x^2$$

$$y = \pm \sqrt{2ax - x^2}$$

The cylinder $x^2 + y^2 = 2ax$ cuts the x -axis when $y = 0$.

$$x^2 = 2ax$$

$$x = 2a, \quad y \text{ varies from } -\sqrt{2ax - x^2} \text{ to } \sqrt{2ax - x^2}$$

x varies from 0 to $2a$.

Half of the volume lies above the XY plane and half below it.

$$\begin{aligned}
\text{Volume} &= 2 \int \int z dy dx \\
&= 2 \int_0^{2a} \int_{-\sqrt{2ax-x^2}}^{\sqrt{2ax-x^2}} \sqrt{2ax} dy dx
\end{aligned}$$

$$\begin{aligned}
&= 2 \int_0^{2a} \left[\sqrt{2ax} y \right]_{-\sqrt{2ax-x^2}}^{\sqrt{2ax-x^2}} dx \\
&= 2\sqrt{2a} \int_0^{2a} \sqrt{x} \left[\sqrt{2ax-x^2} + \sqrt{2ax-x^2} \right] dx \\
&= 4\sqrt{2a} \int_0^{2a} \sqrt{x} \sqrt{2ax-x^2} dx \\
&= 4\sqrt{2a} \int_0^{2a} x \sqrt{2ax} dx
\end{aligned}$$

Put $t = 2a - x$

$x = 0, \quad t = 2a$

$dt = -dx$

$x = 2a, t = 0$

$$\begin{aligned}
\text{Volume} &= 4\sqrt{2a} \int_{2a}^0 (2a-t) \sqrt{t} (-dt) \\
&= 4\sqrt{2a} \int_0^{2a} (2at^{\frac{1}{2}} - t^{\frac{3}{2}}) dt \\
&= 4\sqrt{2a}^{\frac{1}{2}} \left[\frac{2at^{\frac{3}{2}}}{\frac{3}{2}} - \frac{t^{\frac{5}{2}}}{\frac{5}{2}} \right]_0^{2a} \\
&= 4\sqrt{2a}^{\frac{1}{2}} \left[\frac{2 \times 2a \times 2a \times \sqrt{2} \times \sqrt{a}}{3} - \frac{2 \times 4a^2 \times \sqrt{2} \times \sqrt{a}}{5} \right] \\
&= 4\sqrt{2} \sqrt{a} \left[\frac{40\sqrt{2}a^2 \sqrt{a} - 24\sqrt{2}a^2 \sqrt{a}}{15} \right] \\
&= 4\sqrt{2} \sqrt{a} \frac{16\sqrt{2}a^2 \sqrt{a}}{15} = \frac{128a^3}{15} \text{ cubic units.}
\end{aligned}$$

Example 6: Find the volume of the cardioids $r = a(1 + \cos \theta)$ is revolved about the initial line.

Solution:

$$r = a(1 + \cos \theta)$$

$$r \text{ varies from } 0 \text{ to } a(1 + \cos \theta)$$

$$\theta \text{ varies from } 0 \text{ to } \pi$$

$$\begin{aligned} \text{Volume} &= \iint_R (2\pi r^2 \sin \theta) dr d\theta \\ &= 2\pi \int_0^\pi \int_0^{a(1+\cos \theta)} r^2 \sin \theta dr d\theta \\ &= 2\pi \int_0^\pi \left[\frac{r^3}{3} \right]_0^{a(1+\cos \theta)} \sin \theta d\theta \\ &= 2\pi \int_0^\pi \sin \theta \left[\frac{a^3(1+\cos \theta)^3}{3} \right] d\theta \\ &= \frac{2\pi a^3}{3} \int_0^\pi \sin \theta (1+\cos \theta)^3 d\theta \end{aligned}$$

$$\text{Put } 1 + \cos \theta = t \quad \text{when } \theta = 0, t = 2$$

$$-\sin \theta d\theta = dt \quad \theta = \pi, t = 0$$

$$\begin{aligned} \text{Volume} &= \frac{2\pi a^3}{3} \int_2^0 t^3 (-dt) \\ &= \frac{2\pi a^3}{3} \int_0^2 t^3 dt \\ &= \frac{2\pi a^3}{3} \left(\frac{t^4}{4} \right)_0^2 = \frac{2\pi a^3}{3} \left(\frac{16}{4} \right) \end{aligned}$$

$$= \frac{8\pi a^3}{3} \text{ cubic units.}$$

3.8. Volume of a solid as a triple integral

The triple integral of $f(x, y, z)$ on a closed region R is defined by $\iiint_V f(x, y, z) dv$.

If $f(x, y, z) = 1$ then $\iiint_V dv$ represents the volume of the solid.

Example 1: Find the volume of the sphere $x^2 + y^2 + z^2 = a^2$ without transformation.

(April- 2010)

Solution:

Required volume = $8 \times$ volume in an octant.

$$x^2 + y^2 + z^2 = a^2 \quad \dots (1)$$

$$z^2 = a^2 - x^2 - y^2$$

$$z = \pm \sqrt{a^2 - x^2 - y^2}$$

Put $z = 0$ in (1)

$$x^2 + y^2 = a^2 \quad \dots (2)$$

$$y^2 = a^2 - x^2$$

$$y = \pm \sqrt{a^2 - x^2}$$

Put $y = 0$ in (2)

$$x^2 = a^2$$

$$x = \pm a$$

In the first octant,

$$z \text{ varies from } 0 \text{ to } \sqrt{a^2 - x^2 - y^2}$$

y varies from 0 to $\sqrt{a^2 - x^2}$

x varies from 0 to a

$$\begin{aligned}
 \text{Volume} &= 8 \int_0^a \int_0^{\sqrt{a^2 - x^2}} \int_0^{\sqrt{a^2 - x^2 - y^2}} dz \, dy \, dx \\
 &= 8 \int_0^a \int_0^{\sqrt{a^2 - x^2}} [z]_0^{\sqrt{a^2 - x^2 - y^2}} dy \, dx \\
 &= 8 \int_0^a \int_0^{\sqrt{a^2 - x^2}} \sqrt{a^2 - x^2 - y^2} dy \, dx \\
 &= 8 \int_0^a \left[\frac{a^2 - x^2}{2} \sin^{-1} \frac{y}{\sqrt{a^2 - x^2}} + \frac{y}{2} \sqrt{a^2 - x^2 - y^2} \right]_0^{\sqrt{a^2 - x^2}} dx \\
 &= 8 \int_0^a \left[\frac{a^2 - x^2}{2} \sin^{-1}(1) + \frac{\sqrt{a^2 - x^2}}{2} \sqrt{a^2 - x^2 - a^2 + x^2} \right] dx \\
 &= 8 \int_0^a \left[\frac{a^2 - x^2}{2} \frac{\pi}{2} \right] dx \\
 &= \frac{8\pi}{4} \left[a^2 x - \frac{x^3}{3} \right]_0^a = 2\pi \left[a^3 - \frac{a^3}{3} \right] \\
 &= 2\pi \left[\frac{2a^3}{3} \right] = \frac{4\pi a^3}{3} \text{ cubic units.}
 \end{aligned}$$

Example 2: Find the volume of the ellipsoid $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$ which lies in the first octant. (Jan-2010)

Solution:

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1 \quad \dots (1)$$

$$\frac{z^2}{c^2} = 1 - \frac{x^2}{a^2} - \frac{y^2}{b^2}$$

$$z^2 = c^2 \left[1 - \frac{x^2}{a^2} - \frac{y^2}{b^2} \right]$$

$$z = \pm c \sqrt{1 - \frac{x^2}{a^2} - \frac{y^2}{b^2}}$$

Put $z = 0$ in (1)

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \quad \dots (2)$$

$$\frac{y^2}{b^2} = 1 - \frac{x^2}{a^2}$$

$$y^2 = b^2 \left[1 - \frac{x^2}{a^2} \right]$$

$$y = \pm b \sqrt{1 - \frac{x^2}{a^2}}$$

Put $y = 0$ in (2)

$$\frac{x^2}{a^2} = 1$$

$$x^2 = a^2$$

$$x = \pm a$$

$$\text{Volume} = 8 \iiint_R dz \, dy \, dx$$

$$= 8 \int_0^a \int_0^{b\sqrt{1-\frac{x^2}{a^2}}} \int_0^{c\sqrt{1-\frac{x^2}{a^2}-\frac{y^2}{b^2}}} dz \, dy \, dx$$

$$\begin{aligned}
&= 8 \int_0^a \int_0^{b\sqrt{1-\frac{x^2}{a^2}}} [z]_0^{c\sqrt{1-\frac{x^2}{a^2}-\frac{y^2}{b^2}}} dy dx \\
&= 8 \int_0^a \int_0^{b\sqrt{1-\frac{x^2}{a^2}}} c \sqrt{1-\frac{x^2}{a^2}-\frac{y^2}{b^2}} dy dx \\
&= 8c \int_0^a \int_0^{b\sqrt{1-\frac{x^2}{a^2}}} \sqrt{\frac{a^2-x^2}{a^2}-\frac{y^2}{b^2}} dy dx \\
&= \frac{8c}{b} \int_0^a \int_0^{b\sqrt{1-\frac{x^2}{a^2}}} \sqrt{\frac{b^2}{a^2}(a^2-x^2)-y^2} dy dx \\
&=
\end{aligned}$$

$$\begin{aligned}
&\frac{8c}{b} \int_0^a \frac{y}{2} \sqrt{\frac{b^2}{a^2}(a^2-x^2)-y^2} + \frac{1}{2} \frac{b^2}{a^2}(a^2-x^2) \sin^{-1} \left(\frac{y}{b/a \sqrt{(a^2-x^2)}} \right) \bigg|_0^{b\sqrt{1-\frac{x^2}{a^2}}} dx \\
&\left[\because \int \sqrt{a^2-x^2} dx = \frac{x}{2} \sqrt{a^2-x^2} + \frac{a^2}{2} \sin^{-1} \left(\frac{x}{a} \right) \right]
\end{aligned}$$

$$\begin{aligned}
&= \frac{8c}{b} \int_0^a \left(\frac{b^2}{a^2}(a^2-x^2) \sin^{-1}(1) \right) dx \\
&= \frac{8cb^2}{2a^2b} \cdot \frac{\pi}{2} \int_0^a (a^2-x^2) dx = \frac{2cb}{a^2} \left(a^2x - \frac{x^3}{3} \right) \bigg|_0^a \\
&= \frac{2cb}{a^2} \left(a^3 - \frac{a^3}{3} \right) = \frac{2cb}{a^2} \left(\frac{2a^3}{3} \right)
\end{aligned}$$

$$= \frac{4abc}{3} \text{ cubic units.}$$

Example 3: Find the volume of the region bounded by the surfaces $y^2 = 4ax$ and $x^2 = 4ay$ and the plane $z = 0$ and $z = 3$

Solution:

$$\text{Let } y^2 = 4ax \quad \dots (1)$$

$$x^2 = 4ay \quad \dots (2)$$

$$\text{From (2), } y = \frac{x^2}{4a}$$

Sub y in (1)

$$\frac{x^4}{16a^2} = 4ax$$

$$x^3 = 64a^3$$

$$x^3 = (4a)^3$$

$$x = 4a$$

x varies from 0 to $4a$

y varies from $\frac{x^2}{4a}$ to $\sqrt{4ax}$

z varies from 0 to 3.

$$\begin{aligned} \text{Required volume} &= \int_0^{4a} \int_{\frac{x^2}{4a}}^{\sqrt{4ax}} \int_0^3 dz \, dy \, dx \\ &= \int_0^{4a} \int_{\frac{x^2}{4a}}^{\sqrt{4ax}} [z]_0^3 \, dy \, dx \end{aligned}$$

$$\begin{aligned}
&= \int_0^{4a} \int_{\frac{x^2}{4a}}^{\sqrt{4ax}} 3 \, dy \, dx \\
&= \int_0^{4a} [3y]_{\frac{x^2}{4a}}^{\sqrt{4ax}} \, dx \\
&= \int_0^{4a} \left[3\sqrt{4ax} - \frac{3x^2}{4a} \right] \, dx \\
&= \left[\frac{6\sqrt{a}x^{3/2}}{3/2} - \frac{3}{4a} \frac{x^3}{3} \right]_0^{4a} \\
&= \frac{12\sqrt{a}(4a)^{3/2}}{3} - \frac{(4a)^3}{4a} \\
&= \frac{12 \times 4 \times \sqrt{4} \times a^2}{3} - 16a^2 \\
&= 32a^2 - 16a^2 \\
&= 16a^2 \text{ cubic units.}
\end{aligned}$$

Example 4: Find the volume of the tetrahedron bounded by the plane $x = 0, y = 0, z = 0$

and $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$.

(Nov- 2005)

Solution:

$$\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$$

$$\frac{z}{c} = 1 - \frac{x}{a} - \frac{y}{b}$$

$$z = c \left(1 - \frac{x}{a} - \frac{y}{b} \right)$$

z varies from 0 to $c\left(1 - \frac{x}{a} - \frac{y}{b}\right)$

y varies from 0 to $b\left(1 - \frac{x}{a}\right)$

x varies from 0 to a .

$$\begin{aligned}\text{Volume} &= \iiint_R dz \, dy \, dx \\&= \int_0^a \int_0^{b\left(1 - \frac{x}{a}\right)} \int_0^{c\left(1 - \frac{x}{a} - \frac{y}{b}\right)} dz \, dy \, dx \\&= \int_0^a \int_0^{b\left(1 - \frac{x}{a}\right)} [z]_0^{c\left(1 - \frac{x}{a} - \frac{y}{b}\right)} dy \, dx \\&= \int_0^a \int_0^{b\left(1 - \frac{x}{a}\right)} c\left(1 - \frac{x}{a} - \frac{y}{b}\right) dy \, dx \\&= \int_0^a c \left(y - \frac{xy}{a} - \frac{y^2}{2b} \right) \Big|_0^{b\left(1 - \frac{x}{a}\right)} dy \, dx \\&= \int_0^a c \left(\frac{b(a-x)}{a} - \frac{x}{a} \frac{b(a-x)}{a} - \frac{b^2(a-x)^2}{2a^2b} \right) dx \\&= c \int_0^a \left(\frac{ab-bx}{a} - \frac{(abx-x^2b)}{a^2} - \frac{b^2(a^2+x^2-2ax)}{2a^2b} \right) dx \\&= c \int_0^a \left(b - \frac{bx}{a} - \frac{bx}{a} + \frac{x^2b}{a^2} - \frac{b}{2} - \frac{x^2b}{2a^2} + \frac{2xb}{2a} \right) dx\end{aligned}$$

$$\begin{aligned}
&= c \int_0^a \left(\frac{b}{2} - \frac{bx}{a} + \frac{x^2 b}{2a^2} \right) dx \\
&= c \left(\frac{bx}{2} - \frac{bx^2}{2a} + \frac{x^3 b}{6a^2} \right)_0^a \\
&= c \left(\frac{ab}{2} - \frac{a^2 b}{2a} + \frac{a^3 b}{6a^2} \right) \\
&= c \left(\frac{ab}{2} - \frac{ab}{2} + \frac{ab}{6} \right) \\
&= \frac{abc}{6} \text{ cubic units.}
\end{aligned}$$

Example 5: Find the volume bounded by the cylinder $x^2 + y^2 = 4$ and the planes $y + z = 4$ and $z = 0$.

Solution:

$$y + z = 4$$

$$z = 4 - y$$

z varies from 0 to $4 - y$

$$x^2 + y^2 = 4$$

$$y^2 = 4 - x^2$$

$$y = \pm \sqrt{4 - x^2}$$

y varies from $-\sqrt{4 - x^2}$ to $\sqrt{4 - x^2}$

x varies from -2 to 2.

$$\begin{aligned}
\text{Volume} &= \int_{-2}^2 \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} \int_0^{4-y} dz \, dy \, dx \\
&= \int_{-2}^2 \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} [z]_0^{4-y} \, dy \, dx \\
&= \int_{-2}^2 \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} [4-y] \, dy \, dx \\
&= \int_{-2}^2 \left[4y - \frac{y^2}{2} \right]_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} dx \\
&= \int_{-2}^2 \left[4\sqrt{4-x^2} - \frac{(4-x^2)}{2} \right] - \left[-4\sqrt{4-x^2} - \frac{(4-x^2)}{2} \right] dx \\
&= \int_{-2}^2 8\sqrt{(4-x^2)} \, dx \\
&= 16 \int_0^2 \sqrt{(4-x^2)} \, dx \\
&= 16 \left[\frac{x}{2} \sqrt{4-x^2} + \frac{4}{2} \sin^{-1} \left(\frac{x}{2} \right) \right]_0^2 \\
&= 16 \left[2 \frac{\pi}{2} \right] = 16\pi \text{ cubic units.}
\end{aligned}$$

Exercise:

1. Evaluate $\iiint_V dx dy dz$ where V is the volume of the tetrahedron whose vertices are $(0,0,0)$, $(0,1,0)$ and $(0,0,1)$.
2. Find the integral of x over the first octant portion of volume which is common to the cylinders $y^2 + z^2 = 12$ and the parabolic $z = x^2 + y^2$.
3. Evaluate $\iiint_V xyz dx dy dz$, where V is the region of the space inside the tetrahedron bounded by the planes $x=0, y=0, z=0$ and $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$.
4. Evaluate $\iiint xyz dx dy dz$ over the positive octant of the sphere $x^2 + y^2 + z^2 = a^2$.
5. Find the volume of the region bounded by the surface $y = x^2, x = y^2$ and the planes $z = 0, z = 3$.
6. Find the volume of the solid bounded by the surface $x=0, y=0, x+y+z=1$ and $z=0$.
7. Find the volume common to the cylinder $x^2 + y^2 = a^2$ and $x^2 + z^2 = a^2$.
8. Evaluate $\iiint xyz dx dy dz$ taken throughout the volume of the tetrahedron bounded by the planes $x=0, y=0, z=0$ and $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$.
9. Find the volume of the solid enclosed by $x^2 + y^2 + z^2 = a^2$ and $x^2 + y^2 = b^2$ (a, b)

Answers:

$$\begin{array}{llll}
1. \frac{1}{6} & 2. \frac{3}{2}\pi & 3. \frac{1}{720}a^2b^2c^2 & 4. \frac{a^6}{48} \quad 6. \frac{1}{6} \\
7. \frac{16a^3}{3} & 8. \frac{17a^2b^2c^2}{504} & 9. \frac{4\pi}{3} \left[a^3 - (a^2 - b^2)^{3/2} \right] &
\end{array}$$

3.9. Triple integrals in cylindrical and spherical co ordinates

The relation between the Cartesian and cylindrical and rectangular co ordinates are given by the relations $x = r \cos \theta$

$$y = r \sin \theta$$

$$z = z$$

$$\text{Then } \iiint_R f(x, y, z) dv = \iiint f(r \cos \theta, r \sin \theta, z) r dz dr d\theta$$

The relation between the Cartesian and spherical polar co-ordinates of a point are given by the relations

$$x = r \sin \theta \cos \phi$$

$$y = r \sin \theta \sin \phi$$

$$z = r \cos \theta$$

$$dx dy dz = r^2 \sin \theta dr d\theta d\phi.$$

Example 1: Evaluate $\iiint_V xyz dx dy dz$ over the positive octant of the sphere

$$x^2 + y^2 + z^2 = a^2.$$

Solution:

The relation between Cartesian and cylindrical co-ordinates are

$$x = r \cos \theta, y = r \sin \theta, z = z.$$

Then $dx dy dz = r dr d\theta dz$.

Now $x^2 + y^2 + z^2 = a^2$

$$r^2 \cos^2 \theta + r^2 \sin^2 \theta + z^2 = a^2$$

$$r^2 + z^2 = a^2$$

$$z^2 = a^2 - r^2$$

$$z = \sqrt{a^2 - r^2}$$

$\therefore z$ varies from 0 to $\sqrt{a^2 - r^2}$

r varies from 0 to a

θ varies from 0 to $\frac{\pi}{2}$

$$\begin{aligned} \iiint_V xyz \, dx \, dy \, dz &= \int_0^{\frac{\pi}{2}} \int_0^a \int_0^{\sqrt{a^2 - r^2}} ((r \cos \theta)(r \sin \theta)z) r \, dz \, dr \, d\theta \\ &= \int_0^{\frac{\pi}{2}} \int_0^a \left[r^3 \cos \theta \sin \theta \frac{z^2}{2} \right]_0^{\sqrt{a^2 - r^2}} dr \, d\theta \\ &= \int_0^{\frac{\pi}{2}} \int_0^a \left(r^3 \cos \theta \sin \theta \left(\frac{a^2 - r^2}{2} \right) \right) dr \, d\theta \\ &= \int_0^{\frac{\pi}{2}} \int_0^a \frac{\cos \theta \sin \theta}{2} (a^2 r^3 - r^5) dr \, d\theta \\ &= \int_0^{\frac{\pi}{2}} \frac{\cos \theta \sin \theta}{2} \left[\frac{a^2 r^4}{4} - \frac{r^6}{6} \right]_0^a d\theta \\ &= \int_0^{\frac{\pi}{2}} \frac{\cos \theta \sin \theta}{2} \left[\frac{a^6}{4} - \frac{a^6}{6} \right] d\theta \end{aligned}$$

$$\begin{aligned}
&= \int_0^{\frac{\pi}{2}} \frac{\sin 2\theta}{4} \left(\frac{2a^6}{24} \right) d\theta \\
&= \frac{a^6}{48} \left[-\frac{\cos 2\theta}{2} \right]_0^{\frac{\pi}{2}} \\
&= \frac{a^6}{48} \left[-\frac{\cos \pi}{2} + \frac{\cos 0}{2} \right] \\
&= \frac{a^6}{48} \left[\frac{1}{2} + \frac{1}{2} \right] \\
&= \frac{a^6}{48}.
\end{aligned}$$

Example 2: Evaluate the integral $\iiint (x^2 + y^2 + z^2) dx dy dz$ taken over the region of space defined by $x^2 + y^2 \leq 1$ and $0 \leq z \leq 1$.

Solution:

$$x^2 + y^2 = 1$$

$$r^2 \cos^2 \theta + r^2 \sin^2 \theta = 1 \quad (x = r \cos \theta, y = r \sin \theta)$$

$$r^2 = 1$$

$$r = \pm 1$$

$$x^2 + y^2 + z^2 = r^2 \cos^2 \theta + r^2 \sin^2 \theta + z^2$$

$$= r^2 + z^2$$

$$dx dy dz = r dr d\theta dz$$

$\therefore r$ varies from 0 to 1

θ varies from 0 to 2π

z varies from 0 to 1.

$$\begin{aligned}\iiint_V (x^2 + y^2 + z^2) dx dy dz &= \int_0^1 \int_0^{2\pi} \int_0^1 (r^2 + z^2) r dr d\theta dz \\&= \int_0^1 \int_0^{2\pi} \left(\frac{r^4}{4} + z^2 \frac{r^2}{2} \right) d\theta dz \\&= \int_0^1 \int_0^{2\pi} \left(\frac{1}{4} + \frac{z^2}{2} \right) d\theta dz = \int_0^1 \left(\frac{\theta}{4} + \frac{z^2}{2} \theta \right) d\theta dz \\&= \int_0^1 \left(\frac{2\pi}{4} + \frac{2\pi z^2}{2} \right) dz = \left[\frac{\pi z}{2} + \frac{\pi z^3}{3} \right]_0^1 \\&= \frac{\pi}{2} + \frac{\pi}{3} \\&= \frac{5\pi}{6}.\end{aligned}$$

Example 3: Evaluate $\int_0^1 \int_0^{\sqrt{1-x^2}} \int_{\sqrt{x^2+y^2}}^1 \frac{dz dy dx}{\sqrt{x^2 + y^2 + z^2}}.$

Solution:

Let us transform this integral in spherical polar co-ordinates by taking,

$$x = r \sin \theta \cos \phi, y = r \sin \theta \sin \phi, z = r \cos \theta.$$

$$dx dy dz = r^2 \sin \theta dr d\theta d\phi$$

The region of integration is common to the cone $z^2 = x^2 + y^2$ and the cylinder $x^2 + y^2 = 1$ bounded by the plane $z = 1$ in the positive octant.

$\therefore \theta$ varies from 0 to $\frac{\pi}{4}$

r varies from 0 to $\sec \theta$

ϕ varies from 0 to $\frac{\pi}{2}$

$$\begin{aligned}\text{Volume} &= \int_0^{\frac{\pi}{2}} \int_0^{\frac{\pi}{4}} \int_0^{\sec \theta} \frac{1}{r} r^2 \sin \theta \, dr \, d\theta \, d\phi \\&= \int_0^{\frac{\pi}{2}} \int_0^{\frac{\pi}{4}} \int_0^{\sec \theta} r \sin \theta \, dr \, d\theta \, d\phi \\&= \int_0^{\frac{\pi}{2}} \int_0^{\frac{\pi}{4}} \left(\frac{r^2}{2} \sin \theta \right)_0^{\sec \theta} d\theta \, d\phi \\&= \int_0^{\frac{\pi}{2}} \int_0^{\frac{\pi}{4}} \left(\frac{\sec^2 \theta}{2} \sin \theta \right) d\theta \, d\phi \\&= \frac{1}{2} \int_0^{\frac{\pi}{2}} \int_0^{\frac{\pi}{4}} \sec \theta \tan \theta \, d\theta \, d\phi \\&= \frac{1}{2} \int_0^{\frac{\pi}{2}} (\sec \theta)_0^{\frac{\pi}{4}} d\phi \\&= \frac{1}{2} \int_0^{\frac{\pi}{2}} (\sqrt{2} - 1) d\phi \\&= \frac{1}{2} \left[\sqrt{2} \phi - \phi \right]_0^{\frac{\pi}{2}} \\&= \frac{1}{2} \left[\sqrt{2} \frac{\pi}{2} - \frac{\pi}{2} \right]\end{aligned}$$

$$= \frac{\pi}{4} [\sqrt{2} - 1].$$

Example 4: Evaluate the integral $\iiint (x^2 + y^2 + z^2) dx dy dz$ taken over the volume enclosed by the sphere $x^2 + y^2 + z^2 = 1$.

Solution:

Let us convert the given integral into spherical co-ordinates,

$$\text{Let } x = r \sin \theta \cos \phi$$

$$y = r \sin \theta \sin \phi$$

$$z = r \cos \theta$$

$$\begin{aligned} x^2 + y^2 + z^2 &= r^2 \sin^2 \theta \cos^2 \phi + r^2 \sin^2 \theta \sin^2 \phi + r^2 \cos^2 \theta \\ &= r^2. \end{aligned}$$

$$dx dy dz = r^2 \sin^2 \theta dr d\theta d\phi$$

$$\iiint (x^2 + y^2 + z^2) dx dy dz = \int_0^\pi \int_0^{2\pi} \int_0^1 r^2 (r^2 \sin \theta d\theta d\phi dr)$$

$$= \int_0^\pi \int_0^{2\pi} \int_0^1 r^3 \sin \theta d\theta d\phi dr$$

$$= \int_0^{2\pi} d\phi \int_0^\pi \sin \theta d\theta \int_0^1 r^4 dr$$

$$= [\phi]_0^{2\pi} [-\cos \theta]_0^\pi \left[\frac{r^5}{5} \right]_0^1$$

$$= (2\pi)(1+1) \left(\frac{1}{5} \right)$$

$$= \frac{4\pi}{5}.$$

Example 5: Evaluate $\iiint_R \frac{dx dy dz}{x^2 + y^2 + z^2}$ throughout the volume of the sphere

$$x^2 + y^2 + z^2 = a^2.$$

(May- 2007)

Solution:

$$\text{Let } x = r \sin \theta \cos \phi$$

$$y = r \sin \theta \sin \phi$$

$$z = r \cos \theta.$$

$$dx dy dz = r^2 \sin \theta dr d\theta d\phi$$

$$\begin{aligned} x^2 + y^2 + z^2 &= r^2 \sin^2 \theta \cos^2 \phi + r^2 \sin^2 \theta \sin^2 \phi + r^2 \cos^2 \theta \\ &= r^2. \end{aligned}$$

$\therefore r$ varies from 0 to a .

ϕ varies from 0 to 2π .

θ varies from 0 to π .

$$\begin{aligned} \iiint_V \frac{dx dy dz}{x^2 + y^2 + z^2} &= \int_0^{2\pi} \int_0^\pi \int_0^a \frac{r^2 \sin \theta dr d\theta d\phi}{r^2} \\ &= \int_0^a dr \int_0^\pi \sin \theta d\theta \int_0^{2\pi} d\phi \\ &= [r]_0^a [-\cos \theta]_0^\pi [\phi]_0^{2\pi} \\ &= a(1+1)(2\pi) \\ &= 4\pi a. \end{aligned}$$

Example 6: Evaluate $\iiint dx dy dz$ over the volume cut off the sphere $x^2 + y^2 + z^2 = a^2$ by the cone $x^2 + y^2 = z^2$.

Solution.

$$\text{Let } x = r \sin \theta \cos \phi$$

$$y = r \sin \theta \sin \phi$$

$$z = r \cos \theta$$

$$dx dy dz = r^2 \sin^2 \theta dr d\theta d\phi$$

$$x^2 + y^2 + z^2 = a^2$$

$$r^2 = a^2$$

$$r = \pm a$$

$$r = a$$

$$x^2 + y^2 = z^2$$

$$r^2 \sin^2 \theta \cos^2 \phi + r^2 \sin^2 \theta \sin^2 \phi = r^2 \cos^2 \theta$$

$$r^2 \sin^2 \theta = r^2 \cos^2 \theta$$

$$\tan^2 \theta = 1$$

$$\tan \theta = 1$$

$$\theta = \tan^{-1}(1)$$

$$\theta = \frac{\pi}{4}$$

$\therefore r$ varies from 0 to a .

θ varies from 0 to $\frac{\pi}{4}$.

ϕ varies from 0 to 2π .

$$\begin{aligned}
\iiint dx dy dz &= 2 \int_0^{2\pi} \int_0^{\frac{\pi}{4}} \int_0^a r^2 \sin^2 \theta dr d\theta d\phi \\
&= 2 \int_0^{2\pi} d\phi \int_0^{\frac{\pi}{4}} \sin \theta d\theta \int_0^a r^2 dr \\
&= 2 \left[(\phi)_0^{2\pi} (-\cos \theta)_0^{\frac{\pi}{4}} \left(\frac{r^3}{3} \right)_0^a \right] \\
&= 2 \left[(2\pi) \left(-\cos \frac{\pi}{4} - \cos 0 \right) \left(\frac{a^3}{3} \right) \right] \\
&= 2 \left[(2\pi) \left(-\frac{1}{\sqrt{2}} + 1 \right) \left(\frac{a^3}{3} \right) \right] \\
&= 2 \left[2\pi \times \frac{(\sqrt{2}-1)}{\sqrt{2}} \times \frac{a^3}{3} \right] \\
&= (\sqrt{2}-1) \frac{4\pi a^3}{3}.
\end{aligned}$$

Example 7: Find the volume of the solid surrounded by the surface

$$\left(\frac{x}{a} \right)^{\frac{2}{3}} + \left(\frac{y}{b} \right)^{\frac{2}{3}} + \left(\frac{z}{c} \right)^{\frac{2}{3}} = 1.$$

Solution:

$$\text{Let } \left(\frac{x}{a} \right)^{\frac{1}{3}} = x, \left(\frac{y}{b} \right)^{\frac{1}{3}} = y, \left(\frac{z}{c} \right)^{\frac{1}{3}} = z.$$

$$x = ax^3, y = by^3, z = cz^3$$

$$\text{Given } \left(\frac{x}{a}\right)^{2/3} + \left(\frac{y}{b}\right)^{2/3} + \left(\frac{z}{c}\right)^{2/3} = 1$$

$$x^2 + y^2 + z^2 = 1$$

Change x, y, z into spherical polar co ordinates

$$x = r \sin \theta \cos \phi$$

$$y = r \sin \theta \sin \phi$$

$$z = r \cos \theta$$

$$dx dy dz = r^2 \sin \theta dr d\theta d\phi$$

r varies from 0 to 1.

θ varies from 0 to $\frac{\pi}{2}$.

ϕ varies from 0 to $\frac{\pi}{2}$.

$$\begin{aligned} \text{Required Volume} &= \iiint dx dy dz \\ &= 27abc \iiint x^2 y^2 z^2 dx dy dz \\ &= 27abc \int_0^1 r^8 dr \int_0^{\frac{\pi}{2}} \sin^5 \theta \cos^2 \theta d\theta \int_0^{\frac{\pi}{2}} \sin^2 \phi \cos^2 \phi d\phi \\ &= 27abc \left(\frac{r^9}{9}\right)_0^1 \int_0^{\frac{\pi}{2}} \sin^5 \theta (1 - \sin^2 \theta) d\theta \int_0^{\frac{\pi}{2}} \sin^2 \phi (1 - \sin^2 \phi) d\phi \\ &= 27abc \times \frac{1}{9} \int_0^{\frac{\pi}{2}} (\sin^5 \theta - \sin^7 \theta) d\theta \int_0^{\frac{\pi}{2}} (\sin^2 \phi - \sin^4 \phi) d\phi \end{aligned}$$

$$\begin{aligned}
&= 27abc \times \frac{1}{9} \times \left[\left(\frac{5-1}{5} \cdot \frac{5-3}{5-2} \right) - \left(\frac{7-1}{7} \cdot \frac{7-3}{7-2} \cdot \frac{7-5}{7-4} \right) \right] \left[\frac{1}{2} \frac{\pi}{2} - \frac{4-1}{4} \cdot \frac{4-3}{4-2} \cdot \frac{\pi}{2} \right] \\
&= 3abc \times \left[\frac{4}{5} \times \frac{2}{3} - \frac{6}{7} \times \frac{4}{5} \times \frac{2}{3} \right] \left[\frac{\pi}{4} - \frac{3}{4} \times \frac{1}{2} \times \frac{\pi}{2} \right] \\
&= 3abc \times \left[\frac{8}{15} - \frac{16}{35} \right] \times \left[\frac{\pi}{4} - \frac{3\pi}{8} \right] \\
&= 3abc \times \frac{40}{35} \times \frac{\pi}{8} \\
&= \frac{3\pi abc}{7}.
\end{aligned}$$

Exercise

1. Find the volume cut off from the sphere $x^2 + y^2 + z^2 = a^2$ by the cylinder $x^2 + y^2 = ax$ by transforming into cylindrical co-ordinates.
2. Find the volume of the solid bounded by the spherical surface $x^2 + y^2 + z^2 = 4a^2$ and the cylinder $x^2 + y^2 - 2ay = 0$ by transforming into cylindrical co-ordinates.
3. By transforming into cylindrical co ordinates, evaluate the integral $\iiint (x^2 + y^2 + z^2) dx dy dz$ taken over the region of space defined by $x^2 + y^2 \leq 1$ and $0 \leq z \leq 1$.
4. Evaluate $\int_0^1 \int_0^{\sqrt{1-x^2}} \int_{\sqrt{x^2+y^2}}^1 \frac{dx dy dz}{\sqrt{x^2 + y^2 + z^2}}$ by transforming into spherical co ordinates.
5. Evaluate $\iiint z^2 dx dy dz$ through the volume of the sphere $x^2 + y^2 + z^2 = 1$ by changing into spherical polar co ordinates.

6. Evaluate $\iiint_D xyz \left(\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} \right) dx dy dz$ where D is the positive octant of the ellipsoid

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1.$$

Answers.

$$1. \frac{2a^3}{9}(3\pi - 4) \quad 2. \frac{32a^3}{3} \left(\frac{\pi}{2} - \frac{2}{3} \right) \quad 3. \frac{5\pi}{6} \quad 4. \frac{\pi}{4}(\sqrt{2} - 1) \quad 5. \frac{5\pi}{4}.$$

3.10. Mass and centre of Mass as double integrals

If $f(x, y)$ is the density (mass) per unit over a distribution of mass in the xy plane

$\iint_R f(x, y) dx dy$, represents the mass of the region R in the xy plane.

The centre of gravity of the mass “M” at the region R at the xy plane with coordinates

$$(\bar{x}, \bar{y}) \text{ is defined by } \bar{x} = \frac{\iint_R x f(x, y) dx dy}{\iint_R f(x, y) dx dy}$$

$$\bar{y} = \frac{\iint_R y f(x, y) dx dy}{\iint_R f(x, y) dx dy}$$

Moment of Inertia: The moment of inertia of a mass M in the region R in the (x, y) plane

about x and y axes are given by $I_x = \iint_R y^2 f(x, y) dx dy$

$$I_y = \iint_R x^2 f(x, y) dx dy$$

The moment of inertia of the lamina about the axis perpendicular to the xy plane is

$$\iint_R f(x, y)(x^2 + y^2) dx dy .$$

Example 1: Find the centre of gravity of the plane semi circle region $x^2 + y^2 = 4, x \geq 0$.

Solution:

By symmetry centre of gravity lies on the x -axis $\bar{y} = 0$.

$$\begin{aligned} \bar{x} &= \frac{\iint_R x dx dy}{\iint_R dx dy} \\ &= \frac{\int_{-2}^2 \int_0^{\sqrt{4-y^2}} x dx dy}{\int_{-2}^2 \int_0^{\sqrt{4-y^2}} dx dy} \\ &= \frac{\int_{-2}^2 \left[\frac{x^2}{2} \right]_0^{\sqrt{4-y^2}} dy}{\int_{-2}^2 \left[x \right]_0^{\sqrt{4-y^2}} dy} = \frac{\int_{-2}^2 \frac{4-y^2}{2} dy}{\int_{-2}^2 \sqrt{4-y^2} dy} \\ &= \frac{\frac{1}{2} \int_0^2 (4-y^2) dy}{\int_0^2 \sqrt{4-y^2} dy} \\ &= \frac{1}{2} \frac{\left[4y - \frac{y^3}{3} \right]_0^2}{\left[\frac{y}{2} \sqrt{4-y^2} + \frac{4}{2} \sin^{-1} \left(\frac{y}{2} \right) \right]_0^2} \end{aligned}$$

$$\begin{aligned}
 &= \frac{1}{2} \frac{\left(8 - \frac{8}{3}\right)}{\frac{\pi}{2}} \\
 &= \frac{1}{2} \frac{16}{3} \times \frac{1}{\pi} \\
 &= \frac{8}{3\pi}.
 \end{aligned}$$

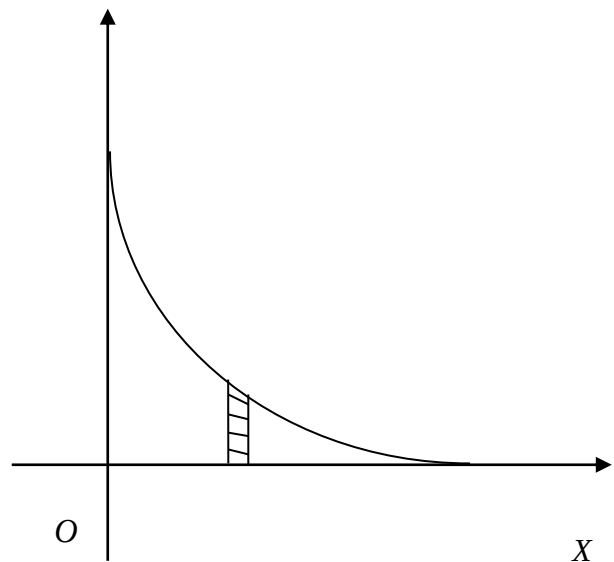
Centre of gravity is the point $\left(\frac{8}{3\pi}, 0\right)$.

Example 2: Using double integration find the centre of gravity of the lamina in the first quadrant of the curve $\left(\frac{x}{a}\right)^{2/3} + \left(\frac{y}{b}\right)^{2/3} = 1$ where the density at any point (x, y) is given by kxy .

Solution:

Let $(\bar{x}, \bar{y})$ be the co ordinates of the area in the first quadrant.

$$\begin{aligned}
 \bar{x} &= \frac{\iint_R f(x, y) x \, dx \, dy}{\iint_R f(x, y) \, dx \, dy} = \frac{\iint_R kxy x \, dy \, dx}{\iint_R kxy \, dy \, dx} \\
 &= \frac{\iint_R x^2 y \, dy \, dx}{\iint_R xy \, dy \, dx}
 \end{aligned}$$



$$= \frac{\int_0^a \int_0^y x^2 y \, dy \, dx}{\int_0^a \int_0^y xy \, dy \, dx}$$

$Q(x, y)$

Fig.3.37

$$= \frac{\int_0^a \left[\frac{x^2 y^2}{2} \right]_0^y dx}{\int_0^a \left[\frac{xy^2}{2} \right]_0^y dx} = \frac{\int_0^a x^2 y^2 \, dx}{\int_0^a xy^2 \, dx}$$

Any point on the curve $x = a \cos^3 \theta, y = b \sin^3 \theta$.

$$dx = 3a \cos^2 \theta \sin \theta \, d\theta.$$

When $x = 0, \theta = \frac{\pi}{2}$

$$x = a, \theta = 0.$$

$$\begin{aligned} \bar{x} &= \frac{\int_0^{\frac{\pi}{2}} a^2 \cos^6 \theta b^2 \sin^6 \theta \, 3a \cos^2 \theta \sin \theta \, d\theta}{\int_0^{\frac{\pi}{2}} a \cos^3 \theta b \sin^6 \theta \, 3a \cos^2 \theta \sin \theta \, d\theta} \\ &= \frac{a \int_0^{\frac{\pi}{2}} \cos^8 \theta \sin^7 \theta \, d\theta}{\int_0^{\frac{\pi}{2}} \cos^5 \theta \sin^7 \theta \, d\theta} \\ &= \frac{a128}{429} \end{aligned}$$

$$\bar{x} = \frac{128}{429}a.$$

By symmetry $\bar{y} = \frac{128}{429}a$

$\therefore$ centre of gravity at lamina is $\left(\frac{128}{429}a, \frac{128}{429}b\right)$.

Example 3: Find the mass of a plate in the shape at the curve $\left(\frac{x}{a}\right)^{2/3} + \left(\frac{y}{b}\right)^{2/3} = 1$ the density being given by $\rho = \mu xy$. (May 2008)

Solution:

Given $\rho = \mu xy$

Elementary area = $dx dy$

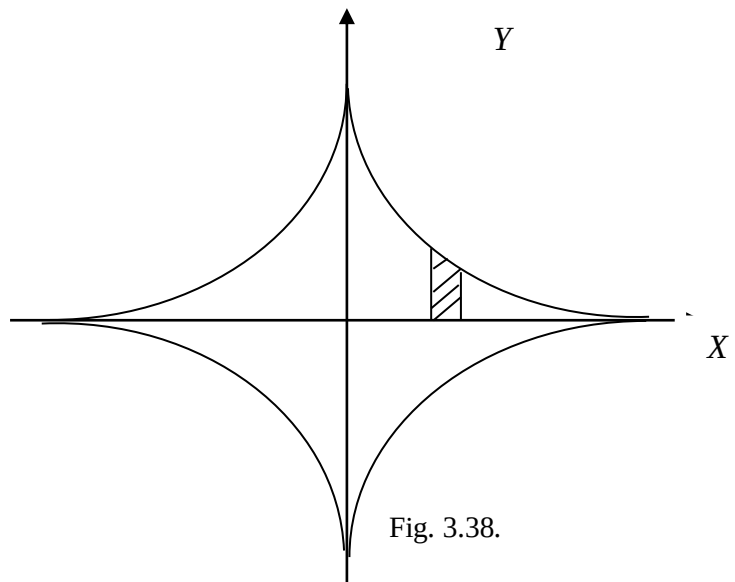
Elementary mass = $\mu xy dx dy$

Total mass I = $4 \iint_R \mu xy dx dy$

$$\left(\frac{x}{a}\right)^{2/3} + \left(\frac{y}{b}\right)^{2/3} = 1$$

$$y = b \left[1 - \left(\frac{x}{a}\right)^{2/3} \right]^{3/2}$$

$$y \text{ varies from } 0 \text{ to } b \left[1 - \left(\frac{x}{a}\right)^{2/3} \right]^{3/2}$$



$$I = 4 \int_0^a \int_0^{b \left[1 - \left(\frac{x}{a} \right)^{2/3} \right]^{3/2}} \mu xy \, dy \, dx$$

$$= 4 \int_0^a \int_0^{y_1} \mu xy \, dy \, dx$$

$$= 4 \int_0^a \left[\frac{\mu xy^2}{2} \right]_0^{y_1} dx$$

$$= \frac{4\mu}{2} \int_0^a (xy_1^2) dx$$

$$= 2\mu \int_0^a xb \left[1 - \left(\frac{x}{a} \right)^{2/3} \right]^{3/2} dx$$

$$\text{where } y_1 = b \left[1 - \left(\frac{x}{a} \right)^{2/3} \right]^{3/2}$$

$$\text{Put } x = a \sin^3 \theta \text{ then } dx = 3a \sin^2 \theta \cos \theta d\theta$$

$$\text{When } x = 0, \theta = 0$$

$$x = a, \theta = \frac{\pi}{2}$$

$$I = 2\mu b^2 \int_0^{\frac{\pi}{2}} a \sin^3 \theta (1 - \sin^2 \theta)^3 3a \sin^2 \theta \cos \theta d\theta$$

$$= 6\mu a^2 b^2 \int_0^{\frac{\pi}{2}} \sin^5 \theta \cos^7 \theta d\theta$$

$$= 6\mu a^2 b^2 \frac{4 \cdot 2 \cdot 6 \cdot 4 \cdot 2}{12 \cdot 10 \cdot 8 \cdot 6 \cdot 4 \cdot 2}$$

$$= \frac{\mu a^2 b^2}{20} .$$

Example 4: Find the mass of a circular plate where the density at any point varies as the square of its distance from a fixed diameter.

(April 2003, Nov 2003)

Solution:

Density at the point $\rho = ky^2$ taking the fixed diameter as the x-axis and k constant.

Elementary area at a point $(x, y) = dx dy$

Hence elementary mass $= ky^2 dx dy$

Total mass $= 4 \iint_R ky^2 dx dy$

The limit for y varies from $y = 0$ to $y = \sqrt{a^2 - x^2}$

x varies from 0 to a .

$$\text{Mass} = 4 \int_0^a \int_0^{\sqrt{a^2 - x^2}} ky^2 dx dy$$

$$= 4 \int_0^a \left[\frac{y^3}{3} \right]_0^{\sqrt{a^2 - x^2}} dx$$

$$= \frac{4k}{3} \int_0^a (a^2 - x^2) \sqrt{a^2 - x^2} dx$$

Put $x = a \sin \theta$

$$dx = a \cos \theta d\theta$$

When $x = 0, \theta = 0$

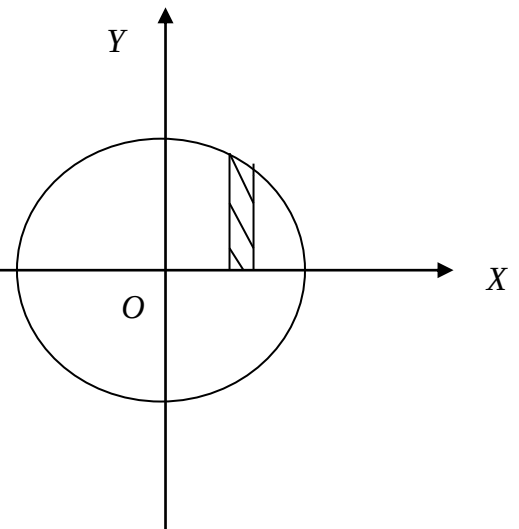


Fig. 3.39.

$$x = a, \theta = \frac{\pi}{2}$$

$$\begin{aligned} \text{Mass} &= \frac{4k}{3} \int_0^{\frac{\pi}{2}} (a^2 - a^2 \sin^2 \theta) \sqrt{a^2 - a^2 \sin^2 \theta} a \cos \theta d\theta \\ &= \frac{4k}{3} \int_0^{\frac{\pi}{2}} a^2 \cos^2 \theta a^2 \cos^2 \theta d\theta \\ &= \frac{4k}{3} a^4 \int_0^{\frac{\pi}{2}} \cos^4 \theta d\theta \\ &= \frac{4ka^4}{3} \cdot \frac{3}{4} \cdot \frac{1}{2} \cdot \frac{\pi}{2} \\ &= \frac{\pi ka^4}{4} \end{aligned}$$

Example 5: Find the mass of the tetrahedron bounded by the co ordinate planes

and the plane $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$ the variable density $\ell = \mu xyz$.

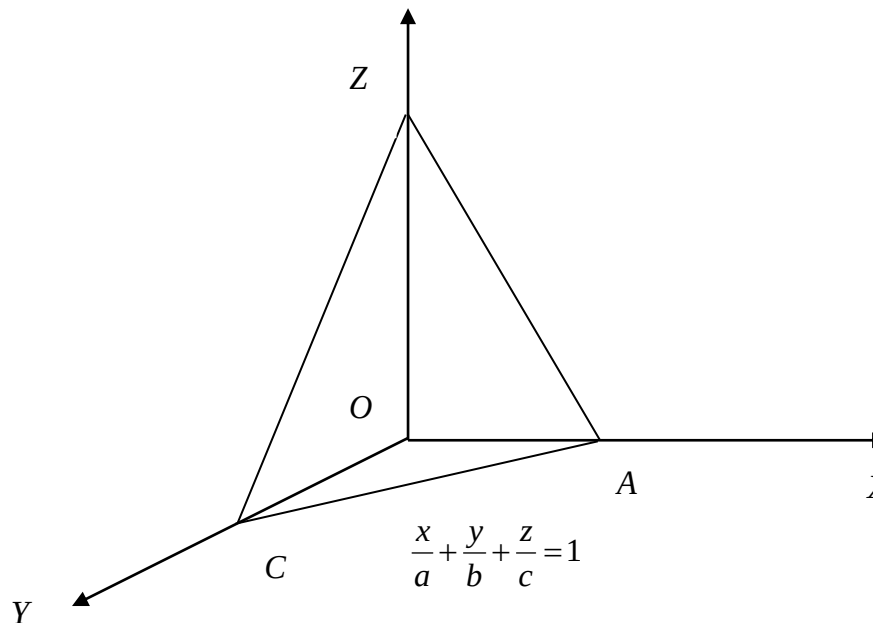
Solution:

The tetrahedron $OABC$ is bounded by the planes $x=0, y=0, z=0$ and $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$.

Variable density $\ell = \mu xyz$

x varies from 0 to $a \left[1 - \frac{y}{b} - \frac{z}{c} \right]$

y varies from 0 to $b \left[1 - \frac{z}{c} \right]$



z varies from 0 to c .

$$\begin{aligned}\text{Required Mass} &= \int_0^c \int_0^{b\left[1-\frac{z}{c}\right]} \int_0^{a\left[1-\frac{y}{b}-\frac{z}{c}\right]} \rho \, dx \, dy \, dz \\ &= \int_0^c \int_0^{b\left[1-\frac{z}{c}\right]} \int_0^{a\left[1-\frac{y}{b}-\frac{z}{c}\right]} \mu xyz \, dx \, dy \, dz\end{aligned}$$

Fig. 3.40.

$$\begin{aligned}&= \int_0^c \int_0^{b\left[1-\frac{z}{c}\right]} \mu xyz \left[\frac{x^2}{2} \right]_0^{a\left[1-\frac{y}{b}-\frac{z}{c}\right]} dy \, dz \\ &= \int_0^c \int_0^{b\left[1-\frac{z}{c}\right]} \frac{\mu xyz a^2}{2} \left[1 - \frac{y}{b} - \frac{z}{c} \right]^2 dy \, dz \\ &= \int_0^c \int_0^{b\left[1-\frac{z}{c}\right]} \frac{\mu yz}{2} a^2 \left[1 - \frac{y}{b} - \frac{z}{c} \right]^2 dy \, dz \\ &= \frac{\mu a^2}{2} \int_0^c \int_0^{b\left[1-\frac{z}{c}\right]} yz \left[1 - \frac{y}{b} - \frac{z}{c} \right]^2 dy \, dz \\ &= \frac{\mu a^2}{2} \int_0^c \int_0^{b\left[1-\frac{z}{c}\right]} yz \left[\left(1 - \frac{z}{c} \right)^2 + \frac{y^2}{b^2} - \frac{2y}{b} \left(1 - \frac{z}{c} \right) \right] dy \, dz \\ &= \frac{\mu a^2}{2} \int_0^c \int_0^{b\left[1-\frac{z}{c}\right]} z \left[\left(y - \frac{z}{c} \right)^2 + \frac{y^3}{b^2} - \frac{2y^2}{b} \left(1 - \frac{z}{c} \right) \right] dy \, dz \\ &= \frac{\mu a^2}{2} \int_0^c z \left[\frac{y^2}{2} \left(1 - \frac{z}{c} \right)^2 + \frac{y^4}{4b^2} - \frac{2y^3}{3b} \left(1 - \frac{z}{c} \right) \right]_0^{b\left[1-\frac{z}{c}\right]} dz\end{aligned}$$

$$\begin{aligned}
&= \frac{\mu a^2}{2} \int_0^c z \left[\frac{b^2}{2} \left(1 - \frac{z}{c}\right)^4 + \frac{b^4 \left(1 - \frac{z}{c}\right)^4}{4b^2} - \frac{2b^3 \left(1 - \frac{z}{c}\right)^4}{3b} \right] dz \\
&= \frac{\mu a^2}{2} \int_0^c z \left(1 - \frac{z}{c}\right)^4 \left(\frac{b^2}{2} + \frac{b^2}{4} - \frac{2b^2}{3} \right) dz \\
&= \frac{\mu a^2}{2} \int_0^c z \left(1 - \frac{z}{c}\right)^4 \left(\frac{6b^2 + 3b^2 - 8b^2}{12} \right) dz \\
&= \frac{\mu a^2 b^2}{24} \int_0^c z \left(1 - \frac{z}{c}\right)^4 dz
\end{aligned}$$

Put $z = c \sin^2 \theta$ when $z = 0, \theta = 0$

$$dz = 2c \sin \theta \cos \theta d\theta \quad z = c, \theta = \frac{\pi}{2}$$

$$\begin{aligned}
\text{Mass} &= \frac{\mu a^2 b^2}{24} \int_0^{\frac{\pi}{2}} c \sin^2 \theta (1 - \sin^2 \theta)^4 2c \sin \theta \cos \theta d\theta \\
&= \frac{2\mu a^2 b^2 c^2}{24} \int_0^{\frac{\pi}{2}} \sin^2 \theta \cos^9 \theta d\theta \\
&= \frac{\mu a^2 b^2 c^2}{12} \cdot \frac{1}{11} \cdot \frac{8}{9} \cdot \frac{6}{7} \cdot \frac{4}{5} \cdot \frac{2}{3} = \frac{\mu a^2 b^2 c^2}{720}
\end{aligned}$$

Example 6: Find the position of the centre of mass of the region bounded by $x = 0, y = 0$ and $x + y = 1$ the density being constant.

Solution:

$$\text{Elementary area} = dx dy$$

Density $\ell = k$

Elementary mass = $k \, dx \, dy$

Total mass = $k \iint_R \ell \, dx \, dy$.

x varies from 0 to 1

y varies from 0 to $1-x$.

$$\text{ss} = k \int_0^1 \int_0^{1-x} dy \, dx$$

$$= k \int_0^1 (y)_0^{1-x} dx$$

$$= k \int_0^1 (1-x) dx$$

$$= k \left(x - \frac{x^2}{2} \right)_0^1$$

$$= k \left(1 - \frac{1}{2} \right)$$

$$\text{Mass} = \frac{k}{2}$$

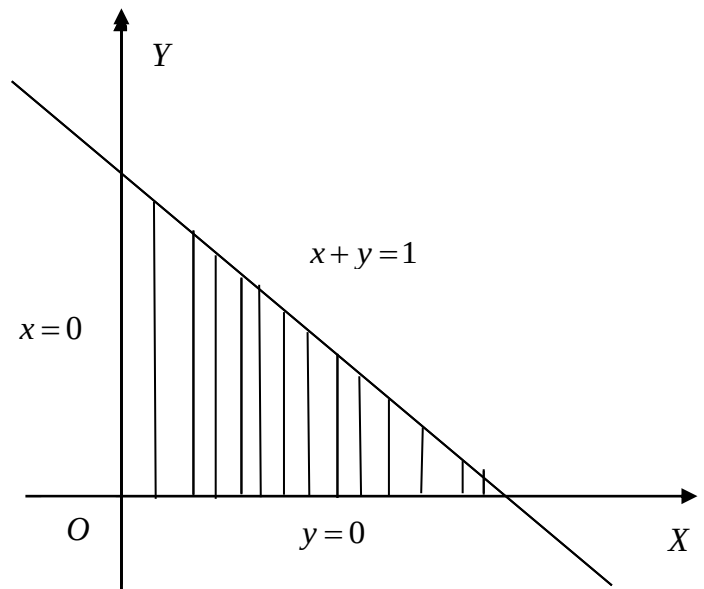


Fig. 3.41.

Example 7: Find the mass of a lamina in the form of the cardioids $r = a(1 + \cos \theta)$

whose density at any point varies as the square of its distance from the initial line.

Solution:

Mass = $2 \times$ Mass above the initial line.

The distance at any point (r, θ) from the initial line is $r \sin \theta$, the density at (r, θ) is given by $\ell = \mu(r \sin \theta)^2 = \mu r^2 \sin^2 \theta$.

θ varies from 0 to π

r varies from 0 to $a(1 + \cos \theta)$.

$$\begin{aligned} \text{Mass} &= 2 \int_0^\pi \int_0^{a(1+\cos \theta)} \mu r^3 \sin^2 \theta \, dr \, d\theta \\ &= 2 \int_0^\pi \mu \sin^2 \theta \left(\frac{r^4}{4} \right)_0^{a(1+\cos \theta)} d\theta \\ &= \frac{2\mu a^4}{4} \int_0^\pi \sin^2 \theta (1 + \cos \theta)^4 d\theta \end{aligned}$$

$$\begin{aligned} &= \frac{4\mu a^4}{2} \int_0^\pi \left(2 \sin \frac{\theta}{2} \cos \frac{\theta}{2} \right)^2 \left(2 \cos^2 \frac{\theta}{2} \right)^4 d\theta \\ &= 32\mu a^4 \int_0^\pi \left(\sin^2 \frac{\theta}{2} \cos^{10} \frac{\theta}{2} \right) d\theta \end{aligned}$$

Put $\frac{\theta}{2} = t$, then

$$d\theta = 2dt$$

When $\theta = 0, t = 0$

$$\theta = \pi, t = \frac{\pi}{2}$$

$$\begin{aligned} \text{Mass} &= 32\mu a^4 \int_0^\pi 2 \sin^2 t + \cos^{10} t \, dt \\ &= 64\mu a^4 \frac{1 \cdot 9 \cdot 7 \cdot 5 \cdot 3 \cdot 1}{12 \cdot 10 \cdot 8 \cdot 6 \cdot 4 \cdot 2} \frac{\pi}{2} \end{aligned}$$

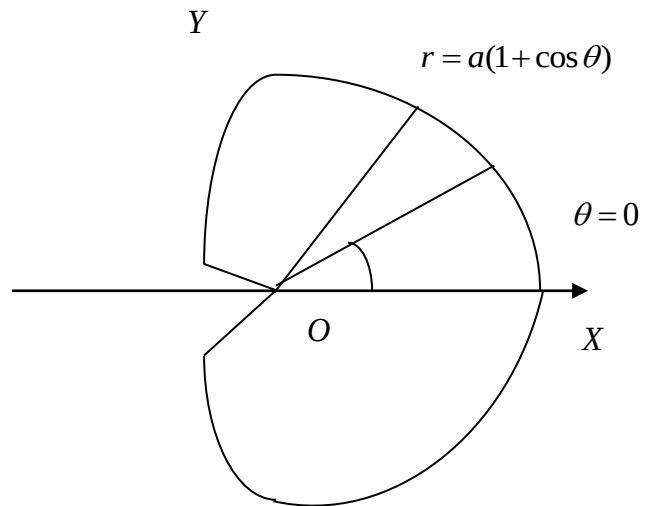


Fig. 3.42.

$$= \frac{21}{32} \mu \pi a^4.$$

Exercise:

1. Find the centre at mass at the third plate in the shape of rectangle $ABCD$ if the density at any point is the product of the distance at the point from two adjacent sides AB and AD .
2. Find the centre of gravity in the first quadrant of the curves $x^{2/3} + y^{2/3} = a^{2/3}$.

Answers:

1. $\left(\frac{2a}{3}, \frac{2b}{3}\right)$
2. $\bar{x} = \bar{y} = \frac{256a}{315y}$

PART A – QUESTIONS AND ANSWERS

1. Define double integral in Cartesian co-ordinates

Solution: The double integral of a function $f(x,y)$ is $\int_{y_1}^{y_2} \int_{x_1}^{x_2} f(x,y) dx dy$, we

first integrate $f(x,y)$ partially with respect to x , keeping y as a constant between the limit x_1 and x_2 and then the resulting expression is to be integrated with respect to y between the limit y_1 and y_2 , x_1 and x_2 will be functions of y_1 and y_2 will be constants.

2. Define double integral in polar co-ordinates.

Solution: The double integral in polar co-ordinates of a function $f(r, \theta)$ is

$$\int_{\theta_1}^{\theta_2} \int_{r_1}^{r_2} f(r, \theta) dr d\theta, \text{ we first integrate } f(r, \theta) \text{ with respect to } r, \text{ keeping } \theta \text{ is}$$

constant, between the limits r_1 and r_2 , and then the resulting expression is to be integrated with respect to θ between the limits θ_1 and θ_2 , r_1 and r_2 will be the functions of θ and θ_1, θ_2 will be constants.

3. Define triple integral.

Solution: The triple integral of a function $f(x, y, z)$ is

$$\int_{z_1}^{z_2} \int_{y_1}^{y_2} \int_{x_1}^{x_2} f(x, y, z) dx dy dz. \text{ we first integrate } f(x, y, z) \text{ with respect to } x,$$

partially, (i.e.) treating y and z as a constant temporarily, between the limits x_1 and x_2 . The resulting function is integrated with respect to y between the limits y_1 and y_2 keeping z as constant. The result thus obtained is finally integrated with respect to z between the limits z_1 and z_2 . In the integral z_1 and z_2 are constants y_1 and y_2 are either constants or functions of z alone and x_1, x_2 are either constants or functions of y, z alone.

(i.e.) $y_1 = f_1(z), y_2 = f_2(z)$ and $x_1 = g_1(y, z), x_2 = g_2(y, z)$.

4. Define Change of order of integration.

Solution: The evaluation of some double integrals may be very difficult. In this case, we may evaluate it easily by changing the order of integration in a given double integral. When we change the order of integration the limits are also changed. However, when the limits for inner integration are functions, of a variable, the change in a order of integration will result in changes in the limits of integration.

Consider $\int_a^b \int_{f_1(x)}^{f_2(x)} f(x, y) dy dx$ here the inner integration is with respect to y

other is with respect to x. When the order of integration is changed will take

the form $\int_c^d \int_{g_1(y)}^{g_2(y)} f(x, y) dx dy$.

5. Define Area as double integral.

Solution: The double integral of $f(x, y)$ over a region R is defined as

$\iint_R f(x, y) dx dy$. If $f(x, y) = 1$, then $\iint_R dx dy$ represents area of the

region R. The area included between the curves

$y = f_1(x)$ and $y = f_2(x)$ and the ordinates $x = a$ and $x = b$ is given by

$$\text{Area} = \int_a^b \int_{f_1(x)}^{f_2(x)} dy dx.$$

6. Transform the integration $\int_0^y \int_0^x dx dy$ to polar co-ordinates.

Solution: The boundaries are $x = 0$, $x = y$, $y = 0$ and $y = \infty$

In polar co-ordinates, $x = r \cos \theta$, $y = r \sin \theta$

$$dx dy = r dr d\theta$$

θ varies from $\theta = \frac{\pi}{4}$ to $\theta = \frac{\pi}{2}$ and r varies from

$r = 0$ to $r = \infty$

$$\text{Hence } \int_0^y \int_0^x dx dy = \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \int_0^{\infty} r dr d\theta.$$

7. When can a double integral be expressed as a product of two single integrals?

Solution: When the integrand is product of two functions of the variables concerned and limits are constants

Example:
$$\int_a^b \int_c^d f_1(x) f_2(y) dx dy = \int_c^d f_1(x) dx \int_a^b f_2(y) dy .$$

8. What is the geometric meaning of $\iint_R dy dz$?

Solution: This gives the volume of the solid given by $z = f(x, y)$. R is the projection of the surface of the solid in the YZ – plane.

9. Define Volume of integral. (May- 2012)

Solution: Let V be a region of space, bounded by a closed surface. Let $f(x, y, z)$ be a continuous function defined at all points of V. Divide V into n sub regions ΔV_r then $\lim_{\substack{n \rightarrow \infty \\ \Delta V_r \rightarrow 0}} \sum_{r=1}^n f(\xi_r, \eta_r, \zeta_r) \Delta V_r$ is called the volume integral of

$f(x, y, z)$ over the region V and denoted by $\iiint_V f(x, y, z) dv$ or $\iiint_V f(x, y, z) dx dy dz$.

10. Evaluate $\int_0^a \int_0^b xy(x-y) dy dx$ (Jan - 2010)

Solution: Let $I = \int_0^a \int_0^b (x^2 y - xy^2) dy dx$

11.

$$\begin{aligned}
&= \int_0^a \left[x^2 \left(\frac{y^2}{2} \right) - x \left(\frac{y^3}{3} \right) \right]_0^b dx \\
&= \int_0^a \left[x^2 \left(\frac{b^2}{2} \right) - x \left(\frac{b^3}{3} \right) \right] dx = \left[\left(\frac{x^3}{3} \right) \left(\frac{b^2}{2} \right) - \left(\frac{x^2}{2} \right) \left(\frac{b^3}{3} \right) \right]_0^a \\
&= \left[\left(\frac{a^3 b^2}{6} \right) - \left(\frac{a^2 b^3}{6} \right) \right]
\end{aligned}$$

12.

$$= \frac{a^2 b^2}{6} [a - b]$$

13.

11. Find the value of $\int_0^1 \int_0^2 \int_1^2 xyz^2 dz dy dx$

(Jan - 2010)

14.

Solution: Let $I = \int_0^1 \int_0^2 \int_1^2 xyz^2 dz dy dx$

$$= \int_0^1 \int_0^2 xy \left[\frac{z^3}{3} \right]_1^2 dy dx$$

$$= \int_0^1 \int_0^2 xy \left[\frac{(8-1)}{3} \right] dy dx = \frac{7}{3} \int_0^1 x \left[\frac{y^2}{2} \right]_0^2 dx$$

1.

$$= \frac{7}{3} \int_0^1 x \left[\frac{4}{2} \right] dx = \frac{14}{3} \left[\frac{x^2}{2} \right]_0^1$$

2.

$$= \frac{14}{3} \left[\frac{1}{2} \right] = \frac{7}{3}$$

3.

12. Evaluate $\int_0^1 \int_1^2 (x^2 + y^2) dx dy$

(Apr -

2010)

Solution: $I = \int_0^1 \int_1^2 (x^2 + y^2) dx dy$

$$= \int_0^1 \left[\frac{x^3}{3} + y^2(x) \right]_1^2 dy$$

a. $= \int_0^1 \left[\left(\frac{8}{3} + 2y^2 \right) - \left(\frac{1}{3} + y^2 \right) \right] dy$

b. $= \left[\frac{8y}{3} + \frac{2y^3}{3} - \frac{y}{3} - \frac{y^3}{3} \right]_0^1 = \left[\frac{8}{3} + \frac{2}{3} - \frac{1}{3} - \frac{1}{3} \right]$

c. $= \left[\frac{10}{3} - \frac{2}{3} \right] = \frac{8}{3}$

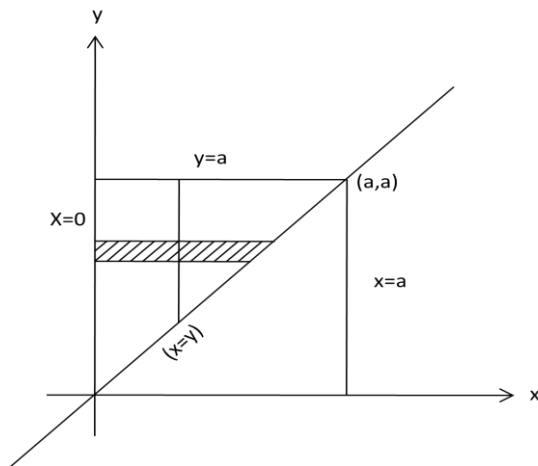
13. Change the order of integration in $\int_0^a \int_x^a (x^2 + y^2) dy dx$ (Apr -

2010)

Solution: In the given integral the region of integration is bounded by the lines $y = x, y = a, x = 0$ and $x = a$

The inner integration is with respect to y , so the strip is parallel to $y -$ axis

The moving area is considered along $x -$ axis.



After changing the order of Integration the strip is parallel to x – axis and the moving area is along y – axis. Now the limits are

x – Limits: x varies from $x = 0$ to $x = y$.

y – Limits: y varies from $y = 0$ to $y = a$

$$\therefore I = \int_0^a \int_0^y (x^2 + y^2) dx dy$$

14. Evaluate $\int_0^1 \int_x^1 x^2 y dy dx$

(Jan -

2011)

Solution: Let $I = \int_0^1 \int_x^1 x^2 y dy dx$

15.
$$= \int_0^1 x^2 \left[\frac{y^2}{2} \right]_x^1 dx$$

16.
$$= \int_0^1 x^2 \left[\frac{1-x^2}{2} \right] dx$$

17.

$$= \frac{1}{2} \int_0^1 [x^2 - x^4] dx = \frac{1}{2} \left[\frac{x^3}{3} - \frac{x^5}{5} \right]_0^1$$

$$= \frac{1}{2} \left[\left(\frac{1}{3} - \frac{1}{5} \right) \right] = \frac{1}{2} \left[\frac{2}{15} \right] = \frac{1}{15}$$

18.

15. If the surface density of the lamina of the points (x, y) is f(x, y) then the
 19. total mass of the lamina is
 (Jan 2011)

Solution: If the surface density of the lamina of the points (x, y) is f(x, y),
 then the total mass of the lamina is $= \iint f(x, y) dx dy$

16. Evaluate $\int_0^{\pi} \int_0^{\sin \theta} r d\theta dr$ (May -
 2011)

Solution: Let $I = \int_0^{\pi} \int_0^{\sin \theta} r d\theta dr$

$$= \int_0^{\pi} \left[\frac{r^2}{2} \right]_0^{\sin \theta} d\theta = \frac{1}{2} \int_0^{\pi} [\sin^2 \theta] d\theta$$

17.

$$= \frac{2}{2} \int_0^{\pi/2} [\sin^2 \theta] d\theta \quad [\text{By Reduction Formula}]$$

18.

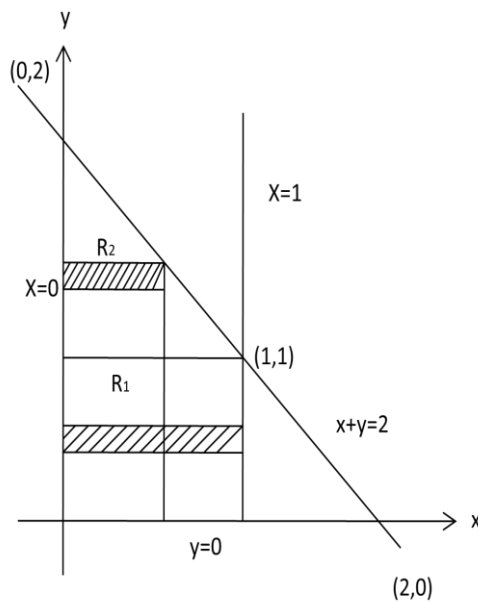
$$= \left[\frac{1}{2} \frac{\pi}{2} \right] = \left[\frac{\pi}{4} \right]$$

19.

17. Change the order of integration in the integral $\int_0^1 \int_0^{2-x} f(x, y) dy dx$ (May -
 2011)

Solution: In the given integral, the region of integration is bounded by the lines

$y = 0$, $y = 2 - x \Rightarrow x + y = 2$, $x = 0$ and $x = 1$. The inner integration is with respect to y , so the strip is parallel to y – axis. The moving area is considered along x – axis.



After changing the order of Integration the strip is parallel to x – axis and the moving area is along y – axis. In y – axis there is two region considered as R_1 and R_2 . Now the limits are

- x – Limits in R_1 : x varies from $x = 0$ to $x = 1$.
- y – Limits in R_1 : y varies from $y = 0$ to $y = 1$
- x – Limits in R_2 : x varies from $x = 0$ to $x = 2 - y$.
- y – Limits in R_2 : y varies from $y = 1$ to $y = 2$

$$\therefore I = \int_0^1 \int_0^1 f(x, y) dx dy + \int_1^2 \int_0^{2-y} f(x, y) dx dy$$

18. Evaluate $\int_0^2 \int_0^1 4xy dx dy$ **(May-2012)**

Solution:
$$I = \int_0^2 \int_0^1 4xy dx dy$$

1.
$$I = 4 \int_0^2 y \left[\frac{x^2}{2} \right]_0^1 dy = 4 \int_0^2 y \left[\frac{1}{2} \right] dy$$

$$I = 2 \int_0^2 y dy = 2 \left[\frac{y^2}{2} \right]_0^2$$

$$I = 2 \left[\frac{4}{2} \right] = 4$$

19. Evaluate $\int_0^a \int_0^x (x^2 + y^2) dy dx$ **(May - 2012)**

Solution: Let
$$I = \int_0^a \int_0^x (x^2 + y^2) dy dx$$

20.
$$I = \int_0^a \left[x^2(y) + \frac{y^3}{3} \right]_0^x dx$$

21.
$$I = \int_0^a \left[x^3 + \frac{x^3}{3} \right] dx$$

22.
$$I = \left[\frac{x^4}{4} + \frac{x^4}{12} \right]_0^a = \frac{a^4}{3}$$

20. Evaluate $\int_0^a \int_0^b \int_0^c (x + y + z) dx dy dz$

(Nov -2012)

Solution:

Let $I = \int_0^a \int_0^b \int_0^c (x + y + z) dx dy dz$

21.
$$= \int_0^a \int_0^b \left[\frac{x^2}{2} + xy + xz \right]_0^c dy dz = \int_0^a \int_0^b \left[\frac{c^2}{2} + cy + cz \right] dy dz$$

22.
$$= \int_0^a \left[\frac{c^2 y}{2} + \frac{cy^2}{2} + czy \right]_0^b dz = \int_0^a \left[\frac{c^2 b}{2} + \frac{cb^2}{2} + czb \right] dz$$

23.
$$= \left[\frac{bc^2 z}{2} + \frac{b^2 cz}{2} + \frac{bcz^2}{2} \right]_0^a = \frac{abc}{2} [a + b + c]$$

21. Evaluate $\int_1^b \int_1^a \frac{dx dy}{xy}$

(Nov -

2012)

Solution: Let $I = \int_1^b \int_1^a \frac{dx dy}{xy}$

$$= \int_1^b \int_1^a \frac{1}{y} [\log x]_1^a dy$$

$$= \int_1^b \frac{1}{y} [\log a - \log 1] dy = \int_1^b \frac{1}{y} [\log a] dy$$

$$= [\log a] \int_1^b \frac{1}{y} dy = [\log a] [\log y]_1^b$$

$$= \log a \log b$$

22. Evaluate $\int_0^1 \int_0^2 \int_0^3 xyz \, dy \, dx \, dz$

(Nov-

2012)

Solution: Let $I = \int_0^1 \int_0^2 \int_0^3 xyz \, dy \, dx \, dz$

$$= \int_0^1 \int_0^2 xz \left[\frac{y^2}{2} \right]_0^3 dx \, dz$$

$$= \int_0^1 \int_0^2 xz \left[\frac{9}{2} \right] dx \, dz = \left[\frac{9}{2} \right] \int_0^1 z \left[\frac{x^2}{2} \right]_0^2 dz$$

$$= \left[\frac{9}{2} \right] \int_0^1 z \left[\frac{4}{2} \right] dz$$

$$= 9 \left[\frac{z^2}{2} \right]_0^1 = \frac{9}{2}$$

23. Define Triple integrals in spherical co-ordinates.

Solution: Spherical co-ordinates (ρ, θ, ϕ) and rectangular co-ordinates (x, y, z) of a point P are related by the equation

$$x = \rho \sin \phi \cos \theta$$

$$y = \rho \sin \phi \sin \theta$$

$$z = \rho \cos \phi$$

$$dx \, dy \, dz = \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta$$

$$\iiint_R f(x, y, z) \, dv = \iiint_R f(\rho \sin \phi \cos \theta, \rho \sin \phi \sin \theta, \rho \cos \phi) \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta$$

24. Write Expression of the mass of the lamina and solid.

Solution: Let the surface density ρ of a plane lamina be a function of the position of a point of the lamina. (ie) $\rho = f(x, y, z)$

Elementary Area of a point $(x, y) = dx dy$

Hence Elementary mass = $\rho dx dy$

$$\text{Total Mass} = \iint \rho dx dy$$

In polar coordinates:

$$\rho = f(r, \theta)$$

$$\text{Mass of lamina} = \iint f(r, \theta) r dr d\theta$$

For a solid, if the density at a point $P(x, y, z)$ be $\rho = f(x, y, z)$, then the total mass of the solid $\iiint_R f(x, y, z) dx dy dz$ with appropriate limits of integration.

25. Find the position of the centre of mass of the region bounded by $x=0$, $y=0$ and $x + y=1$ the density being constant.

Solution: Elementary area = $dx dy$

Density $\rho = k$

Elementary mass = $k dx dy$

$$\text{Total mass} = k \iint_R dx dy$$

x varies from 0 to 1

y varies from 0 to $1-x$

$$\text{Total mass} = k \iint_R dx dy$$

$$= k \int_0^1 \int_0^{1-x} dx dy = k \int_0^1 [y]_0^{1-x} dx$$

$$\begin{aligned}
 &= k \int_0^1 [1-x] dx = k \left[x - \frac{x^2}{2} \right]_0^1 \\
 &= k \left[1 - \frac{1}{2} \right] = \frac{k}{2}
 \end{aligned}$$